Table of Contents

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION

Washington, D.C. 20549



FORM 10-Q

☒ QUARTERLY REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the quarterly period ended June 30, 2019

OR

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the transition period from to

Commission File Number: 1-768

CATERPILLAR INC.

(Exact name of registrant as specified in its charter)

Delaware	**37-0602744**
(State or other jurisdiction of incorporation)	(IRS Employer I.D. No.)
510 Lake Cook Road, Suite 100, Deerfield, Illinois	**60015**
(Address of principal executive offices)	(Zip Code)

Registrant's telephone number, including area code: **(224) 551-4000**

Former Name, Former Address and Former Fiscal Year, if Changed Since Last Report: **N/A**

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol (s)	Name of each exchange on which registered
Common Stock ($1.00 par value)	CAT	New York Stock Exchange
9 3/8% Debentures due March 15, 2021	CAT21	New York Stock Exchange
8% Debentures due February 15, 2023	CAT23	New York Stock Exchange
5.3% Debentures due September 15, 2035	CAT35	New York Stock Exchange

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, smaller reporting company, or an emerging growth company. See definitions of "large accelerated filer," "accelerated filer," "smaller reporting company," and "emerging growth company" in Rule 12b-2 of the Exchange Act.

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

At June 30, 2019, 562,589,191 shares of common stock of the registrant were outstanding.

Table of Contents

Part I. Financial Information

Item 1.	Financial Statements	3
Item 2.	Management's Discussion and Analysis of Financial Condition and Results of Operations	66
Item 3.	Quantitative and Qualitative Disclosures About Market Risk	94
Item 4.	Controls and Procedures	94

Part II. Other Information

Item 1.	Legal Proceedings	95
Item 1A.	Risk Factors	*
Item 2.	Unregistered Sales of Equity Securities and Use of Proceeds	95
Item 3.	Defaults Upon Senior Securities	*
Item 4.	Mine Safety Disclosures	*
Item 5.	Other Information	*
Item 6.	Exhibits	96

* Item omitted because no answer is called for or item is not applicable.

Part I. FINANCIAL INFORMATION

Item 1. Financial Statements

Caterpillar Inc.
Consolidated Statement of Results of Operations
(Unaudited)
(Dollars in millions except per share data)

	Three Months Ended June 30	
	2019	**2018**
Sales and revenues:		
Sales of Machinery, Energy & Transportation	$ 13,671	$ 13,279
Revenues of Financial Products	761	732
Total sales and revenues	14,432	14,011
Operating costs:		
Cost of goods sold	9,941	9,422
Selling, general and administrative expenses	1,309	1,440
Research and development expenses	441	462
Interest expense of Financial Products	192	182
Other operating (income) expenses	336	338
Total operating costs	12,219	11,844
Operating profit	2,213	2,167
Interest expense excluding Financial Products	103	102
Other income (expense)	68	121
Consolidated profit before taxes	2,178	2,186
Provision (benefit) for income taxes	565	490
Profit of consolidated companies	1,613	1,696
Equity in profit (loss) of unconsolidated affiliated companies	6	9
Profit of consolidated and affiliated companies	1,619	1,705
Less: Profit (loss) attributable to noncontrolling interests	(1)	(2)
Profit [1]	$ 1,620	$ 1,707
Profit per common share	$ 2.85	$ 2.86
Profit per common share – diluted [2]	$ 2.83	$ 2.82
Weighted-average common shares outstanding (millions)		
– Basic	567.8	596.2
– Diluted [2]	573.1	604.2

[1] Profit attributable to common shareholders.
[2] Diluted by assumed exercise of stock-based compensation awards using the treasury stock method.

See accompanying notes to Consolidated Financial Statements.

Caterpillar Inc.
Consolidated Statement of Comprehensive Income
(Unaudited)
(Dollars in millions)

	Three Months Ended June 30	
	2019	**2018**
Profit of consolidated and affiliated companies	$ 1,619	$ 1,705
Other comprehensive income (loss), net of tax:		
Foreign currency translation, net of tax (provision)/benefit of: 2019 - $6; 2018 - $(30)	99	(411)
Pension and other postretirement benefits:		
Amortization of prior service (credit) cost, net of tax (provision)/benefit of: 2019 - $3; 2018 - $1	(7)	(7)
Derivative financial instruments:		
Gains (losses) deferred, net of tax (provision)/benefit of: 2019 - $4; 2018 - $(13)	(16)	36
(Gains) losses reclassified to earnings, net of tax (provision)/benefit of: 2019 - $1; 2018 - $30	(1)	(96)
Available-for-sale securities:		
Gains (losses) deferred, net of tax (provision)/benefit of: 2019 - $(2); 2018 - $1	15	(2)
(Gains) losses reclassified to earnings, net of tax (provision)/benefit of: 2019 - $0; 2018 - $0	(1)	—
Total other comprehensive income (loss), net of tax	89	(480)
Comprehensive income	1,708	1,225
Less: comprehensive income attributable to the noncontrolling interests	(1)	(2)
Comprehensive income attributable to shareholders	$ 1,709	$ 1,227

See accompanying notes to Consolidated Financial Statements.

Caterpillar Inc.
Consolidated Statement of Results of Operations
(Unaudited)
(Dollars in millions except per share data)

	Six Months Ended June 30	
	2019	**2018**
Sales and revenues:		
Sales of Machinery, Energy & Transportation	$ 26,395	$ 25,429
Revenues of Financial Products	1,503	1,441
Total sales and revenues	27,898	26,870
Operating costs:		
Cost of goods sold	18,944	17,988
Selling, general and administrative expenses	2,628	2,716
Research and development expenses	876	905
Interest expense of Financial Products	382	348
Other operating (income) expenses	648	638
Total operating costs	23,478	22,595
Operating profit	4,420	4,275
Interest expense excluding Financial Products	206	203
Other income (expense)	228	248
Consolidated profit before taxes	4,442	4,320
Provision (benefit) for income taxes	952	962
Profit of consolidated companies	3,490	3,358
Equity in profit (loss) of unconsolidated affiliated companies	13	14
Profit of consolidated and affiliated companies	3,503	3,372
Less: Profit (loss) attributable to noncontrolling interests	2	—
Profit [1]	$ 3,501	$ 3,372
Profit per common share	$ 6.14	$ 5.65
Profit per common share – diluted [2]	$ 6.08	$ 5.56
Weighted-average common shares outstanding (millions)		
– Basic	569.9	597.0
– Diluted [2]	575.8	606.1

[1] Profit attributable to common shareholders.
[2] Diluted by assumed exercise of stock-based compensation awards using the treasury stock method.

See accompanying notes to Consolidated Financial Statements.

Caterpillar Inc.
Consolidated Statement of Comprehensive Income
(Unaudited)
(Dollars in millions)

	Six Months Ended June 30	
	2019	**2018**
Profit of consolidated and affiliated companies	$ 3,503	$ 3,372
Other comprehensive income (loss), net of tax:		
Foreign currency translation, net of tax (provision)/benefit of: 2019 - $2; 2018 - $(15)	77	(227)
Pension and other postretirement benefits:		
Current year prior service credit (cost), net of tax (provision)/benefit of: 2019 - $0; 2018 - $1	—	(2)
Amortization of prior service (credit) cost, net of tax (provision)/benefit of: 2019 - $6; 2018 - $3	(14)	(14)
Derivative financial instruments:		
Gains (losses) deferred, net of tax (provision)/benefit of: 2019 - $1; 2018 - $(14)	(6)	41
(Gains) losses reclassified to earnings, net of tax (provision)/benefit of: 2019 - $3; 2018 - $24	(10)	(78)
Available-for-sale securities:		
Gains (losses) deferred, net of tax (provision)/benefit of: 2019 - $(8); 2018 - $3	30	(13)
Total other comprehensive income (loss), net of tax	77	(293)
Comprehensive income	3,580	3,079
Less: comprehensive income attributable to the noncontrolling interests	2	—
Comprehensive income attributable to shareholders	$ 3,578	$ 3,079

See accompanying notes to Consolidated Financial Statements.

Caterpillar Inc.
Consolidated Statement of Financial Position
(Unaudited)
(Dollars in millions)

	June 30, 2019	December 31, 2018
Assets		
Current assets:		
Cash and short-term investments	$ 7,429	$ 7,857
Receivables – trade and other	8,996	8,802
Receivables – finance	9,539	8,650
Prepaid expenses and other current assets	1,818	1,765
Inventories	12,007	11,529
Total current assets	39,789	38,603
Property, plant and equipment – net	13,172	13,574
Long-term receivables – trade and other	1,154	1,161
Long-term receivables – finance	12,461	13,286
Noncurrent deferred and refundable income taxes	1,473	1,439
Intangible assets	1,733	1,897
Goodwill	6,211	6,217
Other assets	3,194	2,332
Total assets	$ 79,187	$ 78,509
Liabilities		
Current liabilities:		
Short-term borrowings:		
Machinery, Energy & Transportation	$ —	$ —
Financial Products	5,266	5,723
Accounts payable	7,022	7,051
Accrued expenses	3,789	3,573
Accrued wages, salaries and employee benefits	1,411	2,384
Customer advances	1,263	1,243
Dividends payable	579	495
Other current liabilities	2,157	1,919
Long-term debt due within one year:		
Machinery, Energy & Transportation	13	10
Financial Products	6,235	5,820
Total current liabilities	27,735	28,218
Long-term debt due after one year:		
Machinery, Energy & Transportation	7,657	8,005
Financial Products	17,107	16,995
Liability for postemployment benefits	7,448	7,455
Other liabilities	4,362	3,756
Total liabilities	64,309	64,429
Commitments and contingencies (Notes 11 and 14)		
Shareholders' equity		
Common stock of $1.00 par value:		
Authorized shares: 2,000,000,000 Issued shares: (6/30/19 and 12/31/18 – 814,894,624) at paid-in amount	5,822	5,827
Treasury stock (6/30/19 – 252,305,433 shares; 12/31/18 – 239,351,886 shares) at cost	(22,467)	(20,531)
Profit employed in the business	32,981	30,427
Accumulated other comprehensive income (loss)	(1,499)	(1,684)
Noncontrolling interests	41	41

Total shareholders' equity	14,878	14,080
Total liabilities and shareholders' equity	$ 79,187	$ 78,509

See accompanying notes to Consolidated Financial Statements.

Caterpillar Inc.
Consolidated Statement of Changes in Shareholders' Equity
(Unaudited)
(Dollars in millions)

	Common stock	Treasury stock	Profit employed in the business	Accumulated other comprehensive income (loss)	Noncontrolling interests	Total
Three Months Ended June 30, 2018						
Balance at March 31, 2018	$ 5,640	$ (17,347)	$ 27,929	$ (1,016)	$ 66	$ 15,272
Profit of consolidated and affiliated companies	—	—	1,707	—	(2)	1,705
Foreign currency translation, net of tax	—	—	—	(411)	—	(411)
Pension and other postretirement benefits, net of tax	—	—	—	(7)	—	(7)
Derivative financial instruments, net of tax	—	—	—	(60)	—	(60)
Available-for-sale securities, net of tax	—	—	—	(2)	—	(2)
Dividends declared [1]	—	—	(979)	—	—	(979)
Distribution to noncontrolling interests	—	—	—	—	(1)	(1)
Common shares issued from treasury stock for stock-based compensation: 1,302,281	38	69	—	—	—	107
Stock-based compensation expense	62	—	—	—	—	62
Common shares repurchased: 4,881,793 [2]	—	(750)	—	—	—	(750)
Other	6	—	—	—	—	6
Balance at June 30, 2018	**$ 5,746**	**$ (18,028)**	**$ 28,657**	**$ (1,496)**	**$ 63**	**$ 14,942**
Three Months Ended June 30, 2019						
Balance at March 31, 2019	$ 5,804	$ (21,214)	$ 32,435	$ (1,588)	$ 41	$ 15,478
Profit of consolidated and affiliated companies	—	—	1,620	—	(1)	1,619
Foreign currency translation, net of tax	—	—	—	99	—	99
Pension and other postretirement benefits, net of tax	—	—	—	(7)	—	(7)
Derivative financial instruments, net of tax	—	—	—	(17)	—	(17)
Available-for-sale securities, net of tax	—	—	—	14	—	14
Dividends declared [1]	—	—	(1,074)	—	—	(1,074)
Distribution to noncontrolling interests	—	—	—	—	(1)	(1)
Common shares issued from treasury stock for stock-based compensation: 644,039	11	33	—	—	—	44
Stock-based compensation expense	68	—	—	—	—	68
Common shares repurchased: 9,757,126 [2]	—	(1,286)	—	—	—	(1,286)
Other	(61)	—	—	—	2	(59)
Balance at June 30, 2019	**$ 5,822**	**$ (22,467)**	**$ 32,981**	**$ (1,499)**	**$ 41**	**$ 14,878**

[1] Dividends per share of common stock of $1.89 and $1.64 were declared in the three months ended June 30, 2019 and 2018, respectively.
[2] See Note 12 for additional information.

See accompanying notes to Consolidated Financial Statements.

Caterpillar Inc.
Consolidated Statement of Changes in Shareholders' Equity
(Unaudited)
(Dollars in millions)

	Common stock	Treasury stock	Profit employed in the business	Accumulated other comprehensive income (loss)	Noncontrolling interests	Total
Six Months Ended June 30, 2018						
Balance at December 31, 2017	$ 5,593	$ (17,005)	$ 26,301	$ (1,192)	$ 69	$ 13,766
Adjustments to adopt new accounting guidance						
Revenue recognition	—	—	(12)	—	—	(12)
Tax accounting for intra-entity asset transfers	—	—	(35)	—	—	(35)
Recognition and measurement of financial assets and liabilities	—	—	11	(11)	—	—
Balance at January 1, 2018	5,593	(17,005)	26,265	(1,203)	69	13,719
Profit of consolidated and affiliated companies	—	—	3,372	—	—	3,372
Foreign currency translation, net of tax	—	—	—	(227)	—	(227)
Pension and other postretirement benefits, net of tax	—	—	—	(16)	—	(16)
Derivative financial instruments, net of tax	—	—	—	(37)	—	(37)
Available-for-sale securities, net of tax	—	—	—	(13)	—	(13)
Change in ownership from noncontrolling interests	2	—	—	—	(5)	(3)
Dividends declared [1]	—	—	(980)	—	—	(980)
Distribution to noncontrolling interests	—	—	—	—	(1)	(1)
Common shares issued from treasury stock for stock-based compensation: 4,729,038	29	227	—	—	—	256
Stock-based compensation expense	112	—	—	—	—	112
Common shares repurchased: 8,029,422 [2]	—	(1,250)	—	—	—	(1,250)
Other	10	—	—	—	—	10
Balance at June 30, 2018	**$ 5,746**	**$ (18,028)**	**$ 28,657**	**$ (1,496)**	**$ 63**	**$ 14,942**
Six Months Ended June 30, 2019						
Balance at December 31, 2018	$ 5,827	$ (20,531)	$ 30,427	$ (1,684)	$ 41	$ 14,080
Adjustments to adopt new accounting guidance[3]						
Lease accounting	—	—	235	—	—	235
Reclassification of certain tax effects from accumulated other comprehensive income	—	—	(108)	108	—	—
Balance at January 1, 2019	5,827	(20,531)	30,554	(1,576)	41	14,315
Profit of consolidated and affiliated companies	—	—	3,501	—	2	3,503
Foreign currency translation, net of tax	—	—	—	77	—	77
Pension and other postretirement benefits, net of tax	—	—	—	(14)	—	(14)
Derivative financial instruments, net of tax	—	—	—	(16)	—	(16)
Available-for-sale securities, net of tax	—	—	—	30	—	30
Dividends declared [1]	—	—	(1,074)	—	—	(1,074)
Distribution to noncontrolling interests	—	—	—	—	(2)	(2)
Common shares issued from treasury stock for stock-based compensation: 2,503,104	(62)	101	—	—	—	39
Stock-based compensation expense	113	—	—	—	—	113
Common shares repurchased: 15,456,651 [2]	—	(2,037)	—	—	—	(2,037)
Other	(56)	—	—	—	—	(56)
Balance at June 30, 2019	**$ 5,822**	**$ (22,467)**	**$ 32,981**	**$ (1,499)**	**$ 41**	**$ 14,878**

[1] Dividends per share of common stock of $1.89 and $1.64 were declared in the six months ended June 30, 2019 and 2018, respectively.
[2] See Note 12 for additional information.
[3] See Note 2 for additional information.

See accompanying notes to Consolidated Financial Statements.

Caterpillar Inc.
Consolidated Statement of Cash Flow
(Unaudited)
(Millions of dollars)

	Six Months Ended June 30	
	2019	**2018**
Cash flow from operating activities:		
Profit of consolidated and affiliated companies	$ 3,503	$ 3,372
Adjustments for non-cash items:		
Depreciation and amortization	1,288	1,367
Other	440	446
Changes in assets and liabilities, net of acquisitions and divestitures:		
Receivables – trade and other	(166)	(703)
Inventories	(487)	(1,208)
Accounts payable	134	545
Accrued expenses	151	(31)
Accrued wages, salaries and employee benefits	(979)	(768)
Customer advances	14	(54)
Other assets – net	(221)	174
Other liabilities – net	32	(57)
Net cash provided by (used for) operating activities	3,709	3,083
Cash flow from investing activities:		
Capital expenditures – excluding equipment leased to others	(479)	(645)
Expenditures for equipment leased to others	(746)	(883)
Proceeds from disposals of leased assets and property, plant and equipment	422	539
Additions to finance receivables	(6,181)	(6,143)
Collections of finance receivables	5,902	5,405
Proceeds from sale of finance receivables	119	124
Investments and acquisitions (net of cash acquired)	(3)	(348)
Proceeds from sale of businesses and investments (net of cash sold)	—	12
Proceeds from sale of securities	170	168
Investments in securities	(243)	(318)
Other – net	(40)	21
Net cash provided by (used for) investing activities	(1,079)	(2,068)
Cash flow from financing activities:		
Dividends paid	(986)	(933)
Common stock issued, including treasury shares reissued	39	256
Common shares repurchased	(2,105)	(1,250)
Proceeds from debt issued (original maturities greater than three months):		
Machinery, Energy & Transportation	—	—
Financial Products	5,340	4,307
Payments on debt (original maturities greater than three months):		
Machinery, Energy & Transportation	(4)	(3)
Financial Products	(4,897)	(4,433)
Short-term borrowings – net (original maturities three months or less)	(436)	1,487
Other – net	(2)	(4)
Net cash provided by (used for) financing activities	(3,051)	(573)
Effect of exchange rate changes on cash	(10)	(68)
Increase (decrease) in cash and short-term investments and restricted cash	(431)	374
Cash and short-term investments and restricted cash at beginning of period	7,890	8,320
Cash and short-term investments and restricted cash at end of period	$ 7,459	$ 8,694

All short-term investments, which consist primarily of highly liquid investments with original maturities of three months or less, are considered to be cash equivalents.

See accompanying notes to Consolidated Financial Statements.

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS
(Unaudited)

1. A. Nature of operations

Information in our financial statements and related commentary are presented in the following categories:

Machinery, Energy & Transportation (ME&T) – Represents the aggregate total of Construction Industries, Resource Industries, Energy & Transportation and the All Other operating segment and related corporate items and eliminations.

Financial Products – Primarily includes the company's Financial Products Segment. This category includes Caterpillar Financial Services Corporation (Cat Financial), Caterpillar Insurance Holdings Inc. (Insurance Services) and their respective subsidiaries.

B. Basis of presentation

In the opinion of management, the accompanying unaudited financial statements include all adjustments, consisting only of normal recurring adjustments, necessary for a fair statement of (a) the consolidated results of operations for the three and six months ended June 30, 2019 and 2018, (b) the consolidated comprehensive income for the three and six months ended June 30, 2019 and 2018, (c) the consolidated financial position at June 30, 2019 and December 31, 2018, (d) the consolidated changes in shareholders' equity for the three and six months ended June 30, 2019 and 2018 and (e) the consolidated cash flow for the six months ended June 30, 2019 and 2018. The financial statements have been prepared in conformity with generally accepted accounting principles in the United States of America (U.S. GAAP) and pursuant to the rules and regulations of the Securities and Exchange Commission (SEC).

Interim results are not necessarily indicative of results for a full year. The information included in this Form 10-Q should be read in conjunction with the audited financial statements and notes thereto included in our company's annual report on Form 10-K for the year ended December 31, 2018 (2018 Form 10-K).

The December 31, 2018 financial position data included herein is derived from the audited consolidated financial statements included in the 2018 Form 10-K but does not include all disclosures required by U.S. GAAP. Certain amounts for prior periods have been reclassified to conform to the current period financial statement presentation.

Unconsolidated Variable Interest Entities (VIEs)

We have affiliates, suppliers and dealers that are VIEs of which we are not the primary beneficiary. Although we have provided financial support, we do not have the power to direct the activities that most significantly impact the economic performance of each entity. Our maximum exposure to loss from VIEs for which we are not the primary beneficiary was $148 million and $131 million as of June 30, 2019 and December 31, 2018, respectively.

In addition, Cat Financial has end-user customers that are VIEs of which we are not the primary beneficiary. Although we have provided financial support to these entities and therefore have a variable interest, we do not have the power to direct the activities that most significantly impact their economic performance. Our maximum exposure to loss from our involvement with these VIEs is limited to the credit risk inherently present in the financial support that we have provided. These risks are evaluated and reflected in our financial statements as part of our overall portfolio of finance receivables and related allowance for credit losses.

2. New accounting guidance

A. Adoption of new accounting standards

Lease accounting (Accounting Standards Update (ASU) 2016-02) – In February 2016, the Financial Accounting Standards Board (FASB) issued accounting guidance that revises the accounting for leases. Under the new guidance, lessees are required to recognize a right-of-use asset and a lease liability for substantially all leases. The new guidance will continue to classify leases as either financing or operating, with classification affecting the pattern of expense recognition. The accounting applied by a lessor under the new guidance will be substantially equivalent to current lease accounting guidance. The new guidance was effective January 1, 2019 and was applied using a modified retrospective approach through a cumulative effect adjustment to retained earnings as of January 1, 2019. The prior period comparative information has not been recasted and continues to be reported under the accounting guidance in effect for those periods.

The new guidance provides a number of optional practical expedients in transition. We elected the "package of practical expedients," which allows us not to reassess under the new guidance our prior conclusions about lease identification, lease classification and initial direct costs. We did not elect the use-of-hindsight practical expedient. In addition, the new guidance provides practical expedients for an entity's ongoing lessee accounting. For certain property and information technology equipment leases, we have elected to separate payments for lease components from non-lease components. For all other leases, we have elected not to separate lease and non-lease components. We have elected the short-term lease recognition exemption for all leases that qualify, which means we will not recognize right-of-use assets or lease liabilities for these leases with a term of twelve months or less.

The most significant effects of adoption relate to the recognition of right-of-use assets and lease liabilities on our balance sheet for operating leases and providing new disclosures about our leasing activities. In addition, we derecognized existing assets and debt obligations for a sale-leaseback transaction that qualified for sale accounting under the new guidance. The gain associated with this change in accounting was recognized through opening retained earnings as of January 1, 2019. The adoption did not have a material impact on our results of operations.

In March 2019, the FASB issued Leases - Codification improvements (ASU 2019-01) which amended the new leasing guidance. Under these amendments, lessors that are not manufacturers or dealers will use their cost, less any discounts that may apply, as the fair value of the underlying asset, and lessors within the scope of Financial Services-Depository and Lending guidance will present all principal payments received under leases within investment activities on the statement of cash flows. We adopted the new guidance effective January 1, 2019, and the adoption did not have a material impact to our financial statements.

See Note 10 for additional information.

The cumulative effect of initially applying the new lease guidance to our consolidated financial statements on January 1, 2019 was as follows:

Consolidated Statement of Financial Position (Millions of dollars)	Balance as of December 31, 2018	Cumulative Impact from Adopting New Lease Guidance	Balance as of January 1, 2019
Assets			
Prepaid expenses and other current assets	$ 1,765	$ (17)	$ 1,748
Property, plant and equipment - net	$ 13,574	$ (26)	$ 13,548
Noncurrent deferred and refundable income taxes	$ 1,439	$ (77)	$ 1,362
Other assets	$ 2,332	$ 713	$ 3,045
Liabilities			
Accrued expenses	$ 3,573	$ (27)	$ 3,546
Other current liabilities	$ 1,919	$ 209	$ 2,128
Long-term debt due after one year			
Machinery, Energy & Transportation	$ 8,005	$ (362)	$ 7,643
Other liabilities	$ 3,756	$ 538	$ 4,294
Shareholders' equity			
Profit employed in the business	$ 30,427	$ 235	$ 30,662

The following ASUs were effective January 1, 2019 and did not have a material impact on our financial statements:

ASU	Description
2017-08	Premium amortization on purchased callable debt securities
2017-12	Derivatives and hedging - Targeted improvements
2018-02	Reclassification of certain tax effects from accumulated other comprehensive income

B. Accounting standards issued but not yet adopted

Credit losses (ASU 2016-13) – In June 2016, the FASB issued accounting guidance to introduce a new model for recognizing credit losses on financial instruments based on an estimate of current expected credit losses. The new guidance will apply to loans, accounts receivable, trade receivables, other financial assets measured at amortized cost, loan commitments and other off-balance sheet credit exposures. The new guidance will also apply to debt securities and other financial assets measured at fair value through other comprehensive income. The new guidance is effective January 1, 2020, with early adoption permitted beginning January 1, 2019. An implementation team continues to evaluate data requirements and methodologies and has started designing new processes and controls. The team is also modeling results to assess the effect of the new guidance on our financial statements. We plan to adopt the new guidance effective January 1, 2020.

We consider the applicability and impact of all ASUs. ASUs not listed above were assessed and either determined to be not applicable or not expected to have a material impact on our financial statements.

3. Sales and revenue contract information

Trade receivables represent amounts due from dealers and end users for the sale of our products. In addition, Cat Financial provides wholesale inventory financing for a dealer's purchase of inventory. Wholesale inventory receivables are included in Receivables – trade and other and Long-term receivables – trade and other in the Consolidated Statement of Financial Position. Trade receivables from dealers and end users were $7,870 million and $7,743 million as of June 30, 2019 and December 31, 2018, respectively, and are recognized in Receivables – trade and other in the Consolidated Statement of Financial Position. Long-term trade receivables from dealers and end users were $663 million and $674 million as of June 30, 2019 and December 31, 2018, respectively, and are recognized in Long-term receivables – trade and other in the Consolidated Statement of Financial Position.

We invoice in advance of recognizing the sale of certain products. We recognize advanced customer payments as a contract liability in Customer advances and Other liabilities in the Consolidated Statement of Financial Position. Long-term customer advances recognized in Other liabilities in the Consolidated Statement of Financial Position were $434 million and $437 million as of June 30, 2019 and December 31, 2018, respectively. We reduce the contract liability when revenue is recognized. During the three and six months ended June 30, 2019, we recognized $368 million and $875 million, respectively, of revenue that was recorded as a contract liability at the beginning of 2019.

As of June 30, 2019, we have entered into contracts with dealers and end users for which sales have not been recognized as we have not satisfied our performance obligations and transferred control of the products. The dollar amount of unsatisfied performance obligations for contracts with an original duration greater than one year is $6.0 billion, of which $2.4 billion is expected to be completed and revenue recognized in the twelve months following June 30, 2019. We have elected the practical expedient not to disclose unsatisfied performance obligations with an original contract duration of one year or less. Contracts with an original duration of one year or less are primarily sales to dealers for machinery, engines and replacement parts.

See Note 16 for further disaggregated sales and revenues information.

4. Stock-based compensation

Accounting for stock-based compensation requires that the cost resulting from all stock-based payments be recognized in the financial statements based on the grant date fair value of the award. Our stock-based compensation primarily consists of stock options, restricted stock units (RSUs) and performance-based restricted stock units (PRSUs).

Upon separation from service, if a participant is 55 years of age or older with more than five years of service, the participant meets the criteria for a "Long Service Separation." For PRSU awards granted prior to 2019, only a prorated number of shares may vest at the end of the performance period based upon achievement of the performance target, with the proration based upon the number of months of continuous employment during the three-year performance period. Award terms for the 2019 PRSU grant allow for continued vesting upon achievement of the performance target specified in the award document for employees who meet the criteria for a "Long Service Separation" and fulfill a requisite service period of six months. Compensation expense for the 2019 PRSU grant with respect to employees who have met the criteria for a "Long Service Separation" is recognized over the period from the grant date to the end of the six-month requisite service period. For employees who become eligible for a "Long Service Separation" subsequent to the end date of the six-month requisite service period and prior to the completion of the vesting period, compensation expense is recognized over the period from the grant date to the date eligibility is achieved.

We recognized pretax stock-based compensation expense of $68 million and $113 million for the three and six months ended June 30, 2019, respectively, and $62 million and $112 million for the three and six months ended June 30, 2018, respectively.

The following table illustrates the type and fair value of the stock-based compensation awards granted during the six months ended June 30, 2019 and 2018, respectively:

	Six Months Ended June 30, 2019			Six Months Ended June 30, 2018		
	Shares Granted	Weighted-Average Fair Value Per Share	Weighted-Average Grant Date Stock Price	Shares Granted	Weighted-Average Fair Value Per Share	Weighted-Average Grant Date Stock Price
Stock options	1,499,524	$ 40.98	$ 138.35	1,566,788	$ 46.18	$ 151.12
RSUs	657,389	$ 138.35	$ 138.35	683,339	$ 151.16	$ 151.16
PRSUs	342,097	$ 138.35	$ 138.35	339,559	$ 151.12	$ 151.12

The following table provides the assumptions used in determining the fair value of the stock-based awards for the six months ended June 30, 2019 and 2018, respectively:

	Grant Year	
	2019	2018
Weighted-average dividend yield	2.56%	2.70%
Weighted-average volatility	29.1%	30.2%
Range of volatilities	25.1-38.7%	21.5-33.0%
Range of risk-free interest rates	2.48-2.68%	2.02-2.87%
Weighted-average expected lives	7 years	8 years

As of June 30, 2019, the total remaining unrecognized compensation expense related to nonvested stock-based compensation awards was $261 million, which will be amortized over the weighted-average remaining requisite service periods of approximately 1.8 years.

5. Derivative financial instruments and risk management

Our earnings and cash flow are subject to fluctuations due to changes in foreign currency exchange rates, interest rates and commodity prices. Our Risk Management Policy (policy) allows for the use of derivative financial instruments to prudently manage foreign currency exchange rate, interest rate and commodity price exposures. Our policy specifies that derivatives are not to be used for speculative purposes. Derivatives that we use are primarily foreign currency forward, option and cross currency contracts, interest rate contracts and commodity forward and option contracts. Our derivative activities are subject to the management, direction and control of our senior financial officers. Risk management practices, including the use of financial derivative instruments, are presented to the Audit Committee of the Board of Directors at least annually.

All derivatives are recognized on the Consolidated Statement of Financial Position at their fair value. On the date the derivative contract is entered into, we designate the derivative as (1) a hedge of the fair value of a recognized asset or liability (fair value hedge), (2) a hedge of a forecasted transaction or the variability of cash flow (cash flow hedge) or (3) an undesignated instrument. Changes in the fair value of a derivative that is qualified, designated and highly effective as a fair value hedge, along with the gain or loss on the hedged recognized asset or liability that is attributable to the hedged risk, are recorded in current earnings. Changes in the fair value of a derivative that is qualified, designated and highly effective as a cash flow hedge are recorded in Accumulated other comprehensive income (loss) (AOCI), to the extent effective, on the Consolidated Statement of Financial Position until they are reclassified to earnings in the same period or periods during which the hedged transaction affects earnings. Changes in the fair value of undesignated derivative instruments are reported in current earnings. Cash flows from designated derivative financial instruments are classified within the same category as the item being hedged on the Consolidated Statement of Cash Flow. Cash flows from undesignated derivative financial instruments are included in the investing category on the Consolidated Statement of Cash Flow.

We formally document all relationships between hedging instruments and hedged items, as well as the risk-management objective and strategy for undertaking various hedge transactions. This process includes linking all derivatives that are designated as fair value hedges to specific assets and liabilities on the Consolidated Statement of Financial Position and linking cash flow hedges to specific forecasted transactions or variability of cash flow.

We also formally assess, both at the hedge's inception and on an ongoing basis, whether the designated derivatives that are used in hedging transactions are highly effective in offsetting changes in fair values or cash flow of hedged items. When a derivative is determined not to be highly effective as a hedge or the underlying hedged transaction is no longer probable, we discontinue hedge accounting prospectively, in accordance with the derecognition criteria for hedge accounting.

Foreign Currency Exchange Rate Risk

Foreign currency exchange rate movements create a degree of risk by affecting the U.S. dollar value of sales made and costs incurred in foreign currencies. Movements in foreign currency rates also affect our competitive position as these changes may affect business practices and/or pricing strategies of non-U.S.-based competitors. Additionally, we have balance sheet positions denominated in foreign currencies, thereby creating exposure to movements in exchange rates.

Our Machinery, Energy & Transportation operations purchase, manufacture and sell products in many locations around the world. As we have a diversified revenue and cost base, we manage our future foreign currency cash flow exposure on a net basis. We use foreign currency forward and option contracts to manage unmatched foreign currency cash inflow and outflow. Our objective is to minimize the risk of exchange rate movements that would reduce the U.S. dollar value of our foreign currency cash flow. Our policy allows for managing anticipated foreign currency cash flow for up to five years. As of June 30, 2019, the maximum term of these outstanding contracts was approximately 51 months.

We generally designate as cash flow hedges at inception of the contract any Australian dollar, Brazilian real, British pound, Canadian dollar, Chinese yuan, Indian rupee, Japanese yen, Mexican peso, Singapore dollar or Thailand baht forward or option contracts that meet the requirements for hedge accounting and the maturity extends beyond the current quarter-end. Designation is performed on a specific exposure basis to support hedge accounting. The remainder of Machinery, Energy & Transportation foreign currency contracts are undesignated.

As of June 30, 2019, $6 million of deferred net gains, net of tax, included in equity (AOCI in the Consolidated Statement of Financial Position), are expected to be reclassified to current earnings (Other income (expense) in the Consolidated Statement of Results of Operations) over the next twelve months when earnings are affected by the hedged transactions. The actual amount recorded in Other income (expense) will vary based on exchange rates at the time the hedged transactions impact earnings.

In managing foreign currency risk for our Financial Products operations, our objective is to minimize earnings volatility resulting from conversion and the remeasurement of net foreign currency balance sheet positions and future transactions denominated in foreign currencies. Our policy allows the use of foreign currency forward, option and cross currency contracts to offset the risk of currency mismatch between our assets and liabilities and exchange rate risk associated with future transactions denominated in foreign currencies. Our foreign currency forward and option contracts are primarily undesignated. We designate fixed-to-fixed cross currency contracts as cash flow hedges to protect against movements in exchange rates on foreign currency fixed-rate assets and liabilities.

Interest Rate Risk

Interest rate movements create a degree of risk by affecting the amount of our interest payments and the value of our fixed-rate debt. Our practice is to use interest rate contracts to manage our exposure to interest rate changes.

Our Machinery, Energy & Transportation operations generally use fixed-rate debt as a source of funding. Our objective is to minimize the cost of borrowed funds. Our policy allows us to enter into fixed-to-floating interest rate contracts and forward rate agreements to meet that objective. We designate fixed-to-floating interest rate contracts as fair value hedges at inception of the contract, and we designate certain forward rate agreements as cash flow hedges at inception of the contract.

Financial Products operations has a match-funding policy that addresses interest rate risk by aligning the interest rate profile (fixed or floating rate) of Cat Financial's debt portfolio with the interest rate profile of our receivables portfolio within predetermined ranges on an ongoing basis. In connection with that policy, we use interest rate derivative instruments to modify the debt structure to match assets within the receivables portfolio. This matched funding reduces the volatility of margins between interest-bearing assets and interest-bearing liabilities, regardless of which direction interest rates move.

Our policy allows us to use fixed-to-floating, floating-to-fixed and floating-to-floating interest rate contracts to meet the match-funding objective. We designate fixed-to-floating interest rate contracts as fair value hedges to protect debt against changes in fair value due to changes in the benchmark interest rate. We designate most floating-to-fixed interest rate contracts as cash flow hedges to protect against the variability of cash flows due to changes in the benchmark interest rate.

We have, at certain times, liquidated fixed-to-floating and floating-to-fixed interest rate contracts at both Machinery, Energy & Transportation and Financial Products. The gains or losses associated with these contracts at the time of liquidation are amortized into earnings over the original term of the previously designated hedged item.

Commodity Price Risk

Commodity price movements create a degree of risk by affecting the price we must pay for certain raw material. Our policy is to use commodity forward and option contracts to manage the commodity risk and reduce the cost of purchased materials.

Our Machinery, Energy & Transportation operations purchase base and precious metals embedded in the components we purchase from suppliers. Our suppliers pass on to us price changes in the commodity portion of the component cost. In addition, we are subject to price changes on energy products such as natural gas and diesel fuel purchased for operational use.

Our objective is to minimize volatility in the price of these commodities. Our policy allows us to enter into commodity forward and option contracts to lock in the purchase price of a portion of these commodities within a five-year horizon. All such commodity forward and option contracts are undesignated.

The location and fair value of derivative instruments reported in the Consolidated Statement of Financial Position are as follows:

(Millions of dollars)	Consolidated Statement of Financial Position Location	Asset (Liability) Fair Value	
		June 30, 2019	December 31, 2018
Designated derivatives			
Foreign exchange contracts			
Machinery, Energy & Transportation	Receivables – trade and other	$ 26	$ 16
Machinery, Energy & Transportation	Accrued expenses	(17)	(26)
Machinery, Energy & Transportation	Other liabilities	(4)	(9)
Financial Products	Receivables – trade and other	47	53
Financial Products	Long-term receivables – trade and other	15	35
Financial Products	Accrued expenses	(17)	(9)
Interest rate contracts			
Financial Products	Receivables – trade and other	—	1
Financial Products	Long-term receivables – trade and other	6	3
Financial Products	Accrued expenses	(71)	(40)
		$ (15)	$ 24
Undesignated derivatives			
Foreign exchange contracts			
Machinery, Energy & Transportation	Receivables – trade and other	$ 5	$ 2
Machinery, Energy & Transportation	Accrued expenses	—	(21)
Financial Products	Receivables – trade and other	11	15
Financial Products	Long-term receivables – trade and other	5	5
Financial Products	Accrued expenses	(20)	(14)
Commodity contracts			
Machinery, Energy & Transportation	Receivables – trade and other	4	1
Machinery, Energy & Transportation	Accrued expenses	(8)	(31)
Machinery, Energy & Transportation	Other liabilities	(1)	—
		$ (4)	$ (43)

The total notional amounts of the derivative instruments are as follows:

(Millions of dollars)	June 30, 2019	December 31, 2018
Machinery, Energy & Transportation	$ 2,110	$ 1,834
Financial Products	$ 8,736	$ 10,210

The notional amounts of the derivative financial instruments do not represent amounts exchanged by the parties. The amounts exchanged by the parties are calculated by reference to the notional amounts and by other terms of the derivatives, such as foreign currency exchange rates, interest rates or commodity prices.

The effect of derivatives designated as hedging instruments on the Consolidated Statement of Results of Operations is as follows:

Cash Flow Hedges

	Three Months Ended June 30, 2019			
		Recognized in Earnings		
(Millions of dollars)	Amount of Gains (Losses) Recognized in AOCI	Classification of Gains (Losses)	Amount of Gains (Losses) Reclassified from AOCI	Amount of the line items in the Consolidated Statement of Results of Operations containing hedging gains (losses)
Foreign exchange contracts				
Machinery, Energy & Transportation	$ 4			
		Cost of goods sold	$ (1)	$ 9,941
Financial Products	10			
		Interest expense of Financial Products	7	$ 192
		Other income (expense)	(4)	$ 68
Interest rate contracts				
Financial Products	(34)	Interest expense of Financial Products	—	$ 192
	$ (20)		$ 2	

	Three Months Ended June 30, 2018			
		Recognized in Earnings		
	Amount of Gains (Losses) Recognized in AOCI (Effective Portion)	Classification of Gains (Losses)	Amount of Gains (Losses) Reclassified from AOCI	Recognized in Earnings (Ineffective Portion)
Foreign exchange contracts				
Machinery, Energy & Transportation	$ (79)	Other income (expense)	$ 4	$ —
Financial Products	123	Other income (expense)	119	—
		Interest expense of Financial Products	5	—
Interest rate contracts				
Machinery, Energy & Transportation	—	Interest expense excluding Financial Products	(2)	—
Financial Products	5	Interest expense of Financial Products	—	—
	$ 49		$ 126	$ —

Cash Flow Hedges

(Millions of dollars)	Six Months Ended June 30, 2019			
		Recognized in Earnings		
	Amount of Gains (Losses) Recognized in AOCI	Classification of Gains (Losses)	Amount of Gains (Losses) Reclassified from AOCI	Amount of the line items in the Consolidated Statement of Results of Operations containing hedging gains (losses)
Foreign exchange contracts				
Machinery, Energy & Transportation	$ 21			
		Sales of Machinery, Energy & Transportation	$ 1	$ 26,395
		Cost of goods sold	(4)	$ 18,944
Financial Products	32			
		Interest expense of Financial Products	14	$ 382
		Other income (expense)	2	$ 228
Interest rate contracts				
Machinery, Energy & Transportation	—	Interest expense excluding Financial Products	(1)	$ 206
Financial Products	(60)	Interest expense of Financial Products	1	$ 382
	$ (7)		$ 13	

	Six Months Ended June 30, 2018			
		Recognized in Earnings		
	Amount of Gains (Losses) Recognized in AOCI (Effective Portion)	Classification of Gains (Losses)	Amount of Gains (Losses) Reclassified from AOCI	Recognized in Earnings (Ineffective Portion)
Foreign exchange contracts				
Machinery, Energy & Transportation	$ (40)	Other income (expense)	$ 5	$ —
Financial Products	90	Other income (expense)	90	—
		Interest expense of Financial Products	8	—
Interest rate contracts				
Machinery, Energy & Transportation	—	Interest expense excluding Financial Products	(2)	—
Financial Products	5	Interest expense of Financial Products	1	—
	$ 55		$ 102	$ —

The effect of derivatives not designated as hedging instruments on the Consolidated Statement of Results of Operations is as follows:

(Millions of dollars)	Classification of Gains (Losses)	Three Months Ended June 30, 2019	Three Months Ended June 30, 2018
Foreign exchange contracts			
Machinery, Energy & Transportation	Other income (expense)	$ 7	$ (54)
Financial Products	Other income (expense)	(10)	23
Commodity contracts			
Machinery, Energy & Transportation	Other income (expense)	(7)	9
		$ (10)	$ (22)

	Classification of Gains (Losses)	Six Months Ended June 30, 2019	Six Months Ended June 30, 2018
Foreign exchange contracts			
Machinery, Energy & Transportation	Other income (expense)	$ 13	$ (38)
Financial Products	Other income (expense)	(39)	16
Commodity contracts			
Machinery, Energy & Transportation	Other income (expense)	16	—
		$ (10)	$ (22)

We enter into International Swaps and Derivatives Association (ISDA) master netting agreements within Machinery, Energy & Transportation and Financial Products that permit the net settlement of amounts owed under their respective derivative contracts. Under these master netting agreements, net settlement generally permits the company or the counterparty to determine the net amount payable for contracts due on the same date and in the same currency for similar types of derivative transactions. The master netting agreements generally also provide for net settlement of all outstanding contracts with a counterparty in the case of an event of default or a termination event.

Collateral is generally not required of the counterparties or of our company under the master netting agreements. As of June 30, 2019 and December 31, 2018, no cash collateral was received or pledged under the master netting agreements.

The effect of the net settlement provisions of the master netting agreements on our derivative balances upon an event of default or termination event is as follows:

June 30, 2019 (Millions of dollars)	Gross Amount of Recognized Assets	Gross Amounts Offset in the Statement of Financial Position	Net Amount of Assets Presented in the Statement of Financial Position	Gross Amounts Not Offset in the Statement of Financial Position		Net Amount of Assets
				Financial Instruments	Cash Collateral Received	
Derivatives						
Machinery, Energy & Transportation	$ 35	$ —	$ 35	$ (20)	$ —	$ 15
Financial Products	84	—	84	(19)	—	65
Total	$ 119	$ —	$ 119	$ (39)	$ —	$ 80

June 30, 2019 (Millions of dollars)	Gross Amount of Recognized Liabilities	Gross Amounts Offset in the Statement of Financial Position	Net Amount of Liabilities Presented in the Statement of Financial Position	Gross Amounts Not Offset in the Statement of Financial Position		Net Amount of Liabilities
				Financial Instruments	Cash Collateral Pledged	
Derivatives						
Machinery, Energy & Transportation	$ (30)	$ —	$ (30)	$ 20	$ —	$ (10)
Financial Products	(108)	—	(108)	19	—	(89)
Total	$ (138)	$ —	$ (138)	$ 39	$ —	$ (99)

December 31, 2018 (Millions of dollars)	Gross Amount of Recognized Assets	Gross Amounts Offset in the Statement of Financial Position	Net Amount of Assets Presented in the Statement of Financial Position	Gross Amounts Not Offset in the Statement of Financial Position		Net Amount of Assets
				Financial Instruments	Cash Collateral Received	
Derivatives						
Machinery, Energy & Transportation	$ 19	$ —	$ 19	$ (19)	$ —	$ —
Financial Products	112	—	112	(34)	—	78
Total	$ 131	$ —	$ 131	$ (53)	$ —	$ 78

December 31, 2018 (Millions of dollars)	Gross Amount of Recognized Liabilities	Gross Amounts Offset in the Statement of Financial Position	Net Amount of Liabilities Presented in the Statement of Financial Position	Gross Amounts Not Offset in the Statement of Financial Position		Net Amount of Liabilities
				Financial Instruments	Cash Collateral Pledged	
Derivatives						
Machinery, Energy & Transportation	$ (87)	$ —	$ (87)	$ 19	$ —	$ (68)
Financial Products	(63)	—	(63)	34	—	(29)
Total	$ (150)	$ —	$ (150)	$ 53	$ —	$ (97)

6. Inventories

Inventories (principally using the last-in, first-out (LIFO) method) are comprised of the following:

(Millions of dollars)	June 30, 2019	December 31, 2018
Raw materials	$ 3,593	$ 3,382
Work-in-process	2,574	2,674
Finished goods	5,604	5,241
Supplies	236	232
Total inventories	$ 12,007	$ 11,529

7. Intangible assets and goodwill

A. Intangible assets

Intangible assets are comprised of the following:

(Millions of dollars)	Weighted Amortizable Life (Years)	June 30, 2019 Gross Carrying Amount	Accumulated Amortization	Net
Customer relationships	15	$ 2,460	$ (1,330)	$ 1,130
Intellectual property	12	1,516	(992)	524
Other	12	199	(120)	79
Total finite-lived intangible assets	14	$ 4,175	$ (2,442)	$ 1,733

	Weighted Amortizable Life (Years)	December 31, 2018 Gross Carrying Amount	Accumulated Amortization	Net
Customer relationships	15	$ 2,463	$ (1,249)	$ 1,214
Intellectual property	11	1,557	(965)	592
Other	13	199	(108)	91
Total finite-lived intangible assets	14	$ 4,219	$ (2,322)	$ 1,897

Amortization expense for the three and six months ended June 30, 2019 was $81 million and $163 million, respectively. Amortization expense for the three and six months ended June 30, 2018 was $83 million and $166 million, respectively. Amortization expense related to intangible assets is expected to be:

(Millions of dollars)					
Remaining Six Months of 2019	2020	2021	2022	2023	Thereafter
$162	$311	$293	$274	$216	$477

B. Goodwill

No goodwill was impaired during the six months ended June 30, 2019 or 2018.

The changes in carrying amount of goodwill by reportable segment for the six months ended June 30, 2019 were as follows:

(Millions of dollars)	December 31, 2018	Other Adjustments [1]	June 30, 2019
Construction Industries			
Goodwill	$ 304	$ 4	$ 308
Impairments	(22)	—	(22)
Net goodwill	282	4	286
Resource Industries			
Goodwill	4,172	(7)	4,165
Impairments	(1,175)	—	(1,175)
Net goodwill	2,997	(7)	2,990
Energy & Transportation			
Goodwill	2,882	(4)	2,878
All Other [2]			
Goodwill	56	1	57
Consolidated total			
Goodwill	7,414	(6)	7,408
Impairments	(1,197)	—	(1,197)
Net goodwill	$ 6,217	$ (6)	$ 6,211

[1] Other adjustments are comprised primarily of foreign currency translation.
[2] Includes All Other operating segment (See Note 16).

8. Investments in debt and equity securities

We have investments in certain debt and equity securities, primarily at Insurance Services, which are recorded at fair value and are primarily included in Other assets in the Consolidated Statement of Financial Position.

Debt securities have been classified as available-for-sale, and the unrealized gains and losses arising from the revaluation of these debt securities are included, net of applicable deferred income taxes, in equity (Accumulated other comprehensive income (loss) in the Consolidated Statement of Financial Position). The unrealized gains and losses arising from the revaluation of the equity securities are included in Other income (expense) in the Consolidated Statement of Results of Operations. Realized gains and losses on sales of investments are generally determined using the specific identification method for debt and equity securities and are included in Other income (expense) in the Consolidated Statement of Results of Operations.

The cost basis and fair value of debt securities with unrealized gains and losses included in equity (Accumulated other comprehensive income (loss) in the Consolidated Statement of Financial Position) were as follows:

(Millions of dollars)	June 30, 2019			December 31, 2018		
	Cost Basis	Unrealized Pretax Net Gains (Losses)	Fair Value	Cost Basis	Unrealized Pretax Net Gains (Losses)	Fair Value
Government debt						
U.S. treasury bonds	$ 9	$ —	$ 9	$ 9	$ —	$ 9
Other U.S. and non-U.S. government bonds	47	—	47	42	—	42
Corporate bonds						
Corporate bonds	805	14	819	735	(15)	720
Asset-backed securities	56	—	56	63	—	63
Mortgage-backed debt securities						
U.S. governmental agency	302	3	305	301	(4)	297
Residential	7	—	7	7	—	7
Commercial	36	1	37	14	(1)	13
Total debt securities	$ 1,262	$ 18	$ 1,280	$ 1,171	$ (20)	$ 1,151

Available-for-sale investments in an unrealized loss position that are not other-than-temporarily impaired:

(Millions of dollars)	June 30, 2019					
	Less than 12 months [1]		12 months or more [1]		Total	
	Fair Value	Unrealized Losses	Fair Value	Unrealized Losses	Fair Value	Unrealized Losses
Corporate bonds						
Corporate bonds	$ 33	$ —	$ 134	$ 1	$ 167	$ 1
Mortgage-backed debt securities						
U.S. governmental agency	9	—	57	1	66	1
Total	$ 42	$ —	$ 191	$ 2	$ 233	$ 2

(Millions of dollars)	December 31, 2018					
	Less than 12 months [1]		12 months or more [1]		Total	
	Fair Value	Unrealized Losses	Fair Value	Unrealized Losses	Fair Value	Unrealized Losses
Corporate bonds						
Corporate bonds	$ 280	$ 3	$ 391	$ 11	$ 671	$ 14
Asset-backed securities	6	—	38	1	44	1
Mortgage-backed debt securities						
U.S. governmental agency	52	—	223	5	275	5
Commercial	—	—	14	1	14	1
Total	$ 338	$ 3	$ 666	$ 18	$ 1,004	$ 21

[1] Indicates the length of time that individual securities have been in a continuous unrealized loss position.

Corporate Bonds. The unrealized losses on our investments in corporate bonds relate to changes in interest rates and credit-related yield spreads since time of purchase. We do not intend to sell the investments, and it is not likely that we will be required to sell the investments before recovery of their amortized cost basis. We do not consider these investments to be other-than-temporarily impaired as of June 30, 2019.

Mortgage-Backed Debt Securities. The unrealized losses on our investments in U.S. government agency mortgage-backed securities relate to changes in interest rates and credit-related yield spreads since time of purchase. We do not intend to sell the investments, and it is not likely that we will be required to sell the investments before recovery of their amortized cost basis. We do not consider these investments to be other-than-temporarily impaired as of June 30, 2019.

The cost basis and fair value of the available-for-sale debt securities at June 30, 2019, by contractual maturity, is shown below. Expected maturities will differ from contractual maturities because borrowers may have the right to prepay and creditors may have the right to call obligations.

	June 30, 2019	
(Millions of dollars)	**Cost Basis**	**Fair Value**
Due in one year or less	$ 100	$ 99
Due after one year through five years	664	674
Due after five years through ten years	134	138
Due after ten years	19	20
U.S. governmental agency mortgage-backed securities	302	305
Residential mortgage-backed securities	7	7
Commercial mortgage-backed securities	36	37
Total debt securities – available-for-sale	$ 1,262	$ 1,280

Sales of available-for-sale securities:

	Three Months Ended June 30		Six Months Ended June 30	
(Millions of dollars)	**2019**	**2018**	**2019**	**2018**
Proceeds from the sale of available-for-sale securities	$ 97	$ 67	$ 144	$ 140
Gross gains from the sale of available-for-sale securities	$ 1	$ —	$ 1	$ —
Gross losses from the sale of available-for-sale securities	$ —	$ —	$ 1	$ —

For the three and six months ended June 30, 2019, the net unrealized gains (losses) for equity securities held at June 30, 2019 were $14 million and $53 million, respectively. For each of the three and six months ended June 30, 2018, the net unrealized gains (losses) for equity securities held at June 30, 2018 were $4 million.

9. Postretirement benefits

A. Pension and postretirement benefit costs

(Millions of dollars)	U.S. Pension Benefits June 30		Non-U.S. Pension Benefits June 30		Other Postretirement Benefits June 30	
	2019	2018	2019	2018	2019	2018
For the three months ended:						
Components of net periodic benefit cost:						
Service cost	$ 28	$ 31	$ 20	$ 23	$ 21	$ 21
Interest cost	150	134	24	24	34	31
Expected return on plan assets	(179)	(203)	(38)	(56)	(5)	(8)
Amortization of prior service cost (credit)	—	—	—	—	(10)	(8)
Net periodic benefit cost (benefit) [1]	$ (1)	$ (38)	$ 6	$ (9)	$ 40	$ 36
For the six months ended:						
Components of net periodic benefit cost:						
Service cost	$ 57	$ 63	$ 41	$ 45	$ 41	$ 42
Interest cost	300	267	47	49	68	62
Expected return on plan assets	(360)	(405)	(75)	(112)	(10)	(16)
Amortization of prior service cost (credit)	—	—	—	—	(20)	(17)
Net periodic benefit cost (benefit) [1]	$ (3)	$ (75)	$ 13	$ (18)	$ 79	$ 71

[1] The service cost component of net periodic pension and other postretirement benefits cost (benefit) is included in Operating costs in the Consolidated Statement of Results of Operations. All other components of net periodic pension and other postretirement benefits cost (benefit) are included in Other income (expense) in the Consolidated Statement of Results of Operations.

We made $79 million and $198 million of contributions to our pension and other postretirement plans during the three and six months ended June 30, 2019, respectively. We currently anticipate full-year 2019 contributions of approximately $330 million.

B. Defined contribution benefit costs

Total company costs related to our defined contribution plans were as follows:

(Millions of dollars)	Three Months Ended June 30		Six Months Ended June 30	
	2019	2018	2019	2018
U.S. Plans	$ 93	$ 77	$ 230	$ 150
Non-U.S. Plans	21	21	42	43
	$ 114	$ 98	$ 272	$ 193

10. Leases

A. Lessee arrangements

We lease certain property, information technology equipment, warehouse equipment, vehicles and other equipment through operating leases. We recognize a lease liability and corresponding right-of-use asset based on the present value of lease payments. To determine the present value of lease payments for most of our leases, we use our incremental borrowing rate based on information available on the lease commencement date. For certain property and information technology equipment leases, we have elected to separate payments for lease components from non-lease components. For all other leases, we have elected not to separate payments for lease and non-lease components. Our lease agreements may include options to extend or terminate the lease. When it is reasonably certain that we will exercise that option, we have included the option in the recognition of right-of-use assets and lease liabilities. We have elected not to recognize right-of-use assets or lease liabilities for leases with a term of twelve months or less.

Our finance leases are not significant and therefore are not included in the following disclosures.

The components of lease costs were as follows:

(Millions of dollars)

	Three Months Ended June 30	Six Months Ended June 30
	2019	
Operating lease cost	$ 54	$ 118
Short-term lease cost	$ 20	$ 32

Operating lease right-of-use assets are recognized in Other assets in the Consolidated Statement of Financial Position. The operating lease liabilities are recognized in Other current liabilities and Other liabilities.

Supplemental information related to leases was as follows:

(Millions of dollars)

	June 30, 2019	January 1, 2019
Operating Leases		
Other assets	$ 656	$ 713
Other current liabilities	$ 187	$ 209
Other liabilities	$ 480	$ 511
Weighted average remaining lease term		
Operating leases	7 years	7 years
Weighted average discount rates		
Operating leases	2%	2%

Maturities of operating lease liabilities at June 30, 2019 and minimum payments for operating leases having initial or remaining non-cancelable terms in excess of one year at December 31, 2018 were as follows:

(Millions of dollars)	**June 30, 2019**
Amounts Due In	
Remaining six months of 2019	$ 102
2020	164
2021	122
2022	76
2023	58
Thereafter	196
Total lease payments	718
Less: Imputed interest	(51)
Total	$ 667

	December 31, 2018
Amounts Due In	
2019	$ 205
2020	154
2021	111
2022	67
2023	50
Thereafter	185
Total	$ 772

Supplemental cash flow information related to leases was as follows:

(Millions of dollars)	**Six Months Ended June 30**
	2019
Cash paid for amounts included in the measurement of lease liabilities	
Operating cash flows from operating leases	$ 114
Right-of-use assets obtained in exchange for lease obligations:	
Operating leases	$ 68

B. Lessor arrangements

We lease Caterpillar machinery, engines and other equipment to customers and dealers around the world, primarily through Cat Financial. Cat Financial leases to customers primarily through sales-type (non-tax) leases, where the lessee for tax purposes is considered to be the owner of the equipment during the term of the lease. Cat Financial also offers tax leases that are classified as either operating or direct finance leases for financial accounting purposes, depending on the characteristics of the lease. For tax purposes, Cat Financial is considered the owner of the equipment. Our lease agreements may include options for the lessee to purchase the underlying asset at the end of the lease term for either a stated fixed price or fair market value.

The residual values for Cat Financial's leased assets, which are an estimate of the market value of leased equipment at the end of the lease term, are based on an analysis of historical wholesale market sales prices, projected forward on a level trend line without consideration for inflation or possible future pricing action. At the inception of the lease, residual values are estimated with consideration of the following critical factors: market size and demand, any known significant market/product trends, total expected hours of usage, machine configuration, application, location, model changes, quantities, past remarketing experience, third-party residual guarantees and contractual customer purchase options. Many of these factors are gathered in an application survey that is completed prior to quotation. The lease agreement also clearly defines applicable return conditions and remedies for non-compliance, to ensure that the leased equipment will be in good operating condition upon return. Model changes and updates, as well as market strength and product acceptance, are monitored and adjustments are made to residual values in accordance with the significance of any such changes. Cat Financial's sales staff work closely with customers and dealers to manage the sale of lease returns and the recovery of residual exposure.

During the term of our operating leases, we evaluate the carrying value of our equipment on a regular basis taking into consideration expected residual values at lease termination. Adjustments to depreciation expense reflecting revised estimates of expected residual values at the end of the lease terms are recorded prospectively on a straight-line basis. For finance leases, residual value adjustments are recognized through a reduction of finance revenue.

Contractual maturities of finance lease receivables (sales-type and direct finance leases) were as follows:

(Millions of dollars)	June 30, 2019		
Amounts Due In	**Retail Leases [1]**	**Wholesale Leases [2]**	**Total**
Remaining six months of 2019	$ 1,806	$ 41	$ 1,847
2020	2,562	63	2,625
2021	1,510	46	1,556
2022	691	30	721
2023	285	19	304
Thereafter	116	30	146
Total	6,970	229	7,199
Guaranteed residual value	387	57	444
Unguaranteed residual value	803	34	837
Less: Unearned income	(677)	(40)	(717)
Total	$ 7,483	$ 280	$ 7,763

	December 31, 2018		
Amounts Due In	**Retail Leases [1]**	**Wholesale Leases [2]**	**Total**
2019	$ 2,981	$ 70	$ 3,051
2020	2,026	48	2,074
2021	1,073	30	1,103
2022	453	16	469
2023	166	6	172
Thereafter	56	3	59
Total	6,755	173	6,928
Guaranteed residual value	392	66	458
Unguaranteed residual value	822	35	857
Less: Unearned income	(628)	(16)	(644)
Total	$ 7,341	$ 258	$ 7,599

[1] Included in Receivables - finance and Long-term receivables - finance on the Consolidated Statement of Financial Position.

[2] Included in Receivables - trade and other and Long-term receivables - trade and other in the Consolidated Statement of Financial Position. Wholesale lease receivables are receivables of Cat Financial that arise when Cat Financial provides financing for a dealer's lease of inventory.

Our finance lease receivables generally may be repaid or refinanced without penalty prior to contractual maturity. Accordingly, this presentation should not be regarded as a forecast of future cash collections.

The carrying amount of equipment leased to others, included in Property, plant and equipment - net in the Consolidated Statement of Financial Position, under operating leases was as follows:

(Millions of dollars)	June 30, 2019	December 31, 2018
Equipment leased to others - at original cost	$ 6,520	$ 6,015
Less: Accumulated depreciation	(2,054)	(1,744)
Equipment leased to others - net	$ 4,466	$ 4,271

Payments due for operating leases at June 30, 2019 and scheduled minimum rental payments for operating leases at December 31, 2018 were as follows:

(Millions of dollars)

June 30, 2019

Remaining Six Months of 2019	2020	2021	2022	2023	Thereafter	Total
$580	$696	$406	$208	$98	$100	$ 2,088

December 31, 2018

2019	2020	2021	2022	2023	Thereafter	Total
$896	$574	$314	$158	$71	$69	$ 2,082

Revenues from finance and operating leases, primarily included in Revenues of Financial Products on the Consolidated Statement of Results of Operations, were as follows:

(Millions of dollars)

	Three Months Ended June 30	Six Months Ended June 30
	2019	
Finance lease revenue	$ 135	$ 254
Operating lease revenue	308	624
Total	$ 443	$ 878

Revenues are presented net of sales and other related taxes.

11. Guarantees and product warranty

Caterpillar dealer performance guarantees

We have provided an indemnity to a third-party insurance company for potential losses related to performance bonds issued on behalf of Caterpillar dealers. The bonds have varying terms and are issued to insure governmental agencies against nonperformance by certain dealers. We also provided guarantees to third parties related to the performance of contractual obligations by certain Caterpillar dealers. These guarantees have varying terms and cover potential financial losses incurred by the third parties resulting from the dealers' nonperformance.

In 2016, we provided a guarantee to an end user related to the performance of contractual obligations by a Caterpillar dealer. Under the guarantee, which expires in 2025, non-performance by the Caterpillar dealer could require Caterpillar to satisfy the contractual obligations by providing goods, services or financial compensation to the end user up to an annual designated cap.

Customer loan guarantees

We provide loan guarantees to third-party lenders for financing associated with machinery purchased by customers. These guarantees have varying terms and are secured by the machinery. In addition, Cat Financial participates in standby letters of credit issued to third parties on behalf of their customers. These standby letters of credit have varying terms and beneficiaries and are secured by customer assets.

Supplier consortium performance guarantees

We have provided guarantees to a customer in Brazil and a customer in Europe related to the performance of contractual obligations by supplier consortiums to which our Caterpillar subsidiaries are members. The guarantees cover potential damages incurred by the customers resulting from the supplier consortiums' non-performance. The damages are capped except for failure of the consortiums to meet certain obligations outlined in the contract in the normal course of business. The guarantee for the customer in Europe will expire when the supplier consortium performs all of its contractual obligations, which are expected to be completed in 2022. The agreement with the customer in Brazil was terminated during the second quarter of 2019. No payments were made under the guarantee.

Third-party logistics business lease guarantees

We have provided guarantees to third-party lessors for certain properties leased by a third-party logistics business, formerly Caterpillar Logistics Services LLC, in which we sold our equity interest in 2015. The guarantees are for the possibility that the third-party logistics business would default on real estate lease payments. The guarantees were granted at lease inception and generally will expire at the end of the lease terms.

We have dealer performance guarantees and third-party performance guarantees that do not limit potential payment to end users related to indemnities and other commercial contractual obligations. In addition, we have entered into contracts involving industry standard indemnifications that do not limit potential payment. For these unlimited guarantees, we are unable to estimate a maximum potential amount of future payments that could result from claims made.

No significant loss has been experienced or is anticipated under any of these guarantees. At June 30, 2019 and December 31, 2018, the related liability was $6 million and $8 million, respectively. The maximum potential amount of future payments (undiscounted and without reduction for any amounts that may possibly be recovered under recourse or collateralized provisions) we could be required to make under the guarantees are as follows:

(Millions of dollars)	June 30, 2019	December 31, 2018
Caterpillar dealer performance guarantees	$ 1,252	$ 1,244
Customer loan guarantees	25	31
Supplier consortium performance guarantees	245	527
Third party logistics business lease guarantees	56	60
Other guarantees	125	116
Total guarantees	$ 1,703	$ 1,978

Cat Financial provides guarantees to repurchase certain loans of Caterpillar dealers from a special-purpose corporation (SPC) that qualifies as a variable interest entity. The purpose of the SPC is to provide short-term working capital loans to Caterpillar dealers. This SPC issues commercial paper and uses the proceeds to fund its loan program. Cat Financial has a loan purchase agreement with the SPC that obligates Cat Financial to purchase certain loans that are not paid at maturity. Cat Financial receives a fee for providing this guarantee, which provides a source of liquidity for the SPC. Cat Financial is the primary beneficiary of the SPC as its guarantees result in Cat Financial having both the power to direct the activities that most significantly impact the SPC's economic performance and the obligation to absorb losses, and therefore Cat Financial has consolidated the financial statements of the SPC. As of June 30, 2019 and December 31, 2018, the SPC's assets of $1,368 million and $1,149 million, respectively, were primarily comprised of loans to dealers, and the SPC's liabilities of $1,368 million and $1,148 million, respectively, were primarily comprised of commercial paper. The assets of the SPC are not available to pay Cat Financial's creditors. Cat Financial may be obligated to perform under the guarantee if the SPC experiences losses. No loss has been experienced or is anticipated under this loan purchase agreement.

Our product warranty liability is determined by applying historical claim rate experience to the current field population and dealer inventory. Generally, historical claim rates are based on actual warranty experience for each product by machine model/engine size by customer or dealer location (inside or outside North America). Specific rates are developed for each product shipment month and are updated monthly based on actual warranty claim experience.

(Millions of dollars)	2019
Warranty liability, January 1	$ 1,391
Reduction in liability (payments)	(425)
Increase in liability (new warranties)	490
Warranty liability, June 30	$ 1,456

(Millions of dollars)	2018
Warranty liability, January 1	$ 1,419
Reduction in liability (payments)	(783)
Increase in liability (new warranties)	755
Warranty liability, December 31	$ 1,391

12. Profit per share

Computations of profit per share:	Three Months Ended June 30		Six Months Ended June 30	
(Dollars in millions except per share data)	2019	2018	2019	2018
Profit for the period (A) [1]	$ 1,620	$ 1,707	$ 3,501	$ 3,372
Determination of shares (in millions):				
Weighted-average number of common shares outstanding (B)	567.8	596.2	569.9	597.0
Shares issuable on exercise of stock awards, net of shares assumed to be purchased out of proceeds at average market price	5.3	8.0	5.9	9.1
Average common shares outstanding for fully diluted computation (C) [2]	573.1	604.2	575.8	606.1
Profit per share of common stock:				
Assuming no dilution (A/B)	$ 2.85	$ 2.86	$ 6.14	$ 5.65
Assuming full dilution (A/C) [2]	$ 2.83	$ 2.82	$ 6.08	$ 5.56
Shares outstanding as of June 30 (in millions)			562.6	594.3

[1] Profit attributable to common shareholders.

[2] Diluted by assumed exercise of stock-based compensation awards using the treasury stock method.

Stock options to purchase 2,970,234 and 1,478,726 common shares were outstanding for the three and six months ended June 30, 2019 and 2018, respectively. These stock options were not included in the computation of diluted earnings per share because the effect would have been anti-dilutive.

In July 2018, the Board approved a share repurchase authorization of up to $10.0 billion of Caterpillar common stock effective January 1, 2019, with no expiration (the 2018 Authorization). As of June 30, 2019, approximately $8.0 billion remained available under the 2018 Authorization.

During the first and second quarters of 2019, we repurchased 5.7 million and 9.8 million shares, respectively, of our common stock at an aggregate cost of $751 million and $1.3 billion, respectively. These purchases were made through a combination of accelerated stock repurchase agreements with third-party financial institutions and open market transactions.

13. Accumulated other comprehensive income (loss)

Comprehensive income and its components are presented in the Consolidated Statement of Comprehensive Income. Changes in Accumulated other comprehensive income (loss), net of tax, included in the Consolidated Statement of Changes in Shareholders' Equity, consisted of the following:

(Millions of dollars)	Foreign currency translation	Pension and other postretirement benefits	Derivative financial instruments	Available-for-sale securities	Total
Three Months Ended June 30, 2019					
Balance at March 31, 2019	$ (1,525)	$ 24	$ (88)	$ 1	$ (1,588)
Other comprehensive income (loss) before reclassifications	99	—	(16)	15	98
Amounts reclassified from accumulated other comprehensive (income) loss	—	(7)	(1)	(1)	(9)
Other comprehensive income (loss)	99	(7)	(17)	14	89
Balance at June 30, 2019	$ (1,426)	$ 17	$ (105)	$ 15	$ (1,499)
Three Months Ended June 30, 2018					
Balance at March 31, 2018	$ (1,021)	$ 37	$ (18)	$ (14)	$ (1,016)
Other comprehensive income (loss) before reclassifications	(411)	—	36	(2)	(377)
Amounts reclassified from accumulated other comprehensive (income) loss	—	(7)	(96)	—	(103)
Other comprehensive income (loss)	(411)	(7)	(60)	(2)	(480)
Balance at June 30, 2018	$ (1,432)	$ 30	$ (78)	$ (16)	$ (1,496)

(Millions of dollars)	Foreign currency translation	Pension and other postretirement benefits	Derivative financial instruments	Available-for-sale securities	Total
Six Months Ended June 30, 2019					
Balance at December 31, 2018	$ (1,601)	$ 12	$ (80)	$ (15)	$ (1,684)
Adjustment to adopt new accounting guidance related to reclassification of certain tax effects from accumulated other comprehensive income [1]	98	19	(9)	—	108
Balance at January 1, 2019	(1,503)	31	(89)	(15)	(1,576)
Other comprehensive income (loss) before reclassifications	77	—	(6)	30	101
Amounts reclassified from accumulated other comprehensive (income) loss	—	(14)	(10)	—	(24)
Other comprehensive income (loss)	77	(14)	(16)	30	77
Balance at June 30, 2019	$ (1,426)	$ 17	$ (105)	$ 15	$ (1,499)
Six Months Ended June 30, 2018					
Balance at December 31, 2017	$ (1,205)	$ 46	$ (41)	$ 8	$ (1,192)
Adjustment to adopt recognition and measurement of financial assets and liabilities guidance	—	—	—	(11)	(11)
Balance at January 1, 2018	(1,205)	46	(41)	(3)	(1,203)
Other comprehensive income (loss) before reclassifications	(228)	(2)	41	(13)	(202)
Amounts reclassified from accumulated other comprehensive (income) loss	$ 1	$ (14)	$ (78)	$ —	$ (91)
Other comprehensive income (loss)	$ (227)	$ (16)	$ (37)	$ (13)	$ (293)
Balance at June 30, 2018	$ (1,432)	$ 30	$ (78)	$ (16)	$ (1,496)

[1] See Note 2 for additional information.

The effect of the reclassifications out of Accumulated other comprehensive income (loss) on the Consolidated Statement of Results of Operations is as follows:

(Millions of dollars)	Classification of income (expense)	Three Months Ended June 30 2019	2018
Pension and other postretirement benefits:			
Amortization of prior service credit (cost)	Other income (expense)	$ 10	$ 8
Tax (provision) benefit		(3)	(1)
Reclassifications net of tax		$ 7	$ 7
Derivative financial instruments:			
Foreign exchange contracts	Cost of goods sold	(1)	—
Foreign exchange contracts	Interest expense of Financial Products	7	5
Foreign exchange contracts	Other income (expense)	(4)	123
Interest rate contracts	Interest expense excluding Financial Products	—	(2)
Reclassifications before tax		2	126
Tax (provision) benefit		(1)	(30)
Reclassifications net of tax		$ 1	$ 96
Available-for-sale securities:			
Realized gain (loss)	Other income (expense)	$ 1	$ —
Reclassifications net of tax		$ 1	$ —
Total reclassifications from Accumulated other comprehensive income (loss)		$ 9	$ 103

(Millions of dollars)	Classification of income (expense)	Six Months Ended June 30 2019	2018
Foreign currency translation			
Gain (loss) on foreign currency translation	Other income (expense)	$ —	$ (1)
Reclassifications net of tax		$ —	$ (1)
Pension and other postretirement benefits:			
Amortization of prior service credit (cost)	Other income (expense)	$ 20	$ 17
Tax (provision) benefit		(6)	(3)
Reclassifications net of tax		$ 14	$ 14
Derivative financial instruments:			
Foreign exchange contracts	Sales of Machinery, Energy & Transportation	$ 1	$ —
Foreign exchange contracts	Cost of goods sold	(4)	—
Foreign exchange contracts	Interest expense of Financial Products	14	8
Foreign exchange contracts	Other income (expense)	2	95
Interest rate contracts	Interest expense excluding Financial Products	(1)	(2)
Interest rate contracts	Interest expense of Financial Products	1	1
Reclassifications before tax		13	102
Tax (provision) benefit		(3)	(24)
Reclassifications net of tax		$ 10	$ 78
Total reclassifications from Accumulated other comprehensive income (loss)		$ 24	$ 91

14. Environmental and legal matters

The Company is regulated by federal, state and international environmental laws governing its use, transport and disposal of substances and control of emissions. In addition to governing our manufacturing and other operations, these laws often impact the development of our products, including, but not limited to, required compliance with air emissions standards applicable to internal combustion engines. We have made, and will continue to make, significant research and development and capital expenditures to comply with these emissions standards.

We are engaged in remedial activities at a number of locations, often with other companies, pursuant to federal and state laws. When it is probable we will pay remedial costs at a site, and those costs can be reasonably estimated, the investigation, remediation, and operating and maintenance costs are accrued against our earnings. Costs are accrued based on consideration of currently available data and information with respect to each individual site, including available technologies, current applicable laws and regulations, and prior remediation experience. Where no amount within a range of estimates is more likely, we accrue the minimum. Where multiple potentially responsible parties are involved, we consider our proportionate share of the probable costs. In formulating the estimate of probable costs, we do not consider amounts expected to be recovered from insurance companies or others. We reassess these accrued amounts on a quarterly basis. The amount recorded for environmental remediation is not material and is included in Accrued expenses. We believe there is no more than a remote chance that a material amount for remedial activities at any individual site, or at all the sites in the aggregate, will be required.

On January 7, 2015, the Company received a grand jury subpoena from the U.S. District Court for the Central District of Illinois. The subpoena requested documents and information from the Company relating to, among other things, financial information concerning U.S. and non-U.S. Caterpillar subsidiaries (including undistributed profits of non-U.S. subsidiaries and the movement of cash among U.S. and non-U.S. subsidiaries). The Company has received additional subpoenas relating to this investigation requesting additional documents and information relating to, among other things, the purchase and resale of replacement parts by Caterpillar Inc. and non-U.S. Caterpillar subsidiaries, dividend distributions of certain non-U.S. Caterpillar subsidiaries, and Caterpillar SARL (CSARL) and related structures. On March 2-3, 2017, agents with the Department of Commerce, the Federal Deposit Insurance Corporation and the Internal Revenue Service executed search and seizure warrants at three facilities of the Company in the Peoria, Illinois area, including its former corporate headquarters. The warrants identify, and agents seized, documents and information related to, among other things, the export of products from the United States, the movement of products between the United States and Switzerland, the relationship between Caterpillar Inc. and CSARL, and sales outside the United States. It is the Company's understanding that the warrants, which concern both tax and export activities, are related to the ongoing grand jury investigation. The Company is continuing to cooperate with this investigation. The Company is unable to predict the outcome or reasonably estimate any potential loss; however, we currently believe that this matter will not have a material adverse effect on the Company's consolidated results of operations, financial position or liquidity.

On March 20, 2014, Brazil's Administrative Council for Economic Defense (CADE) published a Technical Opinion which named 18 companies and over 100 individuals as defendants, including two subsidiaries of Caterpillar Inc., MGE - Equipamentos e Serviços Ferroviários Ltda. (MGE) and Caterpillar Brasil Ltda (CBL). The publication of the Technical Opinion opened CADE's official administrative investigation into allegations that the defendants participated in anticompetitive bid activity for the construction and maintenance of metro and train networks in Brazil. While companies cannot be held criminally liable for anticompetitive conduct in Brazil, criminal charges have been brought against one current employee of MGE and two former employees of MGE involving the same conduct alleged by CADE. On July 8, 2019, CADE found MGE, one of its current employees and two of its former employees liable for anticompetitive conduct. CBL was dismissed from the proceeding without any finding of liability. MGE intends to appeal CADE's findings. We currently believe that this matter will not have a material adverse effect on the Company's consolidated results of operations, financial position or liquidity.

In addition, we are involved in other unresolved legal actions that arise in the normal course of business. The most prevalent of these unresolved actions involve disputes related to product design, manufacture and performance liability (including claimed asbestos exposure), contracts, employment issues, environmental matters, intellectual property rights, taxes (other than income taxes) and securities laws. The aggregate range of reasonably possible losses in excess of accrued liabilities, if any, associated with these unresolved legal actions is not material. In some cases, we cannot reasonably estimate a range of loss because there is insufficient information regarding the matter. However, we believe there is no more than a remote chance that any liability arising from these matters would be material. Although it is not possible to predict with certainty the outcome of these unresolved legal actions, we believe that these actions will not individually or in the aggregate have a material adverse effect on our consolidated results of operations, financial position or liquidity.

15. Income taxes

The provision for income taxes for the first six months of 2019 reflected an estimated annual tax rate of 26 percent, compared with 24 percent for the first six months of 2018, excluding the discrete items discussed in the following paragraph. The increase was largely driven by the application of U.S. tax reform provisions to the earnings of certain non-U.S. subsidiaries, which do not have a calendar fiscal year-end. These provisions did not apply to these subsidiaries in 2018.

As a result of final regulations received in 2019 providing additional guidance related to the calculation of the mandatory deemed repatriation of non-U.S. earnings due to U.S. tax reform, we recorded a discrete tax benefit of $178 million in the first six months of 2019 to adjust unrecognized tax benefits. In addition, a discrete tax benefit of $26 million was recorded in the first six months of 2019, compared with $49 million for the first six months of 2018, for the settlement of stock-based compensation awards with associated tax deductions in excess of cumulative U.S. GAAP compensation expense. The provision for income taxes for the first six months of 2018 also included a $25 million benefit for the release of a valuation allowance against the deferred tax assets of a non-U.S. subsidiary.

On January 31, 2018, we received a Revenue Agent's Report from the Internal Revenue Service (IRS) indicating the end of the field examination of our U.S. income tax returns for 2010 to 2012. In the audits of 2007 to 2012 including the impact of a loss carryback to 2005, the IRS has proposed to tax in the United States profits earned from certain parts transactions by CSARL based on the IRS examination team's application of the "substance-over-form" or "assignment-of-income" judicial doctrines. We are vigorously contesting the proposed increases to tax and penalties for these years of approximately $2.3 billion. We believe that the relevant transactions complied with applicable tax laws and did not violate judicial doctrines. We have filed U.S. income tax returns on this same basis for years after 2012. Based on the information currently available, we do not anticipate a significant change to our unrecognized tax benefits for this position within the next 12 months. We currently believe the ultimate disposition of this matter will not have a material adverse effect on our consolidated financial position, liquidity or results of operations.

16. Segment information

A. Basis for segment information

Our Executive Office is comprised of a Chief Executive Officer (CEO), four Group Presidents, a Chief Financial Officer (CFO), a General Counsel & Corporate Secretary and a Chief Human Resources Officer. The Group Presidents and CFO are accountable for a related set of end-to-end businesses that they manage. The General Counsel & Corporate Secretary leads the Law, Security and Public Policy Division. The Chief Human Resources Officer leads the Human Resources Organization. The CEO allocates resources and manages performance at the Group President/CFO level. As such, the CEO serves as our Chief Operating Decision Maker, and operating segments are primarily based on the Group President/CFO reporting structure.

Three of our operating segments, Construction Industries, Resource Industries and Energy & Transportation, are led by Group Presidents. One operating segment, Financial Products, is led by the CFO who also has responsibility for Corporate Services. Corporate Services is a cost center primarily responsible for the performance of certain support functions globally and to provide centralized services; it does not meet the definition of an operating segment. One Group President leads one smaller operating segment that is included in the All Other operating segment. The Law, Security and Public Policy Division and the Human Resources Organization are cost centers and do not meet the definition of an operating segment.

B. Description of segments

We have five operating segments, of which four are reportable segments. Following is a brief description of our reportable segments and the business activities included in the All Other operating segment:

Construction Industries: A segment primarily responsible for supporting customers using machinery in infrastructure, forestry and building construction applications. Responsibilities include business strategy, product design, product management and development, manufacturing, marketing and sales and product support. The product portfolio includes asphalt pavers; backhoe loaders; compactors; cold planers; compact track and multi-terrain loaders; mini, small, medium and large track excavators; forestry excavators; feller bunchers; harvesters; knuckleboom loaders; motor graders; pipelayers; road reclaimers; skidders; skid steer loaders; telehandlers; small and medium track-type tractors; track-type loaders; utility vehicles; wheel excavators; compact, small and medium wheel loaders; and related parts and work tools. Inter-segment sales are a source of revenue for this segment.

Resource Industries: A segment primarily responsible for supporting customers using machinery in mining, quarry and aggregates, waste and material handling applications. Responsibilities include business strategy, product design, product management and development, manufacturing, marketing and sales and product support. The product portfolio includes large track-type tractors, large mining trucks, hard rock vehicles, longwall miners, electric rope shovels, draglines, hydraulic shovels, rotary drills, large wheel loaders, off-highway trucks, articulated trucks, wheel tractor scrapers, wheel dozers, landfill compactors, soil compactors, hard rock continuous mining systems, select work tools, machinery components, electronics and control systems and related parts. In addition to equipment, Resource Industries also develops and sells technology products and services to provide customers fleet management, equipment management analytics and autonomous machine capabilities. Resource Industries also manages areas that provide services to other parts of the company, including integrated manufacturing and research and development. Inter-segment sales are a source of revenue for this segment.

Energy & Transportation: A segment primarily responsible for supporting customers using reciprocating engines, turbines, diesel-electric locomotives and related parts across industries serving Oil and Gas, Power Generation, Industrial and Transportation applications, including marine and rail-related businesses. Responsibilities include business strategy, product design, product management and development, manufacturing, marketing and sales and product support of turbine machinery and integrated systems and solutions and turbine-related services; reciprocating engine-powered generator sets; integrated systems used in the electric power generation industry; reciprocating engines and integrated systems and solutions for the marine and oil and gas industries; reciprocating engines supplied to the industrial industry as well as Cat machinery; the remanufacturing of Caterpillar engines and components and remanufacturing services for other companies; the business strategy, product design, product management and development, manufacturing, remanufacturing, leasing and service of diesel-electric locomotives and components and other rail-related products and services; and product support of on-highway vocational trucks for North America. Inter-segment sales are a source of revenue for this segment.

Financial Products Segment: Provides financing alternatives to customers and dealers around the world for Caterpillar products, as well as financing for vehicles, power generation facilities and marine vessels that, in most cases, incorporate Caterpillar products. Financing plans include operating and finance leases, installment sale contracts, working capital loans and wholesale financing plans. The segment also provides insurance and risk management products and services that help customers and dealers manage their business risk. Insurance and risk management products offered include physical damage insurance, inventory protection plans, extended service coverage for machines and engines, and dealer property and casualty insurance. The various forms of financing, insurance and risk management products offered to customers and dealers help support the purchase and lease of our equipment. The segment also earns revenues from Machinery, Energy & Transportation, but the related costs are not allocated to operating segments.

All Other operating segment: Primarily includes activities such as: business strategy, product management and development, manufacturing and sourcing of filters and fluids, undercarriage, ground-engaging tools, fluid transfer products, precision seals, rubber sealing and connecting components primarily for Cat products; parts distribution; integrated logistics solutions, distribution services responsible for dealer development and administration including a wholly owned dealer in Japan, dealer portfolio management and ensuring the most efficient and effective distribution of machines, engines and parts; and digital investments for new customer and dealer solutions that integrate data analytics with state-of-the-art digital technologies while transforming the buying experience. Results for the All Other operating segment are included as a reconciling item between reportable segments and consolidated external reporting.

C. Segment measurement and reconciliations

There are several methodology differences between our segment reporting and our external reporting. The following is a list of the more significant methodology differences:

- Machinery, Energy & Transportation segment net assets generally include inventories, receivables, property, plant and equipment, goodwill, intangibles, accounts payable and customer advances. Beginning in 2019, operating lease right-of-use assets are included in segment assets. In 2018, the present value of future lease payments for certain Machinery, Energy and Transportation operating leases was included in segment assets while the estimated financing component of the lease payments was excluded. Liabilities other than accounts payable and customer advances are generally managed at the corporate level and are not included in segment operations. Financial Products Segment assets generally include all categories of assets.

- Segment inventories and cost of sales are valued using a current cost methodology.

- Goodwill allocated to segments is amortized using a fixed amount based on a 20 year useful life. This methodology difference only impacts segment assets; no goodwill amortization expense is included in segment profit. In addition, only a portion of goodwill for certain acquisitions made in 2011 or later has been allocated to segments.

- Currency exposures for Machinery, Energy & Transportation are generally managed at the corporate level and the effects of changes in exchange rates on results of operations within the year are not included in segment profit. The net difference created in the translation of revenues and costs between exchange rates used for U.S. GAAP reporting and exchange rates used for segment reporting is reported as a methodology difference.

- Stock-based compensation expense is not included in segment profit.

- Postretirement benefit expenses are split; segments are generally responsible for service costs, with the remaining elements of net periodic benefit cost included as a methodology difference.

- Machinery, Energy & Transportation segment profit is determined on a pretax basis and excludes interest expense and most other income/expense items. Financial Products Segment profit is determined on a pretax basis and includes other income/expense items.

Reconciling items are created based on accounting differences between segment reporting and our consolidated external reporting. Please refer to pages 44 to 52 for financial information regarding significant reconciling items. Most of our reconciling items are self-explanatory given the above explanations. For the reconciliation of profit, we have grouped the reconciling items as follows:

- **Corporate costs:** These costs are related to corporate requirements primarily for compliance and legal functions for the benefit of the entire organization.

- **Restructuring costs:** May include costs for employee separation, long-lived asset impairments and contract terminations. These costs are included in Other operating (income) expenses except for defined-benefit plan curtailment losses and special termination benefits, which are included in Other income (expense). Restructuring costs also include other exit-related costs which may consist of accelerated depreciation, inventory write-downs, building demolition, equipment relocation and project management costs and LIFO inventory decrement benefits from inventory liquidations at closed facilities, all of which are primarily included in Cost of goods sold. Beginning in 2019, only certain restructuring costs are excluded from segment profit. A table, Reconciliation of Restructuring costs on page 49, has been included to illustrate how segment profit would have been impacted by the restructuring costs. See Note 20 for more information.

- **Methodology differences:** See previous discussion of significant accounting differences between segment reporting and consolidated external reporting.

- **Timing:** Timing differences in the recognition of costs between segment reporting and consolidated external reporting. For example, certain costs are reported on the cash basis for segment reporting and the accrual basis for consolidated external reporting.

Reportable Segments
Three Months Ended June 30
(Millions of dollars)

	2019						
	External sales and revenues	Inter-segment sales and revenues	Total sales and revenues	Depreciation and amortization	Segment profit	Segment assets at June 30	Capital expenditures
Construction Industries	$ 6,446	$ 21	$ 6,467	$ 81	$ 1,247	$ 5,004	$ 41
Resource Industries	2,711	89	2,800	104	481	6,456	37
Energy & Transportation	4,524	962	5,486	155	886	8,716	118
Machinery, Energy & Transportation	13,681	1,072	14,753	340	2,614	20,176	196
Financial Products Segment	873 [1]	—	873	207	193	36,584	459
Total	$ 14,554	$ 1,072	$ 15,626	$ 547	$ 2,807	$ 56,760	$ 655

	2018						
	External sales and revenues	Inter-segment sales and revenues	Total sales and revenues	Depreciation and amortization	Segment profit	Segment assets at December 31	Capital expenditures
Construction Industries	$ 6,137	$ 35	$ 6,172	$ 90	$ 1,154	$ 4,902	$ 62
Resource Industries	2,431	95	2,526	115	411	6,442	39
Energy & Transportation	4,714	1,010	5,724	157	1,012	8,386	140
Machinery, Energy & Transportation	13,282	1,140	14,422	362	2,577	19,730	241
Financial Products Segment	829 [1]	—	829	212	134	36,002	533
Total	$ 14,111	$ 1,140	$ 15,251	$ 574	$ 2,711	$ 55,732	$ 774

[1] Includes revenues from Machinery, Energy & Transportation of $136 million and $118 million in the second quarter of 2019 and 2018, respectively.

Reportable Segments
Six Months Ended June 30
(Millions of dollars)

	2019						
	External sales and revenues	Inter-segment sales and revenues	Total sales and revenues	Depreciation and amortization	Segment profit	Segment assets at June 30	Capital expenditures
Construction Industries	$ 12,298	$ 42	$ 12,340	$ 161	$ 2,332	$ 5,004	$ 69
Resource Industries	5,358	169	5,527	209	1,057	6,456	60
Energy & Transportation	8,757	1,939	10,696	307	1,724	8,716	216
Machinery, Energy & Transportation	26,413	2,150	28,563	677	5,113	20,176	345
Financial Products Segment	1,723 [1]	—	1,723	413	404	36,584	705
Total	$ 28,136	$ 2,150	$ 30,286	$ 1,090	$ 5,517	$ 56,760	$ 1,050

	2018						
	External sales and revenues	Inter-segment sales and revenues	Total sales and revenues	Depreciation and amortization	Segment profit	Segment assets at December 31	Capital expenditures
Construction Industries	$ 11,796	$ 53	$ 11,849	$ 179	$ 2,271	$ 4,902	$ 104
Resource Industries	4,639	196	4,835	231	789	6,442	62
Energy & Transportation	8,990	1,953	10,943	315	1,886	8,386	302
Machinery, Energy & Transportation	25,425	2,202	27,627	725	4,946	19,730	468
Financial Products Segment	1,622 [1]	—	1,622	415	275	36,002	894
Total	$ 27,047	$ 2,202	$ 29,249	$ 1,140	$ 5,221	$ 55,732	$ 1,362

[1] Includes revenues from Machinery, Energy & Transportation of $267 million and $223 million in the first half of 2019 and 2018, respectively.

For the three and six months ending June 30, 2019 and 2018, sales and revenues by geographic region reconciled to consolidated sales and revenues were as follows:

Sales and Revenues by Geographic Region

(Millions of dollars)	North America	Latin America	EAME	Asia/ Pacific	External Sales and Revenues
Three Months Ended June 30, 2019					
Construction Industries	$ 3,513	$ 392	$ 1,108	$ 1,433	$ 6,446
Resource Industries	1,058	448	446	759	2,711
Energy & Transportation	2,297	325	1,160	742	4,524
All Other operating segment	14	1	4	15	34
Corporate Items and Eliminations	(39)	(2)	(5)	2	(44)
Machinery, Energy & Transportation Sales	6,843	1,164	2,713	2,951	13,671
Financial Products Segment	563	76	102	132	873 [1]
Corporate Items and Eliminations	(72)	(11)	(9)	(20)	(112)
Financial Products Revenues	491	65	93	112	761
Consolidated Sales and Revenues	$ 7,334	$ 1,229	$ 2,806	$ 3,063	$ 14,432
Three Months Ended June 30, 2018					
Construction Industries	$ 2,739	$ 392	$ 1,171	$ 1,835	$ 6,137
Resource Industries	804	394	569	664	2,431
Energy & Transportation	2,582	287	1,153	692	4,714
All Other operating segment	17	1	4	19	41
Corporate Items and Eliminations	(40)	(3)	—	(1)	(44)
Machinery, Energy & Transportation Sales	6,102	1,071	2,897	3,209	13,279
Financial Products Segment	537	71	101	120	829 [1]
Corporate Items and Eliminations	(57)	(11)	(7)	(22)	(97)
Financial Products Revenues	480	60	94	98	732
Consolidated Sales and Revenues	$ 6,582	$ 1,131	$ 2,991	$ 3,307	$ 14,011

[1] Includes revenues from Machinery, Energy & Transportation of $136 million and $118 million in the second quarter of 2019 and 2018, respectively.

Sales and Revenues by Geographic Region

(Millions of dollars)	North America	Latin America	EAME	Asia/ Pacific	External Sales and Revenues
Six Months Ended June 30, 2019					
Construction Industries	$ 6,478	$ 711	$ 2,114	$ 2,995	$ 12,298
Resource Industries	2,009	871	914	1,564	5,358
Energy & Transportation	4,448	657	2,192	1,460	8,757
All Other operating segment	22	1	15	33	71
Corporate Items and Eliminations	(80)	(1)	(8)	—	(89)
Machinery, Energy & Transportation Sales	12,877	2,239	5,227	6,052	26,395
Financial Products Segment	1,121	146	204	252	1,723 [1]
Corporate Items and Eliminations	(141)	(22)	(18)	(39)	(220)
Financial Products Revenues	980	124	186	213	1,503
Consolidated Sales and Revenues	$ 13,857	$ 2,363	$ 5,413	$ 6,265	$ 27,898
Six Months Ended June 30, 2018					
Construction Industries	$ 5,359	$ 736	$ 2,238	$ 3,463	$ 11,796
Resource Industries	1,602	754	1,089	1,194	4,639
Energy & Transportation	4,807	567	2,245	1,371	8,990
All Other operating segment	32	1	8	37	78
Corporate Items and Eliminations	(68)	(2)	(3)	(1)	(74)
Machinery, Energy & Transportation Sales	11,732	2,056	5,577	6,064	25,429
Financial Products Segment	1,049	145	202	226	1,622 [1]
Corporate Items and Eliminations	(106)	(24)	(12)	(39)	(181)
Financial Products Revenues	943	121	190	187	1,441
Consolidated Sales and Revenues	$ 12,675	$ 2,177	$ 5,767	$ 6,251	$ 26,870

[1] Includes revenues from Machinery, Energy & Transportation of $267 million and $223 million in the first half of 2019 and 2018, respectively.

For the three and six months ending June 30, 2019 and 2018, Energy & Transportation segment sales by end user application were as follows:

Energy & Transportation External Sales

	Three Months Ended June 30	
(Millions of dollars)	**2019**	**2018**
Oil and gas	$ 1,305	$ 1,467
Power generation	1,021	992
Industrial	957	969
Transportation	1,241	1,286
Energy & Transportation External Sales	$ 4,524	$ 4,714

	Six Months Ended June 30	
	2019	**2018**
Oil and gas	$ 2,436	$ 2,682
Power generation	2,057	1,961
Industrial	1,861	1,875
Transportation	2,403	2,472
Energy & Transportation External Sales	$ 8,757	$ 8,990

Reconciliation of Consolidated profit before taxes:

(Millions of dollars)	Machinery, Energy & Transportation	Financial Products	Consolidated Total
Three Months Ended June 30, 2019			
Total profit from reportable segments	$ 2,614	$ 193	$ 2,807
All Other operating segment	11	—	11
Cost centers	17	—	17
Corporate costs	(148)	(1)	(149)
Timing	(58)	—	(58)
Restructuring costs	(72)	(31)	(103)
Methodology differences:			
Inventory/cost of sales	(8)	—	(8)
Postretirement benefit expense	2	—	2
Stock-based compensation expense	(66)	(2)	(68)
Financing costs	(51)	—	(51)
Currency	(92)	—	(92)
Other income/expense methodology differences	(121)	—	(121)
Other methodology differences	(12)	3	(9)
Total consolidated profit before taxes	$ 2,016	$ 162	$ 2,178
Three Months Ended June 30, 2018			
Total profit from reportable segments	$ 2,577	$ 134	$ 2,711
All Other operating segment	23	—	23
Cost centers	(1)	—	(1)
Corporate costs	(178)	—	(178)
Timing	(66)	—	(66)
Restructuring costs	(113)	(1)	(114)
Methodology differences:			
Inventory/cost of sales	31	—	31
Postretirement benefit expense	82	—	82
Stock-based compensation expense	(60)	(2)	(62)
Financing costs	(69)	—	(69)
Currency	(52)	—	(52)
Other income/expense methodology differences	(95)	—	(95)
Other methodology differences	(29)	5	(24)
Total consolidated profit before taxes	$ 2,050	$ 136	$ 2,186

Reconciliation of Consolidated profit before taxes:

(Millions of dollars)	Machinery, Energy & Transportation	Financial Products	Consolidated Total
Six Months Ended June 30, 2019			
Total profit from reportable segments	$ 5,113	$ 404	$ 5,517
All Other operating segment	36	—	36
Cost centers	41	—	41
Corporate costs	(319)	(6)	(325)
Timing	(124)	—	(124)
Restructuring costs	(111)	(31)	(142)
Methodology differences:			
Inventory/cost of sales	(1)	—	(1)
Postretirement benefit expense	(15)	—	(15)
Stock-based compensation expense	(109)	(4)	(113)
Financing costs	(115)	—	(115)
Currency	(48)	—	(48)
Other income/expense methodology differences	(250)	—	(250)
Other methodology differences	(24)	5	(19)
Total consolidated profit before taxes	$ 4,074	$ 368	$ 4,442
Six Months Ended June 30, 2018			
Total profit from reportable segments	$ 4,946	$ 275	$ 5,221
All Other operating segment	80	—	80
Cost centers	26	—	26
Corporate costs	(346)	—	(346)
Timing	(150)	—	(150)
Restructuring costs	(182)	(1)	(183)
Methodology differences:			
Inventory/cost of sales	23	—	23
Postretirement benefit expense	169	—	169
Stock-based compensation expense	(108)	(4)	(112)
Financing costs	(147)	—	(147)
Currency	(49)	—	(49)
Other income/expense methodology differences	(173)	—	(173)
Other methodology differences	(42)	3	(39)
Total consolidated profit before taxes	$ 4,047	$ 273	$ 4,320

Reconciliation of Restructuring costs:

As noted above, certain restructuring costs are a reconciling item between Segment profit and Consolidated profit before taxes. Had we included the amounts in the segments' results, the profit would have been as shown below:

Reconciliation of Restructuring costs:

(Millions of dollars)	Segment profit	Restructuring costs	Segment profit with restructuring costs
Three Months Ended June 30, 2019			
Construction Industries	$ 1,247	$ (45)	$ 1,202
Resource Industries	481	(15)	466
Energy & Transportation	886	(41)	845
Financial Products Segment	193	—	193
All Other operating segment	11	(1)	10
Total	$ 2,818	$ (102)	$ 2,716
Three Months Ended June 30, 2018			
Construction Industries	$ 1,154	$ (29)	$ 1,125
Resource Industries	411	(52)	359
Energy & Transportation	1,012	(24)	988
Financial Products Segment	134	(1)	133
All Other operating segment	23	(5)	18
Total	$ 2,734	$ (111)	$ 2,623

Reconciliation of Restructuring costs:

(Millions of dollars)	Segment profit	Restructuring costs	Segment profit with restructuring costs
Six Months Ended June 30, 2019			
Construction Industries	$ 2,332	$ (54)	$ 2,278
Resource Industries	1,057	(29)	1,028
Energy & Transportation	1,724	(52)	1,672
Financial Products Segment	404	—	404
All Other operating segment	36	(6)	30
Total	$ 5,553	$ (141)	$ 5,412
Six Months Ended June 30, 2018			
Construction Industries	$ 2,271	$ (43)	$ 2,228
Resource Industries	789	(96)	693
Energy & Transportation	1,886	(29)	1,857
Financial Products Segment	275	(1)	274
All Other operating segment	80	(9)	71
Total	$ 5,301	$ (178)	$ 5,123

Reconciliation of Assets:

(Millions of dollars)	Machinery, Energy & Transportation	Financial Products	Consolidating Adjustments	Consolidated Total
June 30, 2019				
Total assets from reportable segments	$ 20,176	$ 36,584	$ —	$ 56,760
All Other operating segment	1,266	—	—	1,266
Items not included in segment assets:				
Cash and short-term investments	6,528	—	—	6,528
Intercompany receivables	1,437	—	(1,437)	—
Investment in Financial Products	3,953	—	(3,953)	—
Deferred income taxes	2,008	—	(647)	1,361
Goodwill and intangible assets	4,457	—	—	4,457
Property, plant and equipment – net and other assets	2,329	—	—	2,329
Inventory methodology differences	(2,406)	—	—	(2,406)
Liabilities included in segment assets	9,445	—	—	9,445
Other	(495)	52	(110)	(553)
Total assets	$ 48,698	$ 36,636	$ (6,147)	$ 79,187
December 31, 2018				
Total assets from reportable segments	$ 19,730	$ 36,002	$ —	$ 55,732
All Other operating segment	1,279	—	—	1,279
Items not included in segment assets:				
Cash and short-term investments	6,968	—	—	6,968
Intercompany receivables	1,633	—	(1,633)	—
Investment in Financial Products	3,672	—	(3,672)	—
Deferred income taxes	2,015	—	(692)	1,323
Goodwill and intangible assets	4,279	—	—	4,279
Property, plant and equipment – net and other assets	1,802	—	—	1,802
Inventory methodology differences	(2,503)	—	—	(2,503)
Liabilities included in segment assets	9,766	—	—	9,766
Other	(166)	66	(37)	(137)
Total assets	$ 48,475	$ 36,068	$ (6,034)	$ 78,509

Reconciliations of Depreciation and amortization:

(Millions of dollars)	Machinery, Energy & Transportation	Financial Products	Consolidated Total
Three Months Ended June 30, 2019			
Total depreciation and amortization from reportable segments	$ 340	$ 207	$ 547
Items not included in segment depreciation and amortization:			
All Other operating segment	53	—	53
Cost centers	33	—	33
Other	5	9	14
Total depreciation and amortization	$ 431	$ 216	$ 647
Three Months Ended June 30, 2018			
Total depreciation and amortization from reportable segments	$ 362	$ 212	$ 574
Items not included in segment depreciation and amortization:			
All Other operating segment	58	—	58
Cost centers	32	—	32
Other	13	9	22
Total depreciation and amortization	$ 465	$ 221	$ 686

Reconciliations of Depreciation and amortization:

(Millions of dollars)	Machinery, Energy & Transportation	Financial Products	Consolidated Total
Six Months Ended June 30, 2019			
Total depreciation and amortization from reportable segments	$ 677	$ 413	$ 1,090
Items not included in segment depreciation and amortization:			
All Other operating segment	105	—	105
Cost centers	65	—	65
Other	8	20	28
Total depreciation and amortization	$ 855	$ 433	$ 1,288
Six Months Ended June 30, 2018			
Total depreciation and amortization from reportable segments	$ 725	$ 415	$ 1,140
Items not included in segment depreciation and amortization:			
All Other operating segment	115	—	115
Cost centers	63	—	63
Other	30	19	49
Total depreciation and amortization	$ 933	$ 434	$ 1,367

Reconciliations of Capital expenditures:

(Millions of dollars)	Machinery, Energy & Transportation	Financial Products	Consolidating Adjustments	Consolidated Total
Three Months Ended June 30, 2019				
Total capital expenditures from reportable segments	$ 196	$ 459	$ —	$ 655
Items not included in segment capital expenditures:				
All Other operating segment	22	—	—	22
Cost centers	29	—	—	29
Timing	(5)	—	—	(5)
Other	(47)	26	(2)	(23)
Total capital expenditures	$ 195	$ 485	$ (2)	$ 678
Three Months Ended June 30, 2018				
Total capital expenditures from reportable segments	$ 241	$ 533	$ —	$ 774
Items not included in segment capital expenditures:				
All Other operating segment	27	—	—	27
Cost centers	26	—	—	26
Timing	(18)	—	—	(18)
Other	(45)	43	(36)	(38)
Total capital expenditures	$ 231	$ 576	$ (36)	$ 771

Reconciliations of Capital expenditures:

(Millions of dollars)	Machinery, Energy & Transportation	Financial Products	Consolidating Adjustments	Consolidated Total
Six Months Ended June 30, 2019				
Total capital expenditures from reportable segments	$ 345	$ 705	$ —	$ 1,050
Items not included in segment capital expenditures:				
All Other operating segment	35	—	—	35
Cost centers	49	—	—	49
Timing	129	—	—	129
Other	(66)	31	(3)	(38)
Total capital expenditures	$ 492	$ 736	$ (3)	$ 1,225
Six Months Ended June 30, 2018				
Total capital expenditures from reportable segments	$ 468	$ 894	$ —	$ 1,362
Items not included in segment capital expenditures:				
All Other operating segment	38	—	—	38
Cost centers	40	—	—	40
Timing	157	—	—	157
Other	(149)	120	(40)	(69)
Total capital expenditures	$ 554	$ 1,014	$ (40)	$ 1,528

17. Cat Financial financing activities

Allowance for credit losses

The allowance for credit losses is an estimate of the losses inherent in Cat Financial's finance receivable portfolio and includes consideration of accounts that have been individually identified as impaired, as well as pools of finance receivables where it is probable that certain receivables in the pool are impaired but the individual accounts cannot yet be identified. In identifying and measuring impairment, management takes into consideration past loss experience, known and inherent risks in the portfolio, adverse situations that may affect the borrower's ability to repay, estimated value of underlying collateral and current economic conditions.

Accounts are identified for individual review based on past-due status and information available about the customer, such as financial statements, news reports and published credit ratings, as well as general information regarding industry trends and the economic environment in which Cat Financial's customers operate. The allowance for credit losses attributable to finance receivables that are individually evaluated and determined to be impaired is based on the present value of expected future cash flows discounted at the receivables' effective interest rate, the fair value of the collateral for collateral-dependent receivables or the observable market price of the receivable. In determining collateral value, Cat Financial estimates the current fair market value of the collateral less selling costs. Cat Financial also considers credit enhancements such as additional collateral and contractual third-party guarantees. The allowance for credit losses attributable to the remaining accounts not yet individually identified as impaired is estimated based on loss forecast models utilizing probabilities of default, our estimate of the loss emergence period and the estimated loss given default. In addition, qualitative factors not able to be fully captured in the loss forecast models including industry trends, macroeconomic factors and model imprecision are considered in the evaluation of the adequacy of the allowance for credit losses. These qualitative factors are subjective and require a degree of management judgment.

Cat Financial's allowance for credit losses is segregated into two portfolio segments:

- Customer - Finance receivables with retail customers.
- Dealer - Finance receivables with Caterpillar dealers.

A portfolio segment is the level at which the company develops a systematic methodology for determining its allowance for credit losses.

Cat Financial further evaluates portfolio segments by the class of finance receivables, which is defined as a level of information (below a portfolio segment) in which the finance receivables have the same initial measurement attribute and a similar method for assessing and monitoring credit risk. Typically, Cat Financial's finance receivables within a geographic area have similar credit risk profiles and methods for assessing and monitoring credit risk. Cat Financial's classes, which align with management reporting for credit losses, are as follows:

- North America - Finance receivables originated in the United States or Canada.
- EAME - Finance receivables originated in Europe, Africa, the Middle East and the Commonwealth of Independent States.
- Asia Pacific - Finance receivables originated in Australia, New Zealand, China, Japan and Southeast Asia.
- Mining - Finance receivables related to large mining customers worldwide and project financing in various countries.
- Latin America - Finance receivables originated in Mexico, and Central and South American countries.
- Caterpillar Power Finance - Finance receivables originated worldwide related to marine vessels with Caterpillar engines and Caterpillar electrical power generation, gas compression and co-generation systems and non-Caterpillar equipment that is powered by these systems.

An analysis of the allowance for credit losses was as follows:

(Millions of dollars)	June 30, 2019		
Allowance for Credit Losses:	**Customer**	**Dealer**	**Total**
Balance at beginning of year	$ 486	$ 21	$ 507
Receivables written off	(125)	—	(125)
Recoveries on receivables previously written off	21	—	21
Provision for credit losses	99	24	123
Other	(8)	—	(8)
Balance at end of period	$ 473	$ 45	$ 518
Individually evaluated for impairment	$ 266	$ 39	$ 305
Collectively evaluated for impairment	207	6	213
Ending Balance	$ 473	$ 45	$ 518
Recorded Investment in Finance Receivables:			
Individually evaluated for impairment	$ 789	$ 78	$ 867
Collectively evaluated for impairment	18,095	3,539	21,634
Ending Balance	$ 18,884	$ 3,617	$ 22,501

(Millions of dollars)	December 31, 2018		
Allowance for Credit Losses:	**Customer**	**Dealer**	**Total**
Balance at beginning of year	$ 353	$ 9	$ 362
Receivables written off	(235)	—	(235)
Recoveries on receivables previously written off	46	—	46
Provision for credit losses	337	12	349
Other	(15)	—	(15)
Balance at end of year	$ 486	$ 21	$ 507
Individually evaluated for impairment	$ 288	$ 14	$ 302
Collectively evaluated for impairment	198	7	205
Ending Balance	$ 486	$ 21	$ 507
Recorded Investment in Finance Receivables:			
Individually evaluated for impairment	$ 858	$ 78	$ 936
Collectively evaluated for impairment	18,152	3,338	21,490
Ending Balance	$ 19,010	$ 3,416	$ 22,426

Credit quality of finance receivables

At origination, Cat Financial evaluates credit risk based on a variety of credit quality factors including prior payment experience, customer financial information, credit-rating agency ratings, loan-to-value ratios and other internal metrics. On an ongoing basis, Cat Financial monitors credit quality based on past-due status and collection experience as there is a meaningful correlation between the past-due status of customers and the risk of loss.

In determining past-due status, Cat Financial considers the entire recorded investment in finance receivables past due when any installment is over 30 days past due. The tables below summarize the recorded investment in finance receivables by aging category.

(Millions of dollars)	June 30, 2019						
	31-60 Days Past Due	61-90 Days Past Due	91+ Days Past Due	Total Past Due	Current	Recorded Investment in Finance Receivables	91+ Still Accruing
Customer							
North America	$ 90	$ 17	$ 44	$ 151	$ 7,805	$ 7,956	$ 12
EAME	29	13	167	209	2,885	3,094	7
Asia Pacific	37	15	25	77	2,511	2,588	12
Mining	3	13	19	35	1,814	1,849	—
Latin America	44	17	85	146	1,352	1,498	—
Caterpillar Power Finance	7	13	296	316	1,583	1,899	—
Dealer							
North America	—	—	—	—	2,108	2,108	—
EAME	—	—	—	—	330	330	—
Asia Pacific	—	—	—	—	461	461	—
Mining	—	—	—	—	4	4	—
Latin America	1	1	79	81	632	713	1
Caterpillar Power Finance	—	—	—	—	1	1	—
Total	$ 211	$ 89	$ 715	$ 1,015	$ 21,486	$ 22,501	$ 32

(Millions of dollars)	December 31, 2018						
	31-60 Days Past Due	61-90 Days Past Due	91+ Days Past Due	Total Past Due	Current	Recorded Investment in Finance Receivables	91+ Still Accruing
Customer							
North America	$ 65	$ 18	$ 84	$ 167	$ 7,825	$ 7,992	$ 14
EAME	19	9	153	181	2,850	3,031	5
Asia Pacific	24	9	8	41	2,409	2,450	5
Mining	28	1	9	38	1,642	1,680	—
Latin America	38	29	71	138	1,421	1,559	—
Caterpillar Power Finance	10	1	384	395	1,903	2,298	—
Dealer							
North America	—	—	—	—	1,895	1,895	—
EAME	—	—	—	—	333	333	—
Asia Pacific	—	—	—	—	466	466	—
Mining	—	—	—	—	4	4	—
Latin America	—	—	78	78	638	716	—
Caterpillar Power Finance	—	—	—	—	2	2	—
Total	$ 184	$ 67	$ 787	$ 1,038	$ 21,388	$ 22,426	$ 24

Impaired finance receivables

For all classes, a finance receivable is considered impaired, based on current information and events, if it is probable that Cat Financial will be unable to collect all amounts due according to the contractual terms. Impaired finance receivables include finance receivables that have been restructured and are considered to be troubled debt restructurings.

There were $78 million of impaired finance receivables with a related allowance of $39 million and $14 million as of June 30, 2019 and December 31, 2018, respectively, for the Dealer portfolio segment, all of which was in Latin America. Cat Financial's recorded investment in impaired finance receivables and the related unpaid principal balances and allowance for the Customer portfolio segment were as follows:

	June 30, 2019			December 31, 2018		
(Millions of dollars)	**Recorded Investment**	**Unpaid Principal Balance**	**Related Allowance**	**Recorded Investment**	**Unpaid Principal Balance**	**Related Allowance**
Impaired Finance Receivables With No Allowance Recorded						
North America	$ 11	$ 11	$ —	$ 10	$ 10	$ —
EAME	1	1	—	1	1	—
Asia Pacific	—	—	—	—	—	—
Mining	26	26	—	33	33	—
Latin America	21	21	—	29	29	—
Caterpillar Power Finance	52	52	—	69	83	—
Total	$ 111	$ 111	$ —	$ 142	$ 156	$ —
Impaired Finance Receivables With An Allowance Recorded						
North America	$ 28	$ 28	$ 10	$ 40	$ 41	$ 14
EAME	101	101	56	92	92	57
Asia Pacific	10	10	3	4	4	2
Mining	63	62	20	56	55	26
Latin America	70	69	26	75	75	25
Caterpillar Power Finance	406	420	151	449	455	164
Total	$ 678	$ 690	$ 266	$ 716	$ 722	$ 288
Total Impaired Finance Receivables						
North America	$ 39	$ 39	$ 10	$ 50	$ 51	$ 14
EAME	102	102	56	93	93	57
Asia Pacific	10	10	3	4	4	2
Mining	89	88	20	89	88	26
Latin America	91	90	26	104	104	25
Caterpillar Power Finance	458	472	151	518	538	164
Total	$ 789	$ 801	$ 266	$ 858	$ 878	$ 288

(Millions of dollars)	Three Months Ended June 30, 2019		Three Months Ended June 30, 2018	
	Average Recorded Investment	Interest Income Recognized	Average Recorded Investment	Interest Income Recognized
Impaired Finance Receivables With No Allowance Recorded				
North America	$ 10	$ —	$ 18	$ 1
EAME	1	—	7	—
Asia Pacific	—	—	28	—
Mining	29	1	45	1
Latin America	20	1	43	—
Caterpillar Power Finance	41	—	189	1
Total	$ 101	$ 2	$ 330	$ 3
Impaired Finance Receivables With An Allowance Recorded				
North America	$ 35	$ —	$ 56	$ —
EAME	94	—	48	1
Asia Pacific	9	—	5	—
Mining	39	—	64	1
Latin America	74	2	65	1
Caterpillar Power Finance	443	4	356	3
Total	$ 694	$ 6	$ 594	$ 6
Total Impaired Finance Receivables				
North America	$ 45	$ —	$ 74	$ 1
EAME	95	—	55	1
Asia Pacific	9	—	33	—
Mining	68	1	109	2
Latin America	94	3	108	1
Caterpillar Power Finance	484	4	545	4
Total	$ 795	$ 8	$ 924	$ 9

(Millions of dollars)	Six Months Ended June 30, 2019		Six Months Ended June 30, 2018	
	Average Recorded Investment	Interest Income Recognized	Average Recorded Investment	Interest Income Recognized
Impaired Finance Receivables With No Allowance Recorded				
Customer				
North America	$ 10	$ —	$ 18	$ 1
EAME	1	—	23	—
Asia Pacific	—	—	30	1
Mining	30	1	78	2
Latin America	23	1	44	1
Caterpillar Power Finance	51	1	178	3
Total	$ 115	$ 3	$ 371	$ 8
Impaired Finance Receivables With An Allowance Recorded				
Customer				
North America	$ 37	$ 1	$ 53	$ 1
EAME	94	1	32	1
Asia Pacific	8	—	5	—
Mining	42	1	36	1
Latin America	75	3	76	2
Caterpillar Power Finance	446	7	355	4
Total	$ 702	$ 13	$ 557	$ 9
Total Impaired Finance Receivables				
Customer				
North America	$ 47	$ 1	$ 71	$ 2
EAME	95	1	55	1
Asia Pacific	8	—	35	1
Mining	72	2	114	3
Latin America	98	4	120	3
Caterpillar Power Finance	497	8	533	7
Total	$ 817	$ 16	$ 928	$ 17

Recognition of income is suspended and the finance receivable is placed on non-accrual status when management determines that collection of future income is not probable (generally after 120 days past due). Recognition is resumed and previously suspended income is recognized when the finance receivable becomes current and collection of remaining amounts is considered probable. Payments received while the finance receivable is on non-accrual status are applied to interest and principal in accordance with the contractual terms.

As of June 30, 2019 and December 31, 2018, there were $78 million of finance receivables on non-accrual status for the Dealer portfolio segment, all of which was in Latin America. The recorded investment in customer finance receivables on non-accrual status was as follows:

(Millions of dollars)	June 30, 2019	December 31, 2018
North America	$ 37	$ 77
EAME	165	154
Asia Pacific	13	4
Mining	20	50
Latin America	98	106
Caterpillar Power Finance	447	416
Total	$ 780	$ 807

Troubled Debt Restructurings

A restructuring of a finance receivable constitutes a troubled debt restructuring (TDR) when the lender grants a concession it would not otherwise consider to a borrower experiencing financial difficulties. Concessions granted may include extended contract maturities, inclusion of interest only periods, below market interest rates, extended skip payment periods and reduction of principal and/or accrued interest.

As of June 30, 2019 and December 31, 2018, there were no additional funds committed to lend to a borrower whose terms have been modified in a TDR.

There were no finance receivables modified as TDRs during the three months ended June 30, 2019 or 2018 for the Dealer portfolio segment. Cat Financial's investment in finance receivables in the Customer portfolio segment modified as TDRs during the three and six months ended June 30, 2019 and 2018 were as follows:

	Three Months Ended June 30, 2019			Three Months Ended June 30, 2018		
(Millions of dollars)	Number of Contracts	Pre-TDR Recorded Investment	Post-TDR Recorded Investment	Number of Contracts	Pre-TDR Recorded Investment	Post-TDR Recorded Investment
North America	8	$ 5	$ 4	17	$ 7	$ 7
EAME	2	10	10	—	—	—
Mining	1	6	6	—	—	—
Latin America	4	2	2	—	—	—
Caterpillar Power Finance	7	47	47	2	50	17
Total	22	$ 70	$ 69	19	$ 57	$ 24

	Six Months Ended June 30, 2019			Six Months Ended June 30, 2018		
	Number of Contracts	Pre-TDR Outstanding Recorded Investment	Post-TDR Outstanding Recorded Investment	Number of Contracts	Pre-TDR Outstanding Recorded Investment	Post-TDR Outstanding Recorded Investment
North America	8	$ 5	$ 4	30	$ 13	$ 13
EAME	21	21	17	—	—	—
Mining	1	6	6	1	29	29
Latin America	4	2	2	1	3	3
Caterpillar Power Finance	15	98	97	5	53	20
Total	49	$ 132	$ 126	37	$ 98	$ 65

18. Fair value disclosures

A. Fair value measurements

The guidance on fair value measurements defines fair value as the exchange price that would be received for an asset or paid to transfer a liability (an exit price) in the principal or most advantageous market for the asset or liability in an orderly transaction between market participants. This guidance also specifies a fair value hierarchy based upon the observability of inputs used in valuation techniques. Observable inputs (highest level) reflect market data obtained from independent sources, while unobservable inputs (lowest level) reflect internally developed market assumptions. In accordance with this guidance, fair value measurements are classified under the following hierarchy:

- **Level 1** – Quoted prices for identical instruments in active markets.
- **Level 2** – Quoted prices for similar instruments in active markets; quoted prices for identical or similar instruments in markets that are not active; and model-derived valuations in which all significant inputs or significant value-drivers are observable in active markets.
- **Level 3** – Model-derived valuations in which one or more significant inputs or significant value-drivers are unobservable.

When available, we use quoted market prices to determine fair value, and we classify such measurements within Level 1. In some cases where market prices are not available, we make use of observable market based inputs to calculate fair value, in which case the measurements are classified within Level 2. If quoted or observable market prices are not available, fair value is based upon valuations in which one or more significant inputs are unobservable, including internally developed models that use, where possible, current market-based parameters such as interest rates, yield curves and currency rates. These measurements are classified within Level 3.

Fair value measurements are classified according to the lowest level input or value-driver that is significant to the valuation. A measurement may therefore be classified within Level 3 even though there may be significant inputs that are readily observable.

Fair value measurement includes the consideration of nonperformance risk. Nonperformance risk refers to the risk that an obligation (either by a counterparty or Caterpillar) will not be fulfilled. For financial assets traded in an active market (Level 1 and certain Level 2), the nonperformance risk is included in the market price. For certain other financial assets and liabilities (certain Level 2 and Level 3), our fair value calculations have been adjusted accordingly.

Investments in debt and equity securities

We have investments in certain debt and equity securities, primarily at Insurance Services, that are recorded at fair value. Fair values for our U.S. treasury bonds and large capitalization value and smaller company growth equity securities are based upon valuations for identical instruments in active markets. Fair values for other government bonds, corporate bonds and mortgage-backed debt securities are based upon models that take into consideration such market-based factors as recent sales, risk-free yield curves and prices of similarly rated bonds.

In addition, Insurance Services has an equity investment in a real estate investment trust (REIT) which is recorded at fair value based on the net asset value (NAV) of the investment and is not classified within the fair value hierarchy.

See Note 8 for additional information on our investments in debt and equity securities.

Derivative financial instruments

The fair value of interest rate contracts is primarily based on models that utilize the appropriate market-based forward swap curves and zero-coupon interest rates to determine discounted cash flows. The fair value of foreign currency and commodity forward, option and cross currency contracts is based on a valuation model that discounts cash flows resulting from the differential between the contract price and the market-based forward rate.

Assets and liabilities measured on a recurring basis at fair value, primarily related to Financial Products, included in our Consolidated Statement of Financial Position as of June 30, 2019 and December 31, 2018 are summarized below:

	June 30, 2019				
(Millions of dollars)	**Level 1**	**Level 2**	**Level 3**	**Measured at NAV**	**Total Assets / Liabilities, at Fair Value**
Assets					
Debt securities					
Government debt					
U.S. treasury bonds	$ 9	$ —	$ —	$ —	$ 9
Other U.S. and non-U.S. government bonds	—	47	—	—	47
Corporate bonds					
Corporate bonds	—	819	—	—	819
Asset-backed securities	—	56	—	—	56
Mortgage-backed debt securities					
U.S. governmental agency	—	305	—	—	305
Residential	—	7	—	—	7
Commercial	—	37	—	—	37
Total debt securities	9	1,271	—	—	1,280
Equity securities					
Large capitalization value	307	—	—	—	307
Smaller company growth	56	—	—	—	56
REIT	—	—	—	122	122
Total equity securities	363	—	—	122	485
Total assets	$ 372	$ 1,271	$ —	$ 122	$ 1,765
Liabilities					
Derivative financial instruments, net	—	19	—	—	19
Total liabilities	$ —	$ 19	$ —	$ —	$ 19

(Millions of dollars)	December 31, 2018				
	Level 1	Level 2	Level 3	Measured at NAV	Total Assets / Liabilities, at Fair Value
Assets					
Debt securities					
Government debt					
U.S. treasury bonds	$ 9	$ —	$ —	$ —	$ 9
Other U.S. and non-U.S. government bonds	—	42	—	—	42
Corporate bonds					
Corporate bonds	—	720	—	—	720
Asset-backed securities	—	63	—	—	63
Mortgage-backed debt securities					
U.S. governmental agency	—	297	—	—	297
Residential	—	7	—	—	7
Commercial	—	13	—	—	13
Total debt securities	9	1,142	—	—	1,151
Equity securities					
Large capitalization value	260	—	—	—	260
Smaller company growth	46	—	—	—	46
REIT	—	—	—	119	119
Total equity securities	306	—	—	119	425
Total assets	$ 315	$ 1,142	$ —	$ 119	$ 1,576
Liabilities					
Derivative financial instruments, net	$ —	$ 19	$ —	$ —	$ 19
Total liabilities	$ —	$ 19	$ —	$ —	$ 19

In addition to the amounts above, Cat Financial impaired loans are subject to measurement at fair value on a nonrecurring basis and are classified as Level 3 measurements. A loan is considered impaired when management determines that collection of contractual amounts due is not probable. In these cases, an allowance for credit losses may be established based either on the present value of expected future cash flows discounted at the receivables' effective interest rate, the fair value of the collateral for collateral-dependent receivables, or the observable market price of the receivable. In determining collateral value, Cat Financial estimates the current fair market value of the collateral less selling costs. Cat Financial had impaired loans with a fair value of $417 million and $469 million as of June 30, 2019 and December 31, 2018, respectively.

B. Fair values of financial instruments

In addition to the methods and assumptions we use to record the fair value of financial instruments as discussed in the Fair value measurements section above, we used the following methods and assumptions to estimate the fair value of our financial instruments:

Cash and short-term investments
Carrying amount approximated fair value.

Restricted cash and short-term investments
Carrying amount approximated fair value. Restricted cash and short-term investments are included in Prepaid expenses and other current assets in the Consolidated Statement of Financial Position.

Finance receivables
Fair value was estimated by discounting the future cash flows using current rates, representative of receivables with similar remaining maturities.

Wholesale inventory receivables
Fair value was estimated by discounting the future cash flows using current rates, representative of receivables with similar remaining maturities.

Short-term borrowings
Carrying amount approximated fair value.

Long-term debt
Fair value for fixed and floating rate debt was estimated based on quoted market prices.

Guarantees
The fair value of guarantees is based upon our estimate of the premium a market participant would require to issue the same guarantee in a stand-alone arms-length transaction with an unrelated party. If quoted or observable market prices are not available, fair value is based upon internally developed models that utilize current market-based assumptions.

Please refer to the table below for the fair values of our financial instruments.

Fair Value of Financial Instruments

(Millions of dollars)	June 30, 2019 Carrying Amount	June 30, 2019 Fair Value	December 31, 2018 Carrying Amount	December 31, 2018 Fair Value	Fair Value Levels	Reference
Assets						
Cash and short-term investments	$ 7,429	$ 7,429	$ 7,857	$ 7,857	1	
Restricted cash and short-term investments	$ 30	$ 30	$ 33	$ 33	1	
Investments in debt and equity securities	$ 1,765	$ 1,765	$ 1,576	$ 1,576	1 & 2	Note 8
Finance receivables – net (excluding finance leases [1])	$ 14,526	$ 14,750	$ 14,714	$ 14,798	3	Note 17
Wholesale inventory receivables – net (excluding finance leases [1])	$ 1,172	$ 1,151	$ 1,050	$ 1,025	3	
Foreign currency contracts – net	$ 51	$ 51	$ 47	$ 47	2	Note 5
Liabilities						
Short-term borrowings	$ 5,266	$ 5,266	$ 5,723	$ 5,723	1	
Long-term debt (including amounts due within one year)						
Machinery, Energy & Transportation	$ 7,670	$ 9,469	$ 8,015	$ 9,046	2	
Financial Products	$ 23,342	$ 23,596	$ 22,815	$ 22,684	2	
Interest rate contracts – net	$ 65	$ 65	$ 36	$ 36	2	Note 5
Commodity contracts – net	$ 5	$ 5	$ 30	$ 30	2	Note 5
Guarantees	$ 6	$ 6	$ 8	$ 8	3	Note 11

[1] Represents finance leases and failed sales leasebacks of $7,758 million at June 30, 2019 and finance leases of $7,463 million at December 31, 2018, respectively.

19. Other income (expense)

(Millions of dollars)	Three Months Ended June 30		Six Months Ended June 30	
	2019	2018	2019	2018
Investment and interest income	$ 54	$ 45	$ 106	$ 81
Foreign exchange gains (losses) [1]	(47)	(60)	(29)	(79)
License fee income	31	36	56	67
Net periodic pension and OPEB income (cost), excluding service cost	24	86	50	172
Gains (losses) on securities	14	7	53	4
Miscellaneous income (loss)	(8)	7	(8)	3
Total	$ 68	$ 121	$ 228	$ 248

[1] Includes gains (losses) from foreign exchange derivative contracts. See Note 5 for further details.

20. Restructuring costs

Our accounting for employee separations is dependent upon how the particular program is designed. For voluntary programs, eligible separation costs are recognized at the time of employee acceptance unless the acceptance requires explicit approval by the company. For involuntary programs, eligible costs are recognized when management has approved the program, the affected employees have been properly notified and the costs are estimable.

Restructuring costs for the three and six months ended June 30, 2019 and 2018 were as follows:

(Millions of dollars)	Three Months Ended June 30	
	2019	2018
Employee separations [1]	$ 10	$ 45
Long-lived asset impairments [1]	29	30
Other [2]	71	39
Total restructuring costs	$ 110	$ 114

	Six Months Ended June 30	
	2019	2018
Employee separations [1]	$ 25	$ 78
Long-lived asset impairments [1]	36	30
Other [2]	97	75
Total restructuring costs	$ 158	$ 183

[1] Recognized in Other operating (income) expenses.

[2] Represents costs related to our restructuring programs, primarily for inventory write-downs, project management costs, accelerated depreciation, equipment relocation and building demolition, all of which are primarily included in Cost of goods sold.

For the six months ended June 30, 2019, the restructuring costs were primarily related to restructuring actions in Construction Industries and Energy & Transportation. For the six months ended June 30, 2018, the restructuring costs were primarily related to ongoing facility closures across the company.

Certain restructuring costs are a reconciling item between Segment profit and Consolidated profit before taxes. See Note 16 for more information.

The following table summarizes the 2018 and 2019 employee separation activity:

(Millions of dollars)		
Liability balance at December 31, 2017	$	249
Increase in liability (separation charges)		112
Reduction in liability (payments)		(276)
Liability balance at December 31, 2018		85
Increase in liability (separation charges)		25
Reduction in liability (payments)		(60)
Liability balance at June 30, 2019	$	50

Most of the liability balance at June 30, 2019 is expected to be paid in 2019 and 2020.

In September 2015, we announced a large scale restructuring plan (the Plan) including a voluntary retirement enhancement program for qualifying U.S. employees, several voluntary separation programs outside of the U.S., additional involuntary programs throughout the company and manufacturing facility consolidations and closures that occurred through 2018. The largest action among those included in the Plan was related to our European manufacturing footprint, which led to the Gosselies, Belgium, facility closure. In the first six months of 2019, we incurred $19 million of restructuring costs related to the Plan. Total restructuring costs incurred since the inception of the Plan were $1,807 million. The remaining costs of approximately $20 million related to the Plan are expected to be recognized in 2019.

Item 2. Management's Discussion and Analysis of Financial Condition and Results of Operations

Overview

Second-quarter 2019 sales and revenues were $14.432 billion, a 3 percent increase from second-quarter 2018 sales and revenues of $14.011 billion. The improvement was primarily due to favorable ***price realization*** and higher ***sales volume*** driven by improved demand for equipment, including a favorable impact from changes in dealer inventories. The increase was partially offset by unfavorable ***currency*** impacts. Sales increased in ***Construction Industries*** and ***Resource Industries***, while ***Energy & Transportation*** sales decreased. Sales increased in North America and ***Latin America***, partially offset by decreases in Asia/Pacific and ***EAME***. Financial Products' revenues also increased. Second-quarter 2019 profit per share was $2.83, compared with $2.82 profit per share in the second quarter of 2018. Profit was $1.620 billion in the second quarter of 2019, a decrease of $87 million from the second quarter of 2018. Profit decreased due to higher ***manufacturing costs***, unfavorable sales volume due to product mix and higher tax expense, partially offset by favorable price realization and lower selling, general and administrative (SG&A) and research and development (R&D) expenses.

Sales and revenues for the six months ended June 30, 2019, were $27.898 billion, up $1.028 billion, or 4 percent, from $26.870 billion for the six months ended June 30, 2018. Profit per share for the six months ended June 30, 2019, was $6.08, up $0.52, or 9 percent, from $5.56 for the same period last year. Profit was $3.501 billion for the six months ended June 30, 2019, up $129 million, or 4 percent, from $3.372 billion for the six months ended June 30, 2018.

Highlights for the second quarter of 2019 include:

- Second-quarter sales and revenues were $14.432 billion, compared with $14.011 billion in the second quarter of 2018. Sales increased in Construction Industries and Resource Industries, while Energy & Transportation sales decreased. Financial Products' revenues also increased.
- Operating profit as a percent of sales and revenues was 15.3 percent in the second quarter of 2019, compared with 15.5 percent in the second quarter of 2018.
- Profit per share was $2.83 in the second quarter of 2019, compared with $2.82 in the second quarter of 2018.
- During the second quarter of 2019, ***Machinery, Energy & Transportation (ME&T)*** operating cash flow was $2.0 billion. In the quarter, the company deployed significant capital through the repurchase of $1.4 billion of Caterpillar common stock and a dividend payment of $492 million. The enterprise cash balance at the end of the second quarter of 2019 was $7.4 billion.

Highlights for the six months ended June 30, 2019, include:

- Sales and revenues for the six months ended June 30, 2019, were $27.898 billion, compared with $26.870 billion for the six months ended June 30, 2018. Sales increased in Resource Industries and Construction Industries, while Energy & Transportation decreased. Financial Products' revenues also increased.
- Operating profit as a percent of sales and revenues was 15.8 percent for the six months ended June 30, 2019, compared with 15.9 percent for the six months ended June 30, 2018.
- Profit per share was $6.08 for the six months ended June 30, 2019, compared with $5.56 for the six months ended June 30, 2018.
- ME&T operating cash flow was $2.8 billion for the six months ended June 30, 2019, compared to $3.0 billion for the six months ended June 30, 2018.

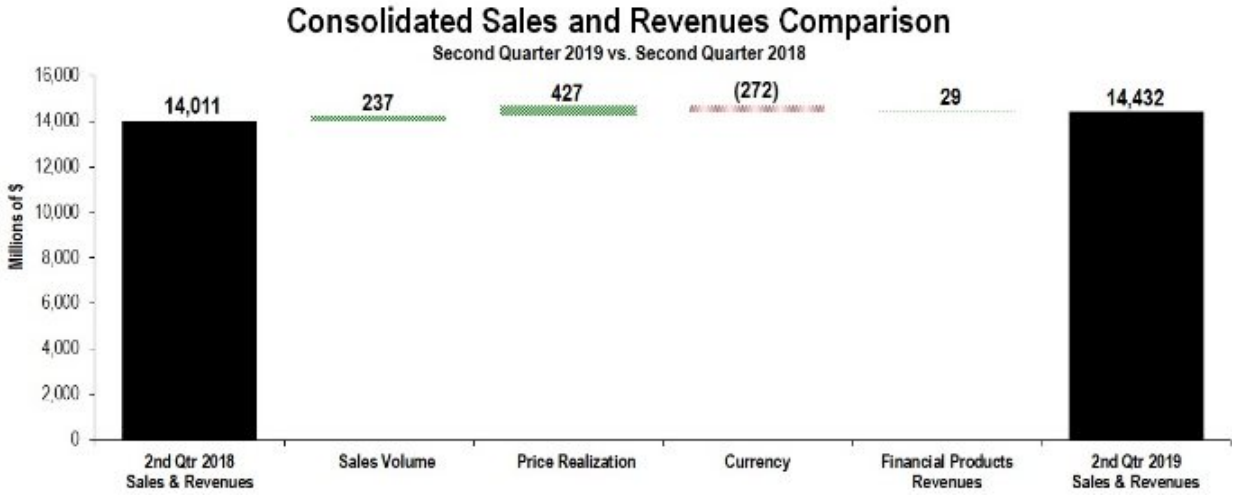
Notes:

- Glossary of terms is included on pages 78-79; first occurrence of terms shown in bold italics.
- Information on non-GAAP financial measures is included on page 84.

Consolidated Results of Operations

THREE MONTHS ENDED JUNE 30, 2019 COMPARED WITH THREE MONTHS ENDED JUNE 30, 2018

CONSOLIDATED SALES AND REVENUES



The chart above graphically illustrates reasons for the change in consolidated sales and revenues between the second quarter of 2018 (at left) and the second quarter of 2019 (at right). Caterpillar management utilizes these charts internally to visually communicate with the company's Board of Directors and employees.

Total sales and revenues were $14.432 billion in the second quarter of 2019, an increase of $421 million, or 3 percent , compared with $14.011 billion in the second quarter of 2018. The improvement was primarily due to favorable price realization and higher sales volume driven by improved demand for equipment, including a favorable impact from changes in dealer inventories. The increase was partially offset by unfavorable currency impacts. Sales increased in Construction Industries and Resource Industries, while Energy & Transportation sales decreased. Financial Products' revenues also increased.

The largest sales increase was in North America, which improved 12 percent as favorable economic conditions in key end markets drove higher demand, including favorable changes in dealer inventories, and favorable price realization. Dealer inventories increased in the second quarter of 2019, compared with a decrease in the second quarter of 2018.

Sales increased 9 percent in Latin America due to improved demand from low levels in a few countries.

EAME sales decreased 6 percent primarily due to economic uncertainty in several countries in the region and a weaker euro, partially offset by favorable price realization.

Asia/Pacific sales decreased 8 percent primarily due to an unfavorable change in dealer inventories and the unfavorable impact of a weaker Australian dollar and Chinese yuan. The decrease in sales was partially offset by favorable price realization. Dealers increased inventories more significantly in the second quarter of 2018, compared with the second quarter of 2019. Lower demand in China due to continued competitive pricing pressures was partially offset by higher demand in other countries in the region.

Dealer machine and engine inventories increased about $500 million during the second quarter of 2019, compared with an increase of about $100 million during the second quarter of 2018.

Sales and Revenues by Segment

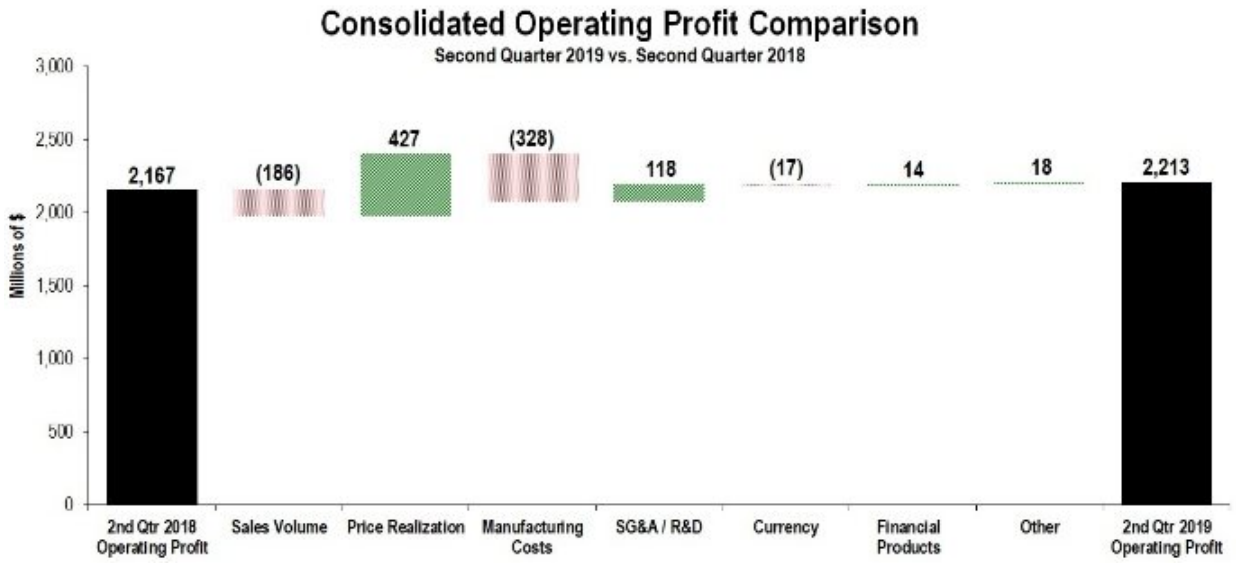
(Millions of dollars)	Second Quarter 2018	Sales Volume	Price Realization	Currency	Inter-Segment / Other	Second Quarter 2019	$ Change	% Change
Construction Industries	$ 6,172	$ 199	$ 253	$ (143)	$ (14)	$ 6,467	$ 295	5%
Resource Industries	2,526	188	129	(37)	(6)	2,800	274	11%
Energy & Transportation	5,724	(143)	44	(91)	(48)	5,486	(238)	(4%)
All Other Segment	124	(6)	—	(1)	8	125	1	1%
Corporate Items and Eliminations	(1,267)	(1)	1	—	60	(1,207)	60	
Machinery, Energy & Transportation Sales	13,279	237	427	(272)	—	13,671	392	3%
Financial Products Segment	829	—	—	—	44	873	44	5%
Corporate Items and Eliminations	(97)	—	—	—	(15)	(112)	(15)	
Financial Products Revenues	732	—	—	—	29	761	29	4%
Consolidated Sales and Revenues	$ 14,011	$ 237	$ 427	$ (272)	$ 29	$ 14,432	$ 421	3%

Sales and Revenues by Geographic Region

(Millions of dollars)	North America		Latin America		EAME		Asia/Pacific		External Sales and Revenues		Inter-Segment		Total Sales and Revenues	
	$	% Chg	$	% Chg	$	% Chg	$	% Chg	$	% Chg	$	% Chg	$	% Chg
Second Quarter 2019														
Construction Industries	$ 3,513	28%	$ 392	—%	$ 1,108	(5%)	$ 1,433	(22%)	$ 6,446	5%	$ 21	(40%)	$ 6,467	5%
Resource Industries	1,058	32%	448	14%	446	(22%)	759	14%	2,711	12%	89	(6%)	2,800	11%
Energy & Transportation	2,297	(11%)	325	13%	1,160	1%	742	7%	4,524	(4%)	962	(5%)	5,486	(4%)
All Other Segment	14	(18%)	1	—%	4	—%	15	(21%)	34	(17%)	91	10%	125	1%
Corporate Items and Eliminations	(39)		(2)		(5)		2		(44)		(1,163)		(1,207)	
Machinery, Energy & Transportation Sales	6,843	12%	1,164	9%	2,713	(6%)	2,951	(8%)	13,671	3%	—	—	13,671	3%
Financial Products Segment	563	5%	76	7%	102	1%	132	10%	873 [1]	5%	—	—	873	5%
Corporate Items and Eliminations	(72)		(11)		(9)		(20)		(112)		—		(112)	
Financial Products Revenues	491	2%	65	8%	93	(1%)	112	14%	761	4%	—	—	761	4%
Consolidated Sales and Revenues	$ 7,334	11%	$ 1,229	9%	$ 2,806	(6%)	$ 3,063	(7%)	$ 14,432	3%	$ —	—	$ 14,432	3%
Second Quarter 2018														
Construction Industries	$ 2,739		$ 392		$ 1,171		$ 1,835		$ 6,137		$ 35		$ 6,172	
Resource Industries	804		394		569		664		2,431		95		2,526	
Energy & Transportation	2,582		287		1,153		692		4,714		1,010		5,724	
All Other Segment	17		1		4		19		41		83		124	
Corporate Items and Eliminations	(40)		(3)		—		(1)		(44)		(1,223)		(1,267)	
Machinery, Energy & Transportation Sales	6,102		1,071		2,897		3,209		13,279		—		13,279	
Financial Products Segment	537		71		101		120		829 [1]		—		829	
Corporate Items and Eliminations	(57)		(11)		(7)		(22)		(97)		—		(97)	
Financial Products Revenues	480		60		94		98		732		—		732	
Consolidated Sales and Revenues	$ 6,582		$ 1,131		$ 2,991		$ 3,307		$ 14,011		$ —		$ 14,011	

[1] Includes revenues from Machinery, Energy & Transportation of $136 million and $118 million in the second quarter of 2019 and 2018, respectively.

CONSOLIDATED OPERATING PROFIT



The chart above graphically illustrates reasons for the change in consolidated operating profit between the second quarter of 2018 (at left) and the second quarter of 2019 (at right). Caterpillar management utilizes these charts internally to visually communicate with the company's Board of Directors and employees. The bar titled Other includes ***consolidating adjustments*** and ***Machinery, Energy & Transportation other operating (income) expenses***.

Operating profit for the second quarter of 2019 was $2.213 billion, up 2 percent compared with $2.167 billion in the second quarter of 2018. The increase of $46 million was due to favorable price realization and lower selling, general and administrative (SG&A) and research and development (R&D) expenses, offset by higher manufacturing costs and unfavorable sales volume due to product mix.

The increase in manufacturing costs was primarily due to higher material costs, including tariffs, variable labor and burden and warranty expense. Favorable SG&A/R&D expenses were mostly due to lower short-term incentive compensation expense.

The change in sales volume was the result of an unfavorable mix of products primarily within Energy & Transportation and Construction Industries.

Short-term incentive compensation expense was about $210 million in the second quarter of 2019, compared to about $360 million in the second quarter of 2018.

Operating profit margin was 15.3 percent for the second quarter of 2019, compared with 15.5 percent for the second quarter of 2018.

Other Profit/Loss and Tax Items

- Interest expense excluding Financial Products in the second quarter of 2019 was $103 million, compared with $102 million in the second quarter of 2018.
- Other income/expense in the second quarter of 2019 was income of $68 million, compared with income of $121 million in the second quarter of 2018. The unfavorable change was primarily a result of the impact from pension and other postretirement benefit plans.
- The provision for income taxes for the second quarter of 2019 reflected an estimated annual tax rate of 26 percent, compared with 24 percent for the second quarter of 2018, excluding the discrete items discussed in the following paragraph. The increase was largely driven by the application of U.S. tax reform provisions to the earnings of certain non-U.S. subsidiaries, which do not have a calendar fiscal year-end. These provisions did not apply to these subsidiaries in 2018.

  A discrete tax benefit of $3 million was recorded in the second quarter of 2019, compared with $9 million in the second quarter of 2018, for the settlement of stock-based compensation awards with associated tax deductions in excess of cumulative U.S. GAAP compensation expense. The provision for income taxes in the second quarter of 2018 also included a $25 million benefit for the release of a valuation allowance against the deferred tax assets of a non-U.S. subsidiary.

Profit by Segment

(Millions of dollars)	Second Quarter 2019	Second Quarter 2018	$ Change	% Change
Construction Industries	$ 1,247	$ 1,154	$ 93	8%
Resource Industries	481	411	70	17%
Energy & Transportation	886	1,012	(126)	(12%)
All Other Segment	11	23	(12)	(52%)
Corporate Items and Eliminations	(441)	(466)	25	
Machinery, Energy & Transportation	2,184	2,134	50	2%
Financial Products Segment	193	134	59	44%
Corporate Items and Eliminations	(50)	(5)	(45)	
Financial Products	143	129	14	11%
Consolidating Adjustments	(114)	(96)	(18)	
Consolidated Operating Profit	$ 2,213	$ 2,167	$ 46	2%

Construction Industries

Construction Industries' total sales were $6.467 billion in the second quarter of 2019, a 5 percent increase compared with $6.172 billion in the second quarter of 2018. Sales increased primarily due to the favorable impact of dealers increasing inventories in the second quarter of 2019, compared with decreasing inventories in the second quarter of 2018. Favorable price realization was partially offset by unfavorable currency impacts.

- In North America, sales increased primarily due to higher demand, including the favorable impact of dealer inventories, and favorable price realization.
- Construction activities remained at low levels in Latin America, leading to flat sales.
- In EAME, the sales decrease was primarily due to a currency impact related to the euro.
- Sales in Asia/Pacific declined due to lower demand mainly in China driven by continued competitive pricing pressures and timing of the selling season as well as unfavorable currency impacts.

Construction Industries' profit was $1.247 billion in the second quarter of 2019, an 8 percent increase compared with $1.154 billion in the second quarter of 2018. The increase in profit was due to favorable price realization, which was mostly offset by higher manufacturing costs. The increase in sales volume was more than offset by an unfavorable mix of products. Higher material costs and variable labor and burden, including the absence of Brazil incentives, were the primary drivers of increased manufacturing costs.

Construction Industries' profit as a percent of total sales was 19.3 percent in the second quarter of 2019, compared with 18.7 percent in the second quarter of 2018.

Resource Industries

Resource Industries' total sales were $2.800 billion in the second quarter of 2019, an 11 percent increase compared with $2.526 billion in the second quarter of 2018. The increase was primarily due to higher equipment demand and favorable price realization. Mining customers increased capital spending to support ongoing mine site operations, which drove higher sales. In addition, sales increased for non-residential construction and quarry and aggregate customers.

Resource Industries' profit was $481 million in the second quarter of 2019, a 17 percent increase compared with $411 million in the second quarter of 2018. The improvement was mostly due to favorable price realization and increased sales volume, partially offset by higher manufacturing costs. Manufacturing costs increased due to unfavorable cost absorption, as well as higher material costs, variable labor and burden and warranty expense. Cost absorption was unfavorable as inventory increased in the second quarter of 2018, compared with a decrease in the second quarter of 2019.

Resource Industries' profit as a percent of total sales was 17.2 percent in the second quarter of 2019, compared with 16.3 percent in the second quarter of 2018.

Energy & Transportation

Sales by Application (Millions of dollars)	Second Quarter 2019	Second Quarter 2018	$ Change	% Change
Oil and Gas	$ 1,305	$ 1,467	$ (162)	(11%)
Power Generation	1,021	992	29	3%
Industrial	957	969	(12)	(1%)
Transportation	1,241	1,286	(45)	(3%)
External Sales	4,524	4,714	(190)	(4%)
Inter-segment	962	1,010	(48)	(5%)
Total Sales	$ 5,486	$ 5,724	$ (238)	(4%)

Energy & Transportation's total sales were $5.486 billion in the second quarter of 2019, a 4 percent decrease compared with $5.724 billion in the second quarter of 2018. Sales decreased primarily due to lower sales volume and unfavorable currency impacts. The decrease was partially offset by favorable price realization.

- Oil and Gas – Sales decreased in North America primarily due to the timing of turbine project deliveries that occurred in the second quarter of 2018 and from lower demand for new equipment in the Permian Basin. This was partially offset by increased sales in EAME primarily due to higher turbine sales for production applications.
- Power Generation – Sales slightly increased mostly due to higher deliveries in North America for both large diesel reciprocating engines and turbines. Increases in North America were partially offset by lower sales in EAME for gas power generation applications.
- Industrial – Sales were slightly down, with a decrease in EAME primarily due to unfavorable currency impacts partially offset by higher volume in Asia Pacific and Latin America.
- Transportation – Sales were slightly lower primarily due to the timing of locomotive deliveries in the second quarter of 2018 and reduced marine activity in North America, partially offset by higher sales for rail services.

Energy & Transportation's profit was $886 million in the second quarter of 2019, compared with $1.012 billion in the second quarter of 2018. The decrease of 12 percent was mostly due to an unfavorable mix of applications and lower volume. Price realization was slightly favorable, offset by slightly higher manufacturing costs.

Energy & Transportation's profit as a percent of total sales was 16.2 percent in the second quarter of 2019, compared with 17.7 percent in the second quarter of 2018.

Financial Products Segment

Financial Products' segment revenues were $873 million in the second quarter of 2019, an increase of $44 million, or 5 percent, from the second quarter of 2018. The increase was primarily due to higher average financing rates in North America and Asia/Pacific and higher average ***earning assets*** in North America.

Financial Products' segment profit was $193 million in the second quarter of 2019, up 44 percent compared with $134 million in the second quarter of 2018. The increase was primarily due to a decrease in the provision for credit losses at Cat Financial and an increase in net yield on average earning assets.

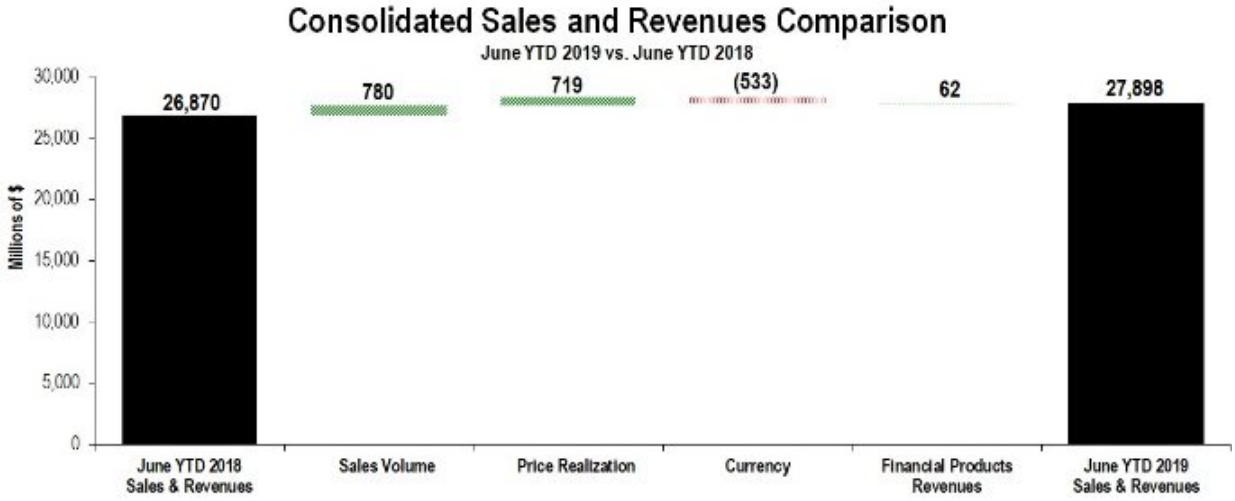
At the end of the second quarter of 2019, past dues at Cat Financial were 3.38 percent, compared with 3.16 percent at the end of the second quarter of 2018. The increase in past dues was primarily driven by EAME. Write-offs, net of recoveries, were $74 million for the second quarter of 2019, compared with $80 million for the second quarter of 2018. As of June 30, 2019, Cat Financial's allowance for credit losses totaled $523 million, or 1.81 percent of our recorded investment in finance receivables, compared with $534 million, or 1.89 percent of our recorded investment in finance receivables at March 31, 2019. The allowance for credit losses at year-end 2018 was $511 million, or 1.80 percent of our recorded investment in finance receivables.

Corporate Items and Eliminations

Expense for corporate items and eliminations was $491 million in the second quarter of 2019, compared with $471 million in the second quarter of 2018.

SIX MONTHS ENDED JUNE 30, 2019 COMPARED WITH SIX MONTHS ENDED JUNE 30, 2018

CONSOLIDATED SALES AND REVENUES



The chart above graphically illustrates reasons for the change in consolidated sales and revenues between the six months ended June 30, 2018 (at left) and the six months ended June 30, 2019 (at right). Caterpillar management utilizes these charts internally to visually communicate with the company's Board of Directors and employees.

Total sales and revenues were $27.898 billion for the six months ended June 30, 2019, an increase of $1.028 billion, or 4 percent, compared with $26.870 billion for the six months ended June 30, 2018. The increase was primarily due to higher sales volume driven by improved demand for equipment and favorable price realization. Sales increased in Resource Industries and Construction Industries, while Energy & Transportation decreased. The sales increase was partially offset by unfavorable currency impacts, primarily from a stronger euro, Australian dollar and Chinese yuan. Financial Products' revenues also increased.

The largest sales increase was in North America, which improved 10 percent driven by higher demand, including a favorable impact from changes in dealer inventories, and favorable price realization. Dealers increased inventories more significantly for the six months ended June 30, 2019, compared with the six months ended June 30, 2018.

Sales increased 9 percent in Latin America due to improved demand from low levels in a few countries.

EAME sales decreased 6 percent primarily due to economic uncertainty in several countries in the region and a weaker euro, partially offset by favorable price realization.

Asia/Pacific sales were about flat as an unfavorable currency impact of a weaker Australian dollar and Chinese yuan was offset by higher demand and favorable price realization. Lower demand in China due to continued competitive pricing pressures was more than offset by higher demand in other countries in the region.

During the six months ended June 30, 2019, dealer machine and engine inventories increased about $1.8 billion, compared to an increase of about $1.3 billion during the six months ended June 30, 2018. Dealers are independent, and there could be many reasons for changes in their inventory levels, including their expectations of future demand and product delivery times. Dealers' demand expectations take into account seasonal changes, macroeconomic conditions, machine rental rates and other factors. Delivery times can vary based on availability of product from Caterpillar factories and product distribution centers. Based on current dealer inventory levels, we now expect dealer inventories will increase for the full year by about $900 million compared to prior year end, driven by Construction Industries and Resource Industries.

Sales and Revenues by Segment

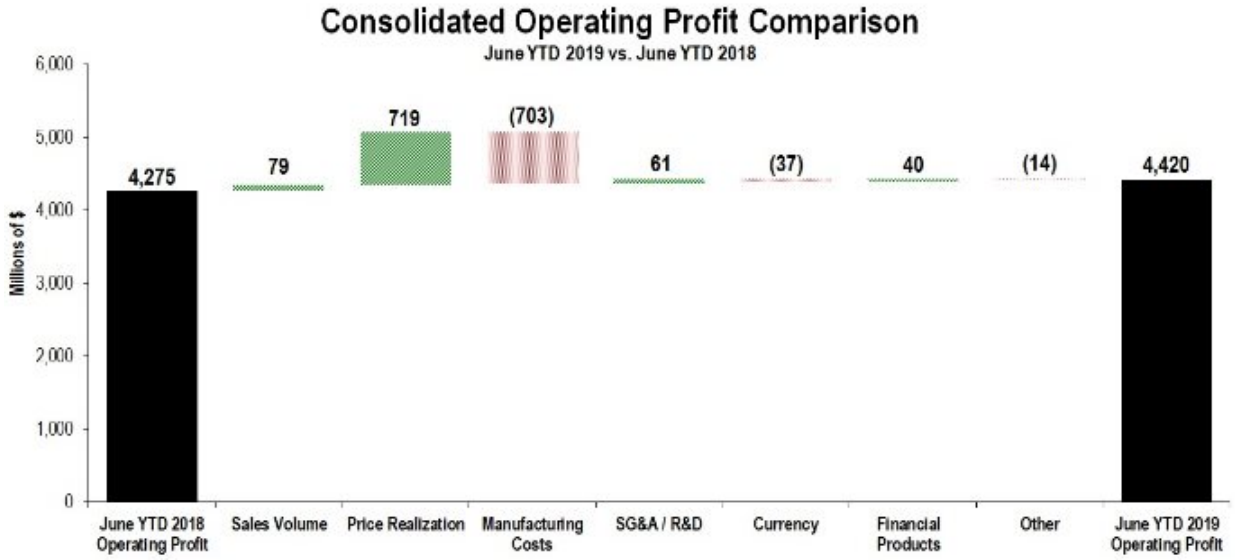
(Millions of dollars)	Six Months Ended June 30, 2018	Sales Volume	Price Realization	Currency	Inter-Segment / Other	Six Months Ended June 30, 2019	$ Change	% Change
Construction Industries	$ 11,849	$ 363	$ 409	$ (270)	$ (11)	$ 12,340	$ 491	4%
Resource Industries	4,835	559	239	(79)	(27)	5,527	692	14%
Energy & Transportation	10,943	(122)	71	(182)	(14)	10,696	(247)	(2%)
All Other Segment	240	(5)	—	(2)	13	246	6	3 %
Corporate Items and Eliminations	(2,438)	(15)	—	—	39	(2,414)	24	
Machinery, Energy & Transportation Sales	25,429	780	719	(533)	—	26,395	966	4%
Financial Products Segment	1,622	—	—	—	101	1,723	101	6%
Corporate Items and Eliminations	(181)	—	—	—	(39)	(220)	(39)	
Financial Products Revenues	1,441	—	—	—	62	1,503	62	4%
Consolidated Sales and Revenues	$ 26,870	$ 780	$ 719	$ (533)	$ 62	$ 27,898	$ 1,028	4%

Sales and Revenues by Geographic Region

(Millions of dollars)	North America		Latin America		EAME		Asia/Pacific		External Sales and Revenues		Inter-Segment		Total Sales and Revenues	
	$	% Chg	$	% Chg	$	% Chg	$	% Chg	$	% Chg	$	% Chg	$	% Chg
Six Months Ended June 30, 2019														
Construction Industries	$ 6,478	21%	$ 711	(3%)	$ 2,114	(6%)	$ 2,995	(14%)	$ 12,298	4%	$ 42	(21%)	$ 12,340	4%
Resource Industries	2,009	25%	871	16%	914	(16%)	1,564	31%	5,358	15%	169	(14%)	5,527	14%
Energy & Transportation	4,448	(7%)	657	16%	2,192	(2%)	1,460	6%	8,757	(3%)	1,939	(1%)	10,696	(2%)
All Other Segment	22	(31%)	1	—%	15	88%	33	(11%)	71	(9%)	175	8%	246	3%
Corporate Items and Eliminations	(80)		(1)		(8)		—		(89)		(2,325)		(2,414)	
Machinery, Energy & Transportation Sales	12,877	10%	2,239	9%	5,227	(6%)	6,052	—%	26,395	4%	—	—%	26,395	4%
Financial Products Segment	1,121	7%	146	1%	204	1%	252	12%	1,723 [1]	6%	—	—%	1,723	6%
Corporate Items and Eliminations	(141)		(22)		(18)		(39)		(220)		—		(220)	
Financial Products Revenues	980	4%	124	2%	186	(2%)	213	14%	1,503	4%	—	—%	1,503	4%
Consolidated Sales and Revenues	$ 13,857	9%	$ 2,363	9%	$ 5,413	(6%)	$ 6,265	—%	$ 27,898	4%	$ —	—%	$ 27,898	4%
Six Months Ended June 30, 2018														
Construction Industries	$ 5,359		$ 736		$ 2,238		$ 3,463		$ 11,796		$ 53		$ 11,849	
Resource Industries	1,602		754		1,089		1,194		4,639		196		4,835	
Energy & Transportation	4,807		567		2,245		1,371		8,990		1,953		10,943	
All Other Segment	32		1		8		37		78		162		240	
Corporate Items and Eliminations	(68)		(2)		(3)		(1)		(74)		(2,364)		(2,438)	
Machinery, Energy & Transportation Sales	11,732		2,056		5,577		6,064		25,429		—		25,429	
Financial Products Segment	1,049		145		202		226		1,622 [1]		—		1,622	
Corporate Items and Eliminations	(106)		(24)		(12)		(39)		(181)		—		(181)	
Financial Products Revenues	943		121		190		187		1,441		—		1,441	
Consolidated Sales and Revenues	$ 12,675		$ 2,177		$ 5,767		$ 6,251		$ 26,870		$ —		$ 26,870	

[1] Includes revenues from Machinery, Energy & Transportation of $267 million and $223 million in the six months ended June 30, 2019 and 2018, respectively.

CONSOLIDATED OPERATING PROFIT



The chart above graphically illustrates reasons for the change in consolidated operating profit between the six months ended June 30, 2018 (at left) and the six months ended June 30, 2019 (at right). Caterpillar management utilizes these charts internally to visually communicate with the company's Board of Directors and employees. The bar titled Other includes consolidating adjustments and Machinery, Energy & Transportation other operating (income) expenses.

Operating profit for the six months ended June 30, 2019, was $4.420 billion, compared with $4.275 billion for the six months ended June 30, 2018. The increase of $145 million was primarily due to favorable price realization, higher sales volume and lower SG&A/R&D expenses, mostly offset by higher manufacturing costs. Financial Products operating profit increased.

The operating profit impact from the change in sales volume was the result of higher demand for equipment, mostly offset by an unfavorable mix of products primarily within Construction Industries and Energy & Transportation.

The increase in manufacturing costs was primarily due to higher variable labor and burden; material costs, including tariffs; and unfavorable cost absorption. Cost absorption was unfavorable as inventory increased more significantly in the first half of 2018 than the first half of 2019. SG&A/R&D expenses were lower primarily due to lower short-term incentive compensation expense, partially offset by increased targeted investments and timing of corporate-level expenses.

Short-term incentive compensation expense is directly related to financial and operational performance, measured against targets set annually. Expense for the six months ended June 30, 2019, was about $430 million, compared with about $720 million for the six months ended June 30, 2018. For 2019, short-term incentive compensation expense is expected to be about $800 million, compared with $1.4 billion in 2018.

Other Profit/Loss Items

- Interest expense excluding Financial Products for the six months ended June 30, 2019, was $206 million, compared with $203 million for the six months ended June 30, 2018.

- Other income/expense for the six months ended June 30, 2019, was income of $228 million, compared with income of $248 million for the six months ended June 30, 2018. The unfavorable change was primarily a result of the impact from pension and other postemployment benefits (OPEB) plans, mostly offset by lower currency translation and hedging net losses and favorable impact from equity securities.

- The provision for income taxes for the six months ended June 30, 2019, reflected an estimated annual tax rate of 26 percent, compared with 24 percent for the six months ended June 30, 2018, excluding the discrete items discussed in the following paragraph. The increase was largely driven by the application of U.S. tax reform provisions to the earnings of certain non-U.S. subsidiaries, which do not have a calendar fiscal year-end. These provisions did not apply to these subsidiaries in 2018.

As a result of final regulations received in 2019 providing additional guidance related to the calculation of the mandatory deemed repatriation of non-U.S. earnings due to U.S. tax reform, we recorded a discrete tax benefit of $178 million for the six months ended June 30, 2019, to adjust unrecognized tax benefits. In addition, a discrete tax benefit of $26 million was recorded for the six months ended June 30, 2019, compared with $49 million for the six months ended June 30, 2018, for the settlement of stock-based compensation awards with associated tax deductions in excess of cumulative U.S. GAAP compensation expense. The provision for income taxes for the first six months of 2018 also included a $25 million benefit for the release of a valuation allowance against the deferred tax assets of a non-U.S. subsidiary.

Profit by Segment

(Millions of dollars)	Six Months Ended June 30, 2019	Six Months Ended June 30, 2018	$ Change	% Change
Construction Industries	$ 2,332	$ 2,271	$ 61	3 %
Resource Industries	1,057	789	268	34 %
Energy & Transportation	1,724	1,886	(162)	(9)%
All Other Segment	36	80	(44)	(55)%
Corporate Items and Eliminations	(816)	(837)	21	
Machinery, Energy & Transportation	4,333	4,189	144	3 %
Financial Products Segment	404	275	129	47 %
Corporate Items and Eliminations	(96)	(7)	(89)	
Financial Products	308	268	40	15 %
Consolidating Adjustments	(221)	(182)	(39)	
Consolidated Operating Profit	$ 4,420	$ 4,275	$ 145	3 %

Construction Industries

Construction Industries' total sales were $12.340 billion for the six months ended June 30, 2019, compared with $11.849 billion for the six months ended June 30, 2018. The increase was mostly due to favorable price realization and higher sales volume for construction equipment. The sales increase was partially offset by unfavorable currency impacts, primarily from a weaker euro and Chinese yuan.

- Sales volume increased primarily due to higher demand for construction equipment and the impact of favorable changes in dealer inventories. Dealer inventories increased significantly more in the six months ended June 30, 2019, than in the six months ended June 30, 2018.

Sales increased in North America and decreased in all other regions.

- In North America, the sales increase was primarily due to higher demand for construction equipment, including favorable changes in dealer inventories, and favorable price realization. Dealer inventories increased significantly more in the six months ended June 30, 2019, than in the six months ended June 30, 2018.
- Latin America sales were slightly lower as construction activities remained at low levels.
- Sales decreased in EAME primarily due to unfavorable currency impacts, mostly from a weaker euro.
- Sales in Asia/Pacific declined due to lower demand mainly in China driven by continued competitive pricing pressures and unfavorable currency impacts, mostly from a weaker Chinese yuan.

Construction Industries' profit was $2.332 billion for the six months ended June 30, 2019, compared with $2.271 billion for the six months ended June 30, 2018. The increase in profit was a result of favorable price realization, mostly offset by higher manufacturing costs. Manufacturing costs increased primarily due to unfavorable variable labor and burden, including the absence of Brazil incentives and increased freight costs, and higher material costs.

Construction Industries' profit as a percent of total sales was 18.9 percent for the six months ended June 30, 2019, compared with 19.2 percent for the six months ended June 30, 2018.

Resource Industries

Resource Industries' total sales were $5.527 billion for the six months ended June 30, 2019, an increase of $692 million from the six months ended June 30, 2018. The increase was primarily due to higher equipment demand and favorable price realization. Higher sales were driven by increased capital investment by mining customers to support ongoing mine site operations. In addition, sales increased for our non-residential construction and quarry and aggregate customers.

Resource Industries' profit was $1.057 billion for the six months ended June 30, 2019, compared with $789 million for the six months ended June 30, 2018. The improvement was mostly due to higher sales volume and favorable price realization, partially offset by higher manufacturing costs. Manufacturing costs increased due to higher variable labor and burden, material costs, unfavorable cost absorption and warranty expense. Cost absorption was unfavorable as inventory increased more significantly in the first half of 2018 than the first half of 2019.

Resource Industries' profit as a percent of total sales was 19.1 percent for the six months ended June 30, 2019, compared with 16.3 percent for the six months ended June 30, 2018.

Energy & Transportation

Sales by Application **(Millions of dollars)**	**Six Months Ended June 30, 2019**	**Six Months Ended June 30, 2018**	**$ Change**	**% Change**
Oil and Gas	$ 2,436	$ 2,682	$ (246)	(9%)
Power Generation	2,057	1,961	96	5%
Industrial	1,861	1,875	(14)	(1%)
Transportation	2,403	2,472	(69)	(3%)
External Sales	8,757	8,990	(233)	(3%)
Inter-Segment	1,939	1,953	(14)	(1%)
Total Sales	$ 10,696	$ 10,943	$ (247)	(2%)

Energy & Transportation's total sales were $10.696 billion for the six months ended June 30, 2019, compared with $10.943 billion for the six months ended June 30, 2018. The decrease was primarily due to unfavorable currency impacts and lower sales volume, partially offset by favorable price realization.

- Oil and Gas – Sales decreased in North America primarily due to the timing of turbine project deliveries that occurred in the six months ended June 30, 2018, and lower new equipment demand for well servicing. Lower sales in North America were partially offset by increases in all other regions.
- Power Generation – Sales increased mostly due to higher deliveries for both large diesel reciprocating engines and turbines in North America and Asia/Pacific. Increases were partially offset by lower sales in EAME driven by gas power generation applications and unfavorable currency impacts.
- Industrial – Sales were about flat, with a decrease in EAME primarily due to unfavorable currency impacts partially offset by higher volume in all other regions.
- Transportation – Sales were slightly lower primarily due to the timing of locomotive deliveries and unfavorable currency impacts, partially offset by higher sales for rail services.

Energy & Transportation's profit was $1.724 billion for the six months ended June 30, 2019, compared with $1.886 billion for the six months ended June 30, 2018. The decrease was mostly due to higher manufacturing costs and an unfavorable mix of applications, partially offset by favorable price realization. Manufacturing costs increased due to higher variable labor and burden, warranty expense and material costs.

Energy & Transportation's profit as a percent of total sales was 16.1 percent for the six months ended June 30, 2019, compared with 17.2 percent for the six months ended June 30, 2018.

Financial Products Segment

Financial Products' segment revenues were $1.723 billion for the six months ended June 30, 2019, an increase of $101 million, or 6 percent, from the six months ended June 30, 2018. The increase was primarily due to higher average financing rates in North America and Asia/Pacific and higher average earning assets in North America.

Financial Products' segment profit was $404 million for the six months ended June 30, 2019, up 47 percent compared with $275 million for the six months ended June 30, 2018. The increase was primarily due to a favorable impact from mark-to-market on equity securities in Insurance Services, a decrease in the provision for credit losses at Cat Financial and an increase in net yield on average earning assets. These favorable impacts were partially offset by an increase in SG&A expenses.

Corporate Items and Eliminations

Expense for corporate items and eliminations was $912 million for the six months ended June 30, 2019, compared with $844 million for the six months ended June 30, 2018, primarily due to an unfavorable change in mark-to-market deferred compensation expense.

RESTRUCTURING COSTS

Restructuring costs for the three and six months ended June 30, 2019 and 2018 were as follows:

(Millions of dollars)	Three Months Ended June 30	
	2019	2018
Employee separations [1]	$ 10	$ 45
Long-lived asset impairments [1]	29	30
Other [2]	71	39
Total restructuring costs	$ 110	$ 114

	Six Months Ended June 30	
	2019	2018
Employee separations [1]	$ 25	$ 78
Long-lived asset impairments [1]	36	30
Other [2]	97	75
Total restructuring costs	$ 158	$ 183

[1] Recognized in Other operating (income) expenses.

[2] Represents costs related to our restructuring programs, primarily for inventory write-downs, project management costs, accelerated depreciation, equipment relocation and building demolition, all of which are primarily included in Cost of goods sold.

For the six months ended June 30, 2019, the restructuring costs were primarily related to restructuring actions in Construction Industries and Energy & Transportation. For the six month ended June 30, 2018, the restructuring costs were primarily related to ongoing facility closures across the company.

Certain restructuring costs are a reconciling item between Segment profit and Consolidated profit before taxes.

The following table summarizes the 2018 and 2019 employee separation activity:

(Millions of dollars)	
Liability balance at December 31, 2017	$ 249
Increase in liability (separation charges)	112
Reduction in liability (payments)	(276)
Liability balance at December 31, 2018	85
Increase in liability (separation charges)	25
Reduction in liability (payments)	(60)
Liability balance at June 30, 2019	$ 50

Most of the liability balance at June 30, 2019 is expected to be paid in 2019 and 2020.

In September 2015, we announced a large scale restructuring plan (the Plan) including a voluntary retirement enhancement program for qualifying U.S. employees, several voluntary separation programs outside of the U.S., additional involuntary programs throughout the company and manufacturing facility consolidations and closures that occurred through 2018. The largest action among those included in the Plan was related to our European manufacturing footprint, which led to the Gosselies, Belgium, facility closure. In the first six months of 2019, we incurred $19 million of restructuring costs related to the Plan. Total restructuring costs incurred since the inception of the Plan were $1,807 million. The remaining costs of approximately $20 million related to the Plan are expected to be recognized in 2019.

We expect that restructuring actions will result in an incremental benefit to operating costs, primarily Cost of goods sold and SG&A expenses of about $300 million in 2019 compared with 2018.

GLOSSARY OF TERMS

1. **All Other Segment** – Primarily includes activities such as: business strategy, product management and development, manufacturing and sourcing of filters and fluids, undercarriage, ground-engaging tools, fluid transfer products, precision seals, rubber sealing and connecting components primarily for Cat® products; parts distribution; integrated logistics solutions, distribution services responsible for dealer development and administration including a wholly owned dealer in Japan, dealer portfolio management and ensuring the most efficient and effective distribution of machines, engines and parts; and digital investments for new customer and dealer solutions that integrate data analytics with state-of-the-art digital technologies while transforming the buying experience.

2. **Consolidating Adjustments** – Elimination of transactions between Machinery, Energy & Transportation and Financial Products.

3. **Construction Industries** – A segment primarily responsible for supporting customers using machinery in infrastructure, forestry and building construction applications. Responsibilities include business strategy, product design, product management and development, manufacturing, marketing and sales and product support. The product portfolio includes asphalt pavers; backhoe loaders; compactors; cold planers; compact track and multi-terrain loaders; mini, small, medium and large track excavators; forestry excavators; feller bunchers; harvesters; knuckleboom loaders; motor graders; pipelayers; road reclaimers; skidders; skid steer loaders; telehandlers; small and medium track-type tractors; track-type loaders; utility vehicles; wheel excavators; compact, small and medium wheel loaders; and related parts and work tools.

4. **Corporate Items and Eliminations** – Includes corporate-level expenses; timing differences, as some expenses are reported in segment profit on a cash basis; methodology differences between segment and consolidated external reporting; restructuring costs; and inter-segment eliminations.

5. **Currency** – With respect to sales and revenues, currency represents the translation impact on sales resulting from changes in foreign currency exchange rates versus the U.S. dollar. With respect to operating profit, currency represents the net translation impact on sales and operating costs resulting from changes in foreign currency exchange rates versus the U.S. dollar. Currency only includes the impact on sales and operating profit for the Machinery, Energy & Transportation lines of business; currency impacts on Financial Products' revenues and operating profit are included in the Financial Products' portions of the respective analyses. With respect to other income/expense, currency represents the effects of forward and option contracts entered into by the company to reduce the risk of fluctuations in exchange rates (hedging) and the net effect of changes in foreign currency exchange rates on our foreign currency assets and liabilities for consolidated results (translation).

6. **EAME** – A geographic region including Europe, Africa, the Middle East and the Commonwealth of Independent States (CIS).

7. **Earning Assets** – Assets consisting primarily of total finance receivables net of unearned income, plus equipment on operating leases, less accumulated depreciation at Cat Financial.

8. **Energy & Transportation** – A segment primarily responsible for supporting customers using reciprocating engines, turbines, diesel-electric locomotives and related parts across industries serving Oil and Gas, Power Generation, Industrial and Transportation applications, including marine and rail-related businesses. Responsibilities include business strategy, product design, product management and development, manufacturing, marketing and sales and product support of turbine machinery and integrated systems and solutions and turbine-related services; reciprocating engine-powered generator sets; integrated systems used in the electric power generation industry; reciprocating engines and integrated systems and solutions for the marine and oil and gas industries; reciprocating engines supplied to the industrial industry as well as Cat machinery; the remanufacturing of Caterpillar engines and components and remanufacturing services for other companies; the business strategy, product design, product management and development, manufacturing, remanufacturing, leasing and service of diesel-electric locomotives and components and other rail-related products and services; and product support of on-highway vocational trucks for North America.

9. **Financial Products Segment** – Provides financing alternatives to customers and dealers around the world for Caterpillar products, as well as financing for vehicles, power generation facilities and marine vessels that, in most cases, incorporate Caterpillar products. Financing plans include operating and finance leases, installment sale contracts, working capital loans and wholesale financing plans. The segment also provides insurance and risk management products and services that help customers and dealers manage their business risk. Insurance and risk management products offered include physical damage insurance, inventory protection plans, extended service coverage for machines and engines, and dealer property and casualty insurance. The various forms of financing, insurance and risk management products offered to customers and dealers help support the purchase and lease of our equipment. The segment also earns revenues from Machinery, Energy & Transportation, but the related costs are not allocated to operating segments. Financial Products' segment profit is determined on a pretax basis and includes other income/expense items.

10. **Latin America** – A geographic region including Central and South American countries and Mexico.

11. **Machinery, Energy & Transportation (ME&T)** – Represents the aggregate total of Construction Industries, Resource Industries, Energy & Transportation, All Other Segment and related corporate items and eliminations.

12. **Machinery, Energy & Transportation Other Operating (Income) Expenses** – Comprised primarily of gains/losses on disposal of long-lived assets, gains/losses on divestitures and legal settlements and accruals.

13. **Manufacturing Costs** – Manufacturing costs exclude the impacts of currency and represent the volume-adjusted change for variable costs and the absolute dollar change for period manufacturing costs. Variable manufacturing costs are defined as having a direct relationship with the volume of production. This includes material costs, direct labor and other costs that vary directly with production volume such as freight, power to operate machines and supplies that are consumed in the manufacturing process. Period manufacturing costs support production but are defined as generally not having a direct relationship to short-term changes in volume. Examples include machinery and equipment repair, depreciation on manufacturing assets, facility support, procurement, factory scheduling, manufacturing planning and operations management.

14. **Price Realization** – The impact of net price changes excluding currency and new product introductions. Price realization includes geographic mix of sales, which is the impact of changes in the relative weighting of sales prices between geographic regions.

15. **Resource Industries** – A segment primarily responsible for supporting customers using machinery in mining, quarry and aggregates, waste and material handling applications. Responsibilities include business strategy, product design, product management and development, manufacturing, marketing and sales and product support. The product portfolio includes large track-type tractors, large mining trucks, hard rock vehicles, longwall miners, electric rope shovels, draglines, hydraulic shovels, rotary drills, large wheel loaders, off-highway trucks, articulated trucks, wheel tractor scrapers, wheel dozers, landfill compactors, soil compactors, hard rock continuous mining systems, select work tools, machinery components, electronics and control systems and related parts. In addition to equipment, Resource Industries also develops and sells technology products and services to provide customers fleet management, equipment management analytics and autonomous machine capabilities. Resource Industries also manages areas that provide services to other parts of the company, including integrated manufacturing and research and development.

16. **Restructuring Costs** – May include costs for employee separation, long-lived asset impairments and contract terminations. These costs are included in Other operating (income) expenses except for defined-benefit plan curtailment losses and special termination benefits, which are included in Other income (expense). Restructuring costs also include other exit-related costs which may consist of accelerated depreciation, inventory write-downs, building demolition, equipment relocation and project management costs and LIFO inventory decrement benefits from inventory liquidations at closed facilities, all of which are primarily included in Cost of goods sold.

17. **Sales Volume** – With respect to sales and revenues, sales volume represents the impact of changes in the quantities sold for Machinery, Energy & Transportation as well as the incremental sales impact of new product introductions, including emissions-related product updates. With respect to operating profit, sales volume represents the impact of changes in the quantities sold for Machinery, Energy & Transportation combined with product mix as well as the net operating profit impact of new product introductions, including emissions-related product updates. Product mix represents the net operating profit impact of changes in the relative weighting of Machinery, Energy & Transportation sales with respect to total sales. The impact of sales volume on segment profit includes inter-segment sales.

18. **Services** – Enterprise services include, but are not limited to, aftermarket parts, Financial Products' revenues and other service-related revenues. Machinery, Energy & Transportation segments exclude Financial Products' revenues.

LIQUIDITY AND CAPITAL RESOURCES

Sources of funds

We generate significant capital resources from operating activities, which are the primary source of funding for our ME&T operations. Funding for these businesses is also available from commercial paper and long-term debt issuances. Financial Products' operations are funded primarily from commercial paper, term debt issuances and collections from its existing portfolio. During the first half of 2019, we experienced favorable global liquidity conditions in both our ME&T and Financial Products' operations. On a consolidated basis, we ended the first half of 2019 with $7.43 billion of cash, a decrease of $428 million from year-end 2018. We intend to maintain a strong cash and liquidity position.

Our cash balances are held in numerous locations throughout the world. As a result of U.S. tax reform legislation enacted in 2017, we expect to be able to use cash held by non-U.S. subsidiaries in the United States in the future with minimal tax consequences.

Consolidated operating cash flow for the first half of 2019 was $3.71 billion, up from $3.08 billion for the same period last year. The increase was primarily due to favorable working capital, which was partially offset by higher cash payments for income taxes and lower profit excluding lower accruals for short-term incentive compensation expense in 2019. Within working capital, changes to inventories, accounts receivable, accrued expenses and customer advances favorably impacted cash flow, but were partially offset by changes in accounts payable. See further discussion of operating cash flow under ME&T and Financial Products.

Total debt as of June 30, 2019 was $36.28 billion, a decrease of $275 million from year-end 2018. Debt related to Financial Products increased $70 million. Debt related to ME&T decreased $345 million in the first half of 2019, primarily due to the impact of new accounting guidance on a previously failed sale-leaseback transaction in Japan. In the first half of 2019, we repurchased $2.11 billion of Caterpillar common stock.

We have three global credit facilities with a syndicate of banks totaling $10.50 billion (Credit Facility) available in the aggregate to both Caterpillar and Cat Financial for general liquidity purposes. Based on management's allocation decision, which can be revised from time to time, the portion of the Credit Facility available to ME&T as of June 30, 2019 was $2.75 billion. Information on our Credit Facility is as follows:

- The 364-day facility of $3.15 billion (of which $0.82 billion is available to ME&T) expires in September 2019.
- The three-year facility, as amended in September 2018, of $2.73 billion (of which $0.72 billion is available to ME&T) expires in September 2021.
- The five-year facility, as amended in September 2018, of $4.62 billion (of which $1.21 billion is available to ME&T) expires in September 2023.

At June 30, 2019, Caterpillar's consolidated net worth was $14.86 billion, which was above the $9.00 billion required under the Credit Facility. The consolidated net worth is defined as the consolidated shareholders' equity including preferred stock but excluding the pension and other postretirement benefits balance within Accumulated other comprehensive income (loss).

At June 30, 2019, Cat Financial's covenant interest coverage ratio was 1.60 to 1. This is above the 1.15 to 1 minimum ratio calculated as (1) profit excluding income taxes, interest expense and net gain/(loss) from interest rate derivatives to (2) interest expense calculated at the end of each calendar quarter for the rolling four quarter period then most recently ended, required by the Credit Facility.

In addition, at June 30, 2019, Cat Financial's six-month covenant leverage ratio was 8.23 to 1. This is below the maximum ratio of debt to net worth of 10 to 1, calculated (1) on a monthly basis as the average of the leverage ratios determined on the last day of each of the six preceding calendar months and (2) at each December 31, required by the Credit Facility.

In the event Caterpillar or Cat Financial does not meet one or more of their respective financial covenants under the Credit Facility in the future (and are unable to obtain a consent or waiver), the syndicate of banks may terminate the commitments allocated to the party that does not meet its covenants. Additionally, in such event, certain of Cat Financial's other lenders under other loan agreements where similar financial covenants or cross default provisions are applicable, may, at their election, choose to pursue remedies under those loan agreements, including accelerating the repayment of outstanding borrowings. At June 30, 2019, there were no borrowings under the Credit Facility.

Our total credit commitments and available credit as of June 30, 2019 were:

(Millions of dollars)	June 30, 2019		
	Consolidated	Machinery, Energy & Transportation	Financial Products
Credit lines available:			
Global credit facilities	$ 10,500	$ 2,750	$ 7,750
Other external	4,841	—	4,841
Total credit lines available	15,341	2,750	12,591
Less: Commercial paper outstanding	(4,434)	—	(4,434)
Less: Utilized credit	(1,104)	—	(1,104)
Available credit	$ 9,803	$ 2,750	$ 7,053

The other external consolidated credit lines with banks as of June 30, 2019 totaled $4.84 billion. These committed and uncommitted credit lines, which may be eligible for renewal at various future dates or have no specified expiration date, are used primarily by our subsidiaries for local funding requirements. Caterpillar or Cat Financial may guarantee subsidiary borrowings under these lines.

We receive debt ratings from the major credit rating agencies. Moody's currently rates our debt as "low-A", while Fitch and S&P maintain a "mid-A" debt rating. To date, this split rating has not had a material impact on our borrowing costs or our overall financial health. However, a downgrade of our credit ratings by any of the major credit rating agencies would result in increased borrowing costs and could make access to certain credit markets more difficult. In the event economic conditions deteriorate such that access to debt markets becomes unavailable, ME&T's operations would rely on cash flow from operations, use of existing cash balances, borrowings from Cat Financial and access to our Credit Facility. Our Financial Products' operations would rely on cash flow from its existing portfolio, existing cash balances, access to our Credit Facility and other credit line facilities of Cat Financial and potential borrowings from Caterpillar. In addition, we maintain a support agreement with Cat Financial, which requires Caterpillar to remain the sole owner of Cat Financial and may, under certain circumstances, require Caterpillar to make payments to Cat Financial should Cat Financial fail to maintain certain financial ratios.

Machinery, Energy & Transportation

Net cash provided by operating activities was $2.82 billion in the first half of 2019, compared with $3.03 billion for the same period in 2018. The decrease was primarily due to higher cash payments for income taxes and lower profit excluding lower accruals for short-term incentive compensation expense in 2019, which was partially offset by favorable working capital.

Net cash used for investing activities in the first half of 2019 was $193 million, compared with net cash used of $664 million in the first half of 2018. The change was primarily due to the acquisitions of ECM S.p.A. and Downer Freight Rail in the first half of 2018, decreased ME&T lending activity with Financial Products and lower capital expenditures.

Net cash used for financing activities during the first half of 2019 was $3.06 billion, compared with net cash used of $1.90 billion in the same period of 2018. In the first half of 2019, we repurchased $2.11 billion of Caterpillar common stock, an increase of $855 million compared to the first half of 2018. Additionally, proceeds from common stock issued from stock options exercised decreased $214 million in the first half of 2019 compared to the prior period.

While our short-term priorities for the use of cash may vary from time to time as business needs and conditions dictate, our long-term cash deployment strategy is focused on the following priorities. Our top priority is to maintain a strong financial position in support of a mid-A rating. Next, we intend to fund operational requirements and commitments. Then, we intend to fund priorities that profitably grow the company and return capital to shareholders through dividend growth and share repurchases. Additional information on cash deployment is as follows:

Strong financial position – Our top priority is to maintain a strong financial position in support of a mid-A rating. We track a diverse group of financial metrics that focus on liquidity, leverage, cash flow and margins which align with our cash deployment actions and the various methodologies used by the major credit rating agencies.

Operational excellence and commitments – Capital expenditures were $492 million during the first half of 2019, compared to $554 million for the same period in 2018. We expect ME&T's capital expenditures in 2019 to be about $1.3 billion. We made $198 million of contributions to our pension and other postretirement benefit plans during the first half of 2019. We currently anticipate full-year 2019 contributions of approximately $330 million. In comparison, we made $227 million of contributions to our pension and other postretirement benefit plans during the first half of 2018.

Fund strategic growth initiatives and return capital to shareholders – We intend to utilize our liquidity and debt capacity to fund targeted investments that drive long-term profitable growth focused in the areas of expanded offerings and services, including acquisitions.

As part of our new capital allocation strategy, ME&T free cash flow is a liquidity measure we will use going forward to determine the cash generated and available for financing activities including debt repayments, dividends and share repurchases. We define ME&T free cash flow as cash from ME&T operations excluding discretionary pension and other postretirement benefit plan contributions less capital expenditures. A goal of our new capital allocation strategy is to return substantially all ME&T free cash flow to shareholders in the form of dividends and share repurchases, while maintaining our mid-A rating.

Our share repurchase plans are subject to the company's cash deployment priorities and are evaluated on an ongoing basis considering the financial condition of the company and the economic outlook, corporate cash flow, the company's liquidity needs, and the health and stability of global credit markets. The timing and amount of future repurchases may vary depending on market conditions and investing priorities. In July 2018, the Board of Directors approved an authorization to repurchase up to $10 billion of Caterpillar common stock (the 2018 Authorization) effective January 1, 2019, with no expiration. In the first half of 2019, we repurchased $2.11 billion of Caterpillar common stock, with $7.89 billion remaining under the 2018 Authorization as of June 30, 2019. Caterpillar's basic shares outstanding as of June 30, 2019 were approximately 563 million.

Each quarter, our Board of Directors reviews the company's dividend for the applicable quarter. The Board evaluates the financial condition of the company and considers the economic outlook, corporate cash flow, the company's liquidity needs, and the health and stability of global credit markets to determine whether to maintain or change the quarterly dividend. Dividends totaled $986 million in the first half of 2019, representing 86 cents per share paid in each of the first and second quarters. Beginning with the dividend to be paid to shareholders in the third quarter of 2019, the Board approved a 20 percent increase in the quarterly dividend rate to $1.03 per share.

Financial Products

Financial Products operating cash flow was $1.04 billion in the first half of 2019, compared with $619 million for the same period a year ago. The increase was primarily due to an overpayment received on a foreign currency exchange settlement. The overpayment received on the last day of June was returned the first week of July. The cash was reflected as accounts payable as of June 30, 2019. Net cash used for investing activities was $818 million for the first half of 2019, compared with cash used of $1.86 billion for the same period in 2018. The change was primarily due to the impact of net intercompany purchased receivables and higher collections of finance receivables. Net cash used for financing activities was $206 million for the first half of 2019, compared with cash provided by financing activities of $1.22 billion for the same period in 2018. The change was primarily due to lower portfolio funding requirements.

RECENT ACCOUNTING PRONOUNCEMENTS

For a discussion of recent accounting pronouncements, see Part I, Item 1. Note 2 - "New accounting guidance."

CRITICAL ACCOUNTING POLICIES

For a discussion of the company's critical accounting policies, see Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations in our 2018 Annual Report on Form 10-K. There have been no significant changes to our critical accounting policies since our 2018 Annual Report on Form 10-K.

OTHER MATTERS

Environmental and Legal Matters

The Company is regulated by federal, state and international environmental laws governing its use, transport and disposal of substances and control of emissions. In addition to governing our manufacturing and other operations, these laws often impact the development of our products, including, but not limited to, required compliance with air emissions standards applicable to internal combustion engines. We have made, and will continue to make, significant research and development and capital expenditures to comply with these emissions standards.

We are engaged in remedial activities at a number of locations, often with other companies, pursuant to federal and state laws. When it is probable we will pay remedial costs at a site, and those costs can be reasonably estimated, the investigation, remediation, and operating and maintenance costs are accrued against our earnings. Costs are accrued based on consideration of currently available data and information with respect to each individual site, including available technologies, current applicable laws and regulations, and prior remediation experience. Where no amount within a range of estimates is more likely, we accrue the minimum. Where multiple potentially responsible parties are involved, we consider our proportionate share of the probable costs. In formulating the estimate of probable costs, we do not consider amounts expected to be recovered from insurance companies or others. We reassess these accrued amounts on a quarterly basis. The amount recorded for environmental remediation is not material and is included in Accrued expenses. We believe there is no more than a remote chance that a material amount for remedial activities at any individual site, or at all the sites in the aggregate, will be required.

On January 7, 2015, the Company received a grand jury subpoena from the U.S. District Court for the Central District of Illinois. The subpoena requested documents and information from the Company relating to, among other things, financial information concerning U.S. and non-U.S. Caterpillar subsidiaries (including undistributed profits of non-U.S. subsidiaries and the movement of cash among U.S. and non-U.S. subsidiaries). The Company has received additional subpoenas relating to this investigation requesting additional documents and information relating to, among other things, the purchase and resale of replacement parts by Caterpillar Inc. and non-U.S. Caterpillar subsidiaries, dividend distributions of certain non-U.S. Caterpillar subsidiaries, and Caterpillar SARL (CSARL) and related structures. On March 2-3, 2017, agents with the Department of Commerce, the Federal Deposit Insurance Corporation and the Internal Revenue Service executed search and seizure warrants at three facilities of the Company in the Peoria, Illinois area, including its former corporate headquarters. The warrants identify, and agents seized, documents and information related to, among other things, the export of products from the United States, the movement of products between the United States and Switzerland, the relationship between Caterpillar Inc. and CSARL, and sales outside the United States. It is the Company’s understanding that the warrants, which concern both tax and export activities, are related to the ongoing grand jury investigation. The Company is continuing to cooperate with this investigation. The Company is unable to predict the outcome or reasonably estimate any potential loss; however, we currently believe that this matter will not have a material adverse effect on the Company’s consolidated results of operations, financial position or liquidity.

On March 20, 2014, Brazil’s Administrative Council for Economic Defense (CADE) published a Technical Opinion which named 18 companies and over 100 individuals as defendants, including two subsidiaries of Caterpillar Inc., MGE - Equipamentos e Serviços Ferroviários Ltda. (MGE) and Caterpillar Brasil Ltda. The publication of the Technical Opinion opened CADE's official administrative investigation into allegations that the defendants participated in anticompetitive bid activity for the construction and maintenance of metro and train networks in Brazil. While companies cannot be held criminally liable for anticompetitive conduct in Brazil, criminal charges have been brought against one current employee of MGE and two former employees of MGE involving the same conduct alleged by CADE. On July 8, 2019, CADE found MGE, one of its current employees and two of its former employees liable for anticompetitive conduct. CBL was dismissed from the proceeding without any finding of liability. MGE intends to appeal CADE's findings. We currently believe that this matter will not have a material adverse effect on the Company's consolidated results of operations, financial position or liquidity.

In addition, we are involved in other unresolved legal actions that arise in the normal course of business. The most prevalent of these unresolved actions involve disputes related to product design, manufacture and performance liability (including claimed asbestos exposure), contracts, employment issues, environmental matters, intellectual property rights, taxes (other than income taxes) and securities laws. The aggregate range of reasonably possible losses in excess of accrued liabilities, if any, associated with these unresolved legal actions is not material. In some cases, we cannot reasonably estimate a range of loss because there is insufficient information regarding the matter. However, we believe there is no more than a remote chance that any liability arising from these matters would be material. Although it is not possible to predict with certainty the outcome of these unresolved legal actions, we believe that these actions will not individually or in the aggregate have a material adverse effect on our consolidated results of operations, financial position or liquidity.

Order Backlog

At the end of the second quarter of 2019, the dollar amount of backlog believed to be firm was approximately $15.0 billion, about $1.9 billion lower than the first quarter of 2019. The order backlog decreased across the three primary segments, with the largest declines in Construction Industries and Resource Industries. In Construction Industries, the backlog decline was the result of slowing orders due to growth in dealer inventory. It is not uncommon for the construction order backlog to decline during the second quarter selling season. In Resource Industries, we improved throughput in factories and improved lead times. In addition, orders can vary due to timing of large projects. Compared with the second quarter of 2018, the order backlog decreased about $2.7 billion with decreases across the three primary segments. Of the total backlog at June 30, 2019, approximately $3.6 billion was not expected to be filled in the following twelve months.

NON-GAAP FINANCIAL MEASURES

The following definitions are provided for the non-GAAP financial measures used in this report. These non-GAAP financial measures have no standardized meaning prescribed by U.S. GAAP and therefore are unlikely to be comparable to the calculation of similar measures for other companies. Management does not intend for these items to be considered in isolation or as a substitute for the related GAAP measures.

We believe it is important to separately quantify the profit impact of several significant items in order for our results to be meaningful to our readers. These items consist of a discrete tax benefit related to U.S. tax reform in the first quarter of 2019 and 2018 restructuring costs, which were incurred to generate longer-term benefits. We do not consider these items indicative of earnings from ongoing business activities and believe the non-GAAP measures will provide investors with useful perspective on underlying business results and trends and aids with assessing our period-over-period results. In addition, we provide a calculation of ME&T free cash flow as we believe it is an important measure for investors to determine the cash generation available for financing activities including debt repayments, dividends and share repurchases.

Reconciliations of adjusted operating profit margin to the most directly comparable GAAP measure, operating profit as a percent of sales and revenues are as follows:

	Three Months Ended June 30		Six Months Ended June 30	
	2019	2018	2019	2018
Operating profit as a percent of total sales and revenues	15.3%	15.5%	15.8%	15.9%
Restructuring costs [1]	—	0.8%	—	0.7%
Adjusted operating profit margin	15.3%	16.3%	15.8%	16.6%

[1] 2019 restructuring costs are not material.

Reconciliations of adjusted profit per share to the most directly comparable GAAP measure, profit per share - diluted are as follows:

	Three Months Ended June 30		Six Months Ended June 30	
	2019	2018	2019	2018
Profit per share - diluted	$ 2.83	$ 2.82	$ 6.08	$ 5.56
Per share U.S. tax reform impact	$ —	$ —	$ (0.31)	$ —
Per share restructuring costs[1]	$ —	$ 0.15	$ —	$ 0.23
Adjusted profit per share	$ 2.83	$ 2.97	$ 5.77	$ 5.79

[1] At estimated annual tax rate of 24 percent. 2019 restructuring costs are not material.

Reconciliations of ME&T free cash flow to the most directly comparable GAAP measure, net cash provided by operating activities are as follows:

Millions of dollars	Six Months Ended June 30	
	2019	2018
ME&T net cash provided by operating activities [1]	$ 2,820	$ 3,032
ME&T capital expenditures	$ (492)	$ (554)
ME&T free cash flow	$ 2,328	$ 2,478

[1] See reconciliation of ME&T net cash provided by operating activities to consolidated net cash provided by operating activities on pages 92-93.

Supplemental Consolidating Data

We are providing supplemental consolidating data for the purpose of additional analysis. The data has been grouped as follows:

Consolidated – Caterpillar Inc. and its subsidiaries.

Machinery, Energy & Transportation – Caterpillar defines Machinery, Energy & Transportation as it is presented in the supplemental data as Caterpillar Inc. and its subsidiaries with Financial Products accounted for on the equity basis. Machinery, Energy & Transportation information relates to the design, manufacturing and marketing of our products. Financial Products' information relates to the financing to customers and dealers for the purchase and lease of Caterpillar and other equipment. The nature of these businesses is different, especially with regard to the financial position and cash flow items. Caterpillar management utilizes this presentation internally to highlight these differences. We also believe this presentation will assist readers in understanding our business.

Financial Products – Our finance and insurance subsidiaries, primarily Cat Financial and Insurance Services.

Consolidating Adjustments – Eliminations of transactions between Machinery, Energy & Transportation and Financial Products.

Pages 86 to 93 reconcile Machinery, Energy & Transportation with Financial Products on the equity basis to Caterpillar Inc. consolidated financial information.

Caterpillar Inc.
Supplemental Data for Results of Operations
For the Three Months Ended June 30, 2019
(Unaudited)
(Millions of dollars)

		Supplemental Consolidating Data			
	Consolidated	Machinery, Energy & Transportation [1]	Financial Products	Consolidating Adjustments	
Sales and revenues:					
Sales of Machinery, Energy & Transportation	$ 13,671	$ 13,671	$ —	$ —	
Revenues of Financial Products	761	—	894	(133)	[2]
Total sales and revenues	14,432	13,671	894	(133)	
Operating costs:					
Cost of goods sold	9,941	9,943	—	(2)	[3]
Selling, general and administrative expenses	1,309	1,102	209	(2)	[3]
Research and development expenses	441	441	—	—	
Interest expense of Financial Products	192	—	201	(9)	[4]
Other operating (income) expenses	336	1	341	(6)	[3]
Total operating costs	12,219	11,487	751	(19)	
Operating profit	2,213	2,184	143	(114)	
Interest expense excluding Financial Products	103	105	—	(2)	[4]
Other income (expense)	68	(63)	19	112	[5]
Consolidated profit before taxes	2,178	2,016	162	—	
Provision (benefit) for income taxes	565	502	63	—	
Profit of consolidated companies	1,613	1,514	99	—	
Equity in profit (loss) of unconsolidated affiliated companies	6	6	—	—	
Equity in profit of Financial Products' subsidiaries	—	94	—	(94)	[6]
Profit of consolidated and affiliated companies	1,619	1,614	99	(94)	
Less: Profit (loss) attributable to noncontrolling interests	(1)	(6)	5	—	
Profit [7]	$ 1,620	$ 1,620	$ 94	$ (94)	

1 Represents Caterpillar Inc. and its subsidiaries with Financial Products accounted for on the equity basis.
2 Elimination of Financial Products' revenues earned from Machinery, Energy & Transportation.
3 Elimination of net expenses recorded by Machinery, Energy & Transportation paid to Financial Products.
4 Elimination of interest expense recorded between Financial Products and Machinery, Energy & Transportation.
5 Elimination of discount recorded by Machinery, Energy & Transportation on receivables sold to Financial Products and of interest earned between Machinery, Energy & Transportation and Financial Products.
6 Elimination of Financial Products' profit due to equity method of accounting.
7 Profit attributable to common shareholders.

Caterpillar Inc.
Supplemental Data for Results of Operations
For the Six Months Ended June 30, 2019
(Unaudited)
(Millions of dollars)

		Supplemental Consolidating Data		
	Consolidated	Machinery, Energy & Transportation [1]	Financial Products	Consolidating Adjustments
Sales and revenues:				
Sales of Machinery, Energy & Transportation	$ 26,395	$ 26,395	$ —	$ —
Revenues of Financial Products	1,503	—	1,764	(261) [2]
Total sales and revenues	27,898	26,395	1,764	(261)
Operating costs:				
Cost of goods sold	18,944	18,946	—	(2) [3]
Selling, general and administrative expenses	2,628	2,229	401	(2) [3]
Research and development expenses	876	876	—	—
Interest expense of Financial Products	382	—	401	(19) [4]
Other operating (income) expenses	648	11	654	(17) [3]
Total operating costs	23,478	22,062	1,456	(40)
Operating profit	4,420	4,333	308	(221)
Interest expense excluding Financial Products	206	215	—	(9) [4]
Other income (expense)	228	(44)	60	212 [5]
Consolidated profit before taxes	4,442	4,074	368	—
Provision (benefit) for income taxes	952	837	115	—
Profit of consolidated companies	3,490	3,237	253	—
Equity in profit (loss) of unconsolidated affiliated companies	13	13	—	—
Equity in profit of Financial Products' subsidiaries	—	242	—	(242) [6]
Profit of consolidated and affiliated companies	3,503	3,492	253	(242)
Less: Profit (loss) attributable to noncontrolling interests	2	(9)	11	—
Profit [7]	$ 3,501	$ 3,501	$ 242	$ (242)

1 Represents Caterpillar Inc. and its subsidiaries with Financial Products accounted for on the equity basis.
2 Elimination of Financial Products' revenues earned from Machinery, Energy & Transportation.
3 Elimination of net expenses recorded by Machinery, Energy & Transportation paid to Financial Products.
4 Elimination of interest expense recorded between Financial Products and Machinery, Energy & Transportation.
5 Elimination of discount recorded by Machinery, Energy & Transportation on receivables sold to Financial Products and of interest earned between Machinery, Energy & Transportation and Financial Products.
6 Elimination of Financial Products' profit due to equity method of accounting.
7 Profit attributable to common shareholders.

Caterpillar Inc.
Supplemental Data for Results of Operations
For the Three Months Ended June 30, 2018
(Unaudited)
(Millions of dollars)

		Supplemental Consolidating Data			
	Consolidated	Machinery, Energy & Transportation [1]	Financial Products	Consolidating Adjustments	
Sales and revenues:					
Sales of Machinery, Energy & Transportation	$ 13,279	$ 13,279	$ —	$ —	
Revenues of Financial Products	732	—	849	(117)	[2]
Total sales and revenues	14,011	13,279	849	(117)	
Operating costs:					
Cost of goods sold	9,422	9,422	—	—	
Selling, general and administrative expenses	1,440	1,223	223	(6)	[3]
Research and development expenses	462	462	—	—	
Interest expense of Financial Products	182	—	191	(9)	[4]
Other operating (income) expenses	338	38	306	(6)	[3]
Total operating costs	11,844	11,145	720	(21)	
Operating profit	2,167	2,134	129	(96)	
Interest expense excluding Financial Products	102	111	—	(9)	[4]
Other income (expense)	121	27	7	87	[5]
Consolidated profit before taxes	2,186	2,050	136	—	
Provision (benefit) for income taxes	490	457	33	—	
Profit of consolidated companies	1,696	1,593	103	—	
Equity in profit (loss) of unconsolidated affiliated companies	9	9	—	—	
Equity in profit of Financial Products' subsidiaries	—	98	—	(98)	[6]
Profit of consolidated and affiliated companies	1,705	1,700	103	(98)	
Less: Profit (loss) attributable to noncontrolling interests	(2)	(7)	5	—	
Profit [7]	$ 1,707	$ 1,707	$ 98	$ (98)	

1 Represents Caterpillar Inc. and its subsidiaries with Financial Products accounted for on the equity basis.
2 Elimination of Financial Products' revenues earned from Machinery, Energy & Transportation.
3 Elimination of net expenses recorded by Machinery, Energy & Transportation paid to Financial Products.
4 Elimination of interest expense recorded between Financial Products and Machinery, Energy & Transportation.
5 Elimination of discount recorded by Machinery, Energy & Transportation on receivables sold to Financial Products and of interest earned between Machinery, Energy & Transportation and Financial Products.
6 Elimination of Financial Products' profit due to equity method of accounting.
7 Profit attributable to common shareholders.

Caterpillar Inc.
Supplemental Data for Results of Operations
For the Six Months Ended June 30, 2018
(Unaudited)
(Millions of dollars)

		Supplemental Consolidating Data		
	Consolidated	Machinery, Energy & Transportation [1]	Financial Products	Consolidating Adjustments
Sales and revenues:				
Sales of Machinery, Energy & Transportation	$ 25,429	$ 25,429	$ —	$ —
Revenues of Financial Products	1,441	—	1,660	(219) [2]
Total sales and revenues	26,870	25,429	1,660	(219)
Operating costs:				
Cost of goods sold	17,988	17,988	—	—
Selling, general and administrative expenses	2,716	2,310	412	(6) [3]
Research and development expenses	905	905	—	—
Interest expense of Financial Products	348	—	364	(16) [4]
Other operating (income) expenses	638	37	616	(15) [3]
Total operating costs	22,595	21,240	1,392	(37)
Operating profit	4,275	4,189	268	(182)
Interest expense excluding Financial Products	203	223	—	(20) [4]
Other income (expense)	248	81	5	162 [5]
Consolidated profit before taxes	4,320	4,047	273	—
Provision (benefit) for income taxes	962	898	64	—
Profit of consolidated companies	3,358	3,149	209	—
Equity in profit (loss) of unconsolidated affiliated companies	14	14	—	—
Equity in profit of Financial Products' subsidiaries	—	200	—	(200) [6]
Profit of consolidated and affiliated companies	3,372	3,363	209	(200)
Less: Profit (loss) attributable to noncontrolling interests	—	(9)	9	—
Profit [7]	$ 3,372	$ 3,372	$ 200	$ (200)

1 Represents Caterpillar Inc. and its subsidiaries with Financial Products accounted for on the equity basis.
2 Elimination of Financial Products' revenues earned from Machinery, Energy & Transportation.
3 Elimination of net expenses recorded by Machinery, Energy & Transportation paid to Financial Products.
4 Elimination of interest expense recorded between Financial Products and Machinery, Energy & Transportation.
5 Elimination of discount recorded by Machinery, Energy & Transportation on receivables sold to Financial Products and of interest earned between Machinery, Energy & Transportation and Financial Products.
6 Elimination of Financial Products' profit due to equity method of accounting.
7 Profit attributable to common shareholders.

Caterpillar Inc.
Supplemental Data for Financial Position
At June 30, 2019
(Unaudited)
(Millions of dollars)

		Supplemental Consolidating Data		
	Consolidated	Machinery, Energy & Transportation [1]	Financial Products	Consolidating Adjustments
Assets				
Current assets:				
Cash and short-term investments	$ 7,429	$ 6,528	$ 901	$ —
Receivables – trade and other	8,996	4,312	488	4,196 [2,3]
Receivables – finance	9,539	—	15,184	(5,645) [3]
Prepaid expenses and other current assets	1,818	1,376	482	(40) [4]
Inventories	12,007	12,007	—	—
Total current assets	39,789	24,223	17,055	(1,489)
Property, plant and equipment – net	13,172	8,671	4,501	—
Long-term receivables – trade and other	1,154	313	181	660 [2,3]
Long-term receivables – finance	12,461	—	13,129	(668) [3]
Investments in Financial Products subsidiaries	—	3,953	—	(3,953) [5]
Noncurrent deferred and refundable income taxes	1,473	2,008	112	(647) [6]
Intangible assets	1,733	1,733	—	—
Goodwill	6,211	6,211	—	—
Other assets	3,194	1,586	1,658	(50) [7]
Total assets	$ 79,187	$ 48,698	$ 36,636	$ (6,147)
Liabilities				
Current liabilities:				
Short-term borrowings	$ 5,266	$ —	$ 5,266	$ —
Short-term borrowings with consolidated companies	—	—	1,287	(1,287) [8]
Accounts payable	7,022	6,690	494	(162) [9]
Accrued expenses	3,789	3,388	401	—
Accrued wages, salaries and employee benefits	1,411	1,378	33	—
Customer advances	1,263	1,263	—	—
Dividends payable	579	579	—	—
Other current liabilities	2,157	1,716	493	(52) [6,10]
Long-term debt due within one year	6,248	13	6,235	—
Total current liabilities	27,735	15,027	14,209	(1,501)
Long-term debt due after one year	24,764	7,665	17,107	(8) [8]
Liability for postemployment benefits	7,448	7,448	—	—
Other liabilities	4,362	3,680	1,367	(685) [6]
Total liabilities	64,309	33,820	32,683	(2,194)
Commitments and contingencies				
Shareholders' equity				
Common stock	5,822	5,822	919	(919) [5]
Treasury stock	(22,467)	(22,467)	—	—
Profit employed in the business	32,981	32,981	3,689	(3,689) [5]
Accumulated other comprehensive income (loss)	(1,499)	(1,499)	(819)	819 [5]
Noncontrolling interests	41	41	164	(164) [5]
Total shareholders' equity	14,878	14,878	3,953	(3,953)
Total liabilities and shareholders' equity	$ 79,187	$ 48,698	$ 36,636	$ (6,147)

[1] Represents Caterpillar Inc. and its subsidiaries with Financial Products accounted for on the equity basis.
[2] Elimination of receivables between Machinery, Energy & Transportation and Financial Products.

[3] Reclassification of Machinery, Energy & Transportation's trade receivables purchased by Financial Products and Financial Products' wholesale inventory receivables.
[4] Elimination of Machinery, Energy & Transportation's insurance premiums that are prepaid to Financial Products.
[5] Elimination of Financial Products' equity which is accounted for by Machinery, Energy & Transportation on the equity basis.
[6] Reclassification reflecting required netting of deferred tax assets / liabilities by taxing jurisdiction.
[7] Elimination of other intercompany assets between Machinery, Energy & Transportation and Financial Products.
[8] Elimination of debt between Machinery, Energy & Transportation and Financial Products.
[9] Elimination of payables between Machinery, Energy & Transportation and Financial Products.
[10] Elimination of prepaid insurance in Financial Products' other liabilities.

Caterpillar Inc.
Supplemental Data for Financial Position
At December 31, 2018
(Unaudited)
(Millions of dollars)

		Supplemental Consolidating Data		
	Consolidated	Machinery, Energy & Transportation [1]	Financial Products	Consolidating Adjustments
Assets				
Current assets:				
Cash and short-term investments	$ 7,857	$ 6,968	$ 889	$ —
Receivables – trade and other	8,802	4,677	401	3,724 [2,3]
Receivables – finance	8,650	—	13,989	(5,339) [3]
Prepaid expenses and other current assets	1,765	1,227	583	(45) [4]
Inventories	11,529	11,529	—	—
Total current assets	38,603	24,401	15,862	(1,660)
Property, plant and equipment – net	13,574	9,085	4,489	—
Long-term receivables – trade and other	1,161	302	204	655 [2,3]
Long-term receivables – finance	13,286	—	13,951	(665) [3]
Investments in Financial Products subsidiaries	—	3,672	—	(3,672) [5]
Noncurrent deferred and refundable income taxes	1,439	2,015	116	(692) [6]
Intangible assets	1,897	1,897	—	—
Goodwill	6,217	6,217	—	—
Other assets	2,332	886	1,446	—
Total assets	$ 78,509	$ 48,475	$ 36,068	$ (6,034)
Liabilities				
Current liabilities:				
Short-term borrowings	$ 5,723	$ —	$ 5,723	$ —
Short-term borrowings with consolidated companies	—	—	1,500	(1,500) [7]
Accounts payable	7,051	6,972	194	(115) [8]
Accrued expenses	3,573	3,212	361	—
Accrued wages, salaries and employee benefits	2,384	2,350	34	—
Customer advances	1,243	1,243	—	—
Dividends payable	495	495	—	—
Other current liabilities	1,919	1,532	433	(46) [6,9]
Long-term debt due within one year	5,830	10	5,820	—
Total current liabilities	28,218	15,814	14,065	(1,661)
Long-term debt due after one year	25,000	8,015	16,995	(10) [7]
Liability for postemployment benefits	7,455	7,455	—	—
Other liabilities	3,756	3,111	1,336	(691) [6]
Total liabilities	64,429	34,395	32,396	(2,362)
Commitments and contingencies				
Shareholders' equity				
Common stock	5,827	5,827	919	(919) [5]
Treasury stock	(20,531)	(20,531)	—	—
Profit employed in the business	30,427	30,427	3,543	(3,543) [5]
Accumulated other comprehensive income (loss)	(1,684)	(1,684)	(943)	943 [5]
Noncontrolling interests	41	41	153	(153) [5]
Total shareholders' equity	14,080	14,080	3,672	(3,672)
Total liabilities and shareholders' equity	$ 78,509	$ 48,475	$ 36,068	$ (6,034)

[1] Represents Caterpillar Inc. and its subsidiaries with Financial Products accounted for on the equity basis.

[2] Elimination of receivables between Machinery, Energy & Transportation and Financial Products.
[3] Reclassification of Machinery, Energy & Transportation's trade receivables purchased by Financial Products and Financial Products' wholesale inventory receivables.
[4] Elimination of Machinery, Energy & Transportation's insurance premiums that are prepaid to Financial Products.
[5] Elimination of Financial Products' equity which is accounted for by Machinery, Energy & Transportation on the equity basis.
[6] Reclassification reflecting required netting of deferred tax assets / liabilities by taxing jurisdiction.
[7] Elimination of debt between Machinery, Energy & Transportation and Financial Products.
[8] Elimination of payables between Machinery, Energy & Transportation and Financial Products.
[9] Elimination of prepaid insurance in Financial Products' other liabilities.

Caterpillar Inc.
Supplemental Data for Cash Flow
For the Six Months Ended June 30, 2019
(Unaudited)
(Millions of dollars)

		Supplemental Consolidating Data		
	Consolidated	Machinery, Energy & Transportation [1]	Financial Products	Consolidating Adjustments
Cash flow from operating activities:				
Profit of consolidated and affiliated companies	$ 3,503	$ 3,492	$ 253	$ (242) [2]
Adjustments for non-cash items:				
Depreciation and amortization	1,288	855	433	—
Undistributed profit of Financial Products	—	(242)	—	242 [3]
Other	440	264	(51)	227 [4]
Changes in assets and liabilities, net of acquisitions and divestitures:				
Receivables – trade and other	(166)	161	(5)	(322) [4, 5]
Inventories	(487)	(489)	—	2 [4]
Accounts payable	134	(94)	279	(51) [4]
Accrued expenses	151	158	(5)	(2) [4]
Accrued wages, salaries and employee benefits	(979)	(978)	(1)	—
Customer advances	14	18	—	(4) [4]
Other assets – net	(221)	(228)	57	(50) [4]
Other liabilities – net	32	(97)	75	54 [4]
Net cash provided by (used for) operating activities	3,709	2,820	1,035	(146)
Cash flow from investing activities:				
Capital expenditures – excluding equipment leased to others	(479)	(468)	(11)	—
Expenditures for equipment leased to others	(746)	(24)	(725)	3 [4]
Proceeds from disposals of leased assets and property, plant and equipment	422	97	367	(42) [4]
Additions to finance receivables	(6,181)	—	(7,027)	846 [5]
Collections of finance receivables	5,902	—	6,548	(646) [5]
Net intercompany purchased receivables	—	—	15	(15) [5]
Proceeds from sale of finance receivables	119	—	119	—
Net intercompany borrowings	—	213	1	(214) [6]
Investments and acquisitions (net of cash acquired)	(3)	(3)	—	—
Proceeds from sale of securities	170	13	157	—
Investments in securities	(243)	(12)	(231)	—
Other – net	(40)	(9)	(31)	—
Net cash provided by (used for) investing activities	(1,079)	(193)	(818)	(68)
Cash flow from financing activities:				
Dividends paid	(986)	(986)	—	—
Common stock issued, including treasury shares reissued	39	39	—	—
Common shares repurchased	(2,105)	(2,105)	—	—
Net intercompany borrowings	—	(1)	(213)	214 [6]
Proceeds from debt issued (original maturities greater than three months)	5,340	—	5,340	—
Payments on debt (original maturities greater than three months)	(4,901)	(4)	(4,897)	—
Short-term borrowings – net (original maturities three months or less)	(436)	—	(436)	—
Other – net	(2)	(2)	—	—
Net cash provided by (used for) financing activities	(3,051)	(3,059)	(206)	214
Effect of exchange rate changes on cash	(10)	(8)	(2)	—
Increase (decrease) in cash and short-term investments and restricted cash	(431)	(440)	9	—
Cash and short-term investments and restricted cash at beginning of period	7,890	6,994	896	—
Cash and short-term investments and restricted cash at end of period	$ 7,459	$ 6,554	$ 905	$ —

[1] Represents Caterpillar Inc. and its subsidiaries with Financial Products accounted for on the equity basis.
[2] Elimination of Financial Products' profit after tax due to equity method of accounting.
[3] Elimination of non-cash adjustment for the undistributed earnings from Financial Products.
[4] Elimination of non-cash adjustments and changes in assets and liabilities related to consolidated reporting.
[5] Reclassification of Financial Products' cash flow activity from investing to operating for receivables that arose from the sale of inventory.
[6] Elimination of net proceeds and payments to/from Machinery, Energy & Transportation and Financial Products.

Caterpillar Inc.
Supplemental Data for Cash Flow
For the Six Months Ended June 30, 2018
(Unaudited)
(Millions of dollars)

		Supplemental Consolidating Data		
	Consolidated	Machinery, Energy & Transportation [1]	Financial Products	Consolidating Adjustments
Cash flow from operating activities:				
Profit of consolidated and affiliated companies	$ 3,372	$ 3,363	$ 209	$ (200) [2]
Adjustments for non-cash items:				
Depreciation and amortization	1,367	933	434	—
Undistributed profit of Financial Products	—	(200)	—	200 [3]
Other	446	197	61	188 [4]
Changes in assets and liabilities, net of acquisitions and divestitures:				
Receivables – trade and other	(703)	136	(74)	(765) [4, 5]
Inventories	(1,208)	(1,186)	—	(22) [4]
Accounts payable	545	570	(56)	31 [4]
Accrued expenses	(31)	(40)	9	—
Accrued wages, salaries and employee benefits	(768)	(745)	(23)	—
Customer advances	(54)	(54)	—	—
Other assets – net	174	176	(10)	8 [4]
Other liabilities – net	(57)	(118)	69	(8) [4]
Net cash provided by (used for) operating activities	3,083	3,032	619	(568)
Cash flow from investing activities:				
Capital expenditures – excluding equipment leased to others	(645)	(550)	(95)	—
Expenditures for equipment leased to others	(883)	(4)	(919)	40 [4]
Proceeds from disposals of leased assets and property, plant and equipment	539	93	461	(15) [4]
Additions to finance receivables	(6,143)	—	(6,823)	680 [5, 7]
Collections of finance receivables	5,405	—	6,144	(739) [5]
Net intercompany purchased receivables	—	—	(608)	608 [5]
Proceeds from sale of finance receivables	124	—	124	—
Net intercompany borrowings	—	112	—	(112) [6]
Investments and acquisitions (net of cash acquired)	(348)	(348)	—	—
Proceeds from sale of businesses and investments (net of cash sold)	12	18	—	(6) [7]
Proceeds from sale of securities	168	10	158	—
Investments in securities	(318)	(19)	(299)	—
Other – net	21	24	(4)	1 [8]
Net cash provided by (used for) investing activities	(2,068)	(664)	(1,861)	457
Cash flow from financing activities:				
Dividends paid	(933)	(933)	—	—
Common stock issued, including treasury shares reissued	256	256	1	(1) [8]
Common shares repurchased	(1,250)	(1,250)	—	—
Net intercompany borrowings	—	—	(112)	112 [6]
Proceeds from debt issued (original maturities greater than three months)	4,307	—	4,307	—
Payments on debt (original maturities greater than three months)	(4,436)	(3)	(4,433)	—
Short-term borrowings – net (original maturities three months or less)	1,487	34	1,453	—
Other – net	(4)	(4)	—	—
Net cash provided by (used for) financing activities	(573)	(1,900)	1,216	111
Effect of exchange rate changes on cash	(68)	(61)	(7)	—
Increase (decrease) in cash and short-term investments and restricted cash	374	407	(33)	—
Cash and short-term investments and restricted cash at beginning of period	8,320	7,416	904	—
Cash and short-term investments and restricted cash at end of period	$ 8,694	$ 7,823	$ 871	$ —

[1] Represents Caterpillar Inc. and its subsidiaries with Financial Products accounted for on the equity basis.
[2] Elimination of Financial Products' profit after tax due to equity method of accounting.
[3] Elimination of non-cash adjustment for the undistributed earnings from Financial Products.
[4] Elimination of non-cash adjustments and changes in assets and liabilities related to consolidated reporting.
[5] Reclassification of Financial Products' cash flow activity from investing to operating for receivables that arose from the sale of inventory.
[6] Elimination of net proceeds and payments to/from Machinery, Energy & Transportation and Financial Products.
[7] Elimination of proceeds received from Financial Products related to Machinery, Energy & Transportation's sale of businesses and investments.
[8] Elimination of change in investment and common stock related to Financial Products.

Forward-looking Statements

Certain statements in this Form 10-Q relate to future events and expectations and are forward-looking statements within the meaning of the Private Securities Litigation Reform Act of 1995. Words such as “believe,” “estimate,” “will be,” “will,” “would,” “expect,” “anticipate,” “plan,” “forecast,” “target,” “guide,” “project,” “intend,” “could,” “should” or other similar words or expressions often identify forward-looking statements. All statements other than statements of historical fact are forward-looking statements, including, without limitation, statements regarding our outlook, projections, forecasts or trend descriptions. These statements do not guarantee future performance and speak only as of the date they are made, and we do not undertake to update our forward-looking statements.

Caterpillar’s actual results may differ materially from those described or implied in our forward-looking statements based on a number of factors, including, but not limited to: (i) global and regional economic conditions and economic conditions in the industries we serve; (ii) commodity price changes, material price increases, fluctuations in demand for our products or significant shortages of material; (iii) government monetary or fiscal policies; (iv) political and economic risks, commercial instability and events beyond our control in the countries in which we operate; (v) international trade policies and their impact on demand for our products and our competitive position, including the imposition of new tariffs or changes in existing tariff rates; (vi) our ability to develop, produce and market quality products that meet our customers’ needs; (vii) the impact of the highly competitive environment in which we operate on our sales and pricing; (viii) information technology security threats and computer crime; (ix) inventory management decisions and sourcing practices of our dealers and our OEM customers; (x) a failure to realize, or a delay in realizing, all of the anticipated benefits of our acquisitions, joint ventures or divestitures; (xi) union disputes or other employee relations issues; (xii) adverse effects of unexpected events including natural disasters; (xiii) disruptions or volatility in global financial markets limiting our sources of liquidity or the liquidity of our customers, dealers and suppliers; (xiv) failure to maintain our credit ratings and potential resulting increases to our cost of borrowing and adverse effects on our cost of funds, liquidity, competitive position and access to capital markets; (xv) our Financial Products segment’s risks associated with the financial services industry; (xvi) changes in interest rates or market liquidity conditions; (xvii) an increase in delinquencies, repossessions or net losses of Cat Financial’s customers; (xviii) currency fluctuations; (xix) our or Cat Financial’s compliance with financial and other restrictive covenants in debt agreements; (xx) increased pension plan funding obligations; (xxi) alleged or actual violations of trade or anti-corruption laws and regulations; (xxii) additional tax expense or exposure, including the impact of U.S. tax reform; (xxiii) significant legal proceedings, claims, lawsuits or government investigations; (xxiv) new regulations or changes in financial services regulations; (xxv) compliance with environmental laws and regulations; and (xxvi) other factors described in more detail in Caterpillar’s Forms 10-Q, 10-K and other filings with the Securities and Exchange Commission.

Item 3. Quantitative and Qualitative Disclosures About Market Risk

The information required by this Item is incorporated by reference from Note 5 – “Derivative financial instruments and risk management” included in Part I, Item 1 and Management’s Discussion and Analysis included in Part I, Item 2 of this Form 10-Q.

Item 4. Controls and Procedures

Evaluation of disclosure controls and procedures

An evaluation was performed under the supervision and with the participation of the company’s management, including the Chief Executive Officer (CEO) and Chief Financial Officer (CFO), of the effectiveness of the design and operation of the company’s disclosure controls and procedures, as that term is defined in Rule 13a-15(e) under the Securities Exchange Act of 1934, as amended, as of the end of the period covered by this quarterly report. Based on that evaluation, the CEO and CFO concluded that the company’s disclosure controls and procedures were effective as of the end of the period covered by this quarterly report.

Changes in internal control over financial reporting

During the second quarter of 2019, there has been no change in the company's internal control over financial reporting that has materially affected, or is reasonably likely to materially affect, the company’s internal control over financial reporting.

PART II. OTHER INFORMATION

Item 1. Legal Proceedings

The information required by this Item is incorporated by reference from Note 14 – "Environmental and legal matters" included in Part I, Item 1 of this Form 10-Q.

Item 2. Unregistered Sales of Equity Securities and Use of Proceeds

Issuer Purchases of Equity Securities

Period	Total Number of Shares Purchased[2,3]	Average Price Paid per Share[2,3]	Total Number of Shares Purchased as Part of Publicly Announced Program	Approximate Dollar Value of Shares that may yet be Purchased under the Program (in billions)[1]
April 1-30, 2019	—	$ —	—	$ 9.249
May 1-31, 2019	7,374,648	$ 133.05	7,374,648	$ 8.268
June 1-30, 2019	2,382,478	$ 127.78	2,382,478	$ 7.963
Total	9,757,126	$ 131.76	9,757,126	

[1] In July 2018, the Board approved a share repurchase authorization of up to $10.0 billion of Caterpillar common stock effective January 1, 2019, with no expiration (the 2018 Authorization). As of June 30, 2019, approximately $8.0 billion remained available under the 2018 Authorization.

[2] During the second quarter of 2019, we entered into an accelerated stock repurchase (ASR) with a third-party financial institution to purchase $750 million of our common stock. In May 2019, upon payment of the $750 million to the financial institution, we received 5.0 million shares. In July 2019, upon final settlement of the ASR, we received an additional 0.8 million shares. In total, we repurchased 5.8 million shares under this ASR at an average price per share of $128.29.

[3] In May and June of 2019, we repurchased 2.4 million and 2.4 million shares, respectively, for an aggregate of $604 million in open market transactions at an average price per share of $126.47 and $127.78, respectively.

Non-U.S. Employee Stock Purchase Plans

As of June 30, 2019, we had 26 employee stock purchase plans (the "EIP Plans") that are administered outside the United States for our non-U.S. employees, which had approximately 12,000 active participants in the aggregate. During the second quarter of 2019, approximately 105,000 shares of Caterpillar common stock were purchased by the EIP Plans pursuant to the terms of such plans.

Item 6. Exhibits

31.1	Certification of D. James Umpleby III, Chief Executive Officer of Caterpillar Inc., as required pursuant to Section 302 of the Sarbanes-Oxley Act of 2002.
31.2	Certification of Andrew R.J. Bonfield, Chief Financial Officer of Caterpillar Inc., as required pursuant to Section 302 of the Sarbanes-Oxley Act of 2002.
32	Certification of D. James Umpleby III, Chief Executive Officer of Caterpillar Inc. and Andrew R.J. Bonfield, Chief Financial Officer of Caterpillar Inc., as required pursuant to Section 906 of the Sarbanes-Oxley Act of 2002.
101.SCH	XBRL Taxonomy Extension Schema Document
101.CAL	XBRL Taxonomy Extension Calculation Linkbase Document
101.DEF	XBRL Taxonomy Extension Definition Linkbase Document
101.LAB	XBRL Taxonomy Extension Label Linkbase Document
101.PRE	XBRL Taxonomy Extension Presentation Linkbase Document

SIGNATURES

Pursuant to the requirements of the Securities Exchange Act of 1934, the Registrant has duly caused this report to be signed on its behalf by the undersigned thereunto duly authorized.

CATERPILLAR INC.

August 1, 2019	*/s/ D. James Umpleby III* D. James Umpleby III	Chairman & Chief Executive Officer
August 1, 2019	*/s/ Andrew R.J. Bonfield* Andrew R.J. Bonfield	Chief Financial Officer
August 1, 2019	*/s/ Suzette M. Long* Suzette M. Long	General Counsel & Corporate Secretary
August 1, 2019	*/s/ G. Michael Marvel* G. Michael Marvel	Chief Accounting Officer

EXHIBIT 31.1

SECTION 302 CERTIFICATION

I, D. James Umpleby III, certify that:

1. I have reviewed this quarterly report on Form 10-Q of Caterpillar Inc.;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;

4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:

   a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

   b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

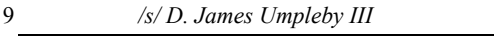
   c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

   d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and

5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent function):

   a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and

   b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

August 1, 2019 *  * Chief Executive Officer

D. James Umpleby III

EXHIBIT 31.2

SECTION 302 CERTIFICATION

I, Andrew R.J. Bonfield, certify that:

1. I have reviewed this quarterly report on Form 10-Q of Caterpillar Inc.;

2. Based on my knowledge, this report does not contain any untrue statement of a material fact or omit to state a material fact necessary to make the statements made, in light of the circumstances under which such statements were made, not misleading with respect to the period covered by this report;

3. Based on my knowledge, the financial statements, and other financial information included in this report, fairly present in all material respects the financial condition, results of operations and cash flows of the registrant as of, and for, the periods presented in this report;

4. The registrant's other certifying officer and I are responsible for establishing and maintaining disclosure controls and procedures (as defined in Exchange Act Rules 13a-15(e) and 15d-15(e)) and internal control over financial reporting (as defined in Exchange Act Rules 13a-15(f) and 15d-15(f)) for the registrant and have:

   a) designed such disclosure controls and procedures, or caused such disclosure controls and procedures to be designed under our supervision, to ensure that material information relating to the registrant, including its consolidated subsidiaries, is made known to us by others within those entities, particularly during the period in which this report is being prepared;

   b) designed such internal control over financial reporting, or caused such internal control over financial reporting to be designed under our supervision, to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles;

   c) evaluated the effectiveness of the registrant's disclosure controls and procedures and presented in this report our conclusions about the effectiveness of the disclosure controls and procedures, as of the end of the period covered by this report based on such evaluation; and

   d) disclosed in this report any change in the registrant's internal control over financial reporting that occurred during the registrant's most recent fiscal quarter (the registrant's fourth fiscal quarter in the case of an annual report) that has materially affected, or is reasonably likely to materially affect, the registrant's internal control over financial reporting; and

5. The registrant's other certifying officer and I have disclosed, based on our most recent evaluation of internal control over financial reporting, to the registrant's auditors and the audit committee of the registrant's board of directors (or persons performing the equivalent function):

   a) all significant deficiencies and material weaknesses in the design or operation of internal control over financial reporting which are reasonably likely to adversely affect the registrant's ability to record, process, summarize and report financial information; and

   b) any fraud, whether or not material, that involves management or other employees who have a significant role in the registrant's internal control over financial reporting.

August 1, 2019 ** Chief Financial Officer

Andrew R.J. Bonfield

EXHIBIT 32

CERTIFICATION PURSUANT TO
18 U.S.C. SECTION 1350
AS ADOPTED PURSUANT TO
SECTION 906 OF THE SARBANES-OXLEY ACT OF 2002

In connection with the quarterly report of Caterpillar Inc. (the "Company") on Form 10-Q for the period ending June 30, 2019 as filed with the Securities and Exchange Commission on the date hereof (the "Report"), the undersigned hereby certify pursuant to 18 U.S.C. Section 1350, as adopted pursuant to Section 906 of the Sarbanes-Oxley Act of 2002, that to the best of our knowledge:

(1) The Report fully complies with the requirements of section 13(a) or 15(d) of the Securities Exchange Act of 1934; and

(2) The information contained in the Report fairly presents, in all material respects, the financial condition and results of operations of the Company.

August 1, 2019 *\/s\/ D. James Umpleby III* Chief Executive Officer

D. James Umpleby III

August 1, 2019 *\/s\/ Andrew R.J. Bonfield* Chief Financial Officer

Andrew R.J. Bonfield

A signed original of this written statement required by Section 906 has been provided to Caterpillar Inc. and will be retained by Caterpillar Inc. and furnished to the Securities and Exchange Commission or its staff upon request.